

The Universal Right-Triangle Reflection Sequence: a word-length metric, effective universality, and the lamplighter structure of A396406

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Abstract

For a right triangle T in the plane with unequal rational legs, the group $W(T)$ generated by the three side-reflections acts on the orbit of T , and the orbit-growth sequence u_d^T (the number of triangles at word-distance d) is the subject of OEIS entry A396406. We prove that u_d^T is independent of the shape T through depth 32, and that this is sharp: the $(1, 2)$ triangle deviates at depth 33. The common sequence is the orbit growth of the *generic* right triangle, whose abstract reflection group is the group $W = \langle R_0, R_1, R_2 \mid R_i^2 = (R_0 R_1)^2 = 1 \rangle$. We show that the subgroup of W on which the linear part is a rotation is the wreath product $\mathbb{Z} \wr \mathbb{Z}$ of index 4, so the generic group is virtually $\mathbb{Z} \wr \mathbb{Z}$; the translation lattice is computed exactly as $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$ via a Mayer–Vietoris/Crowell identification of the translation module with the cyclic $\mathbb{Z}[Q]$ -ideal generated by $h = (c-1)(Y-1)$. From the resulting lamp model we derive a closed-form word-length metric whose geodesic length equals a relaxed length plus twice an explicit connectivity penalty. A triangle-independent symbolic normal form reduces every extra collision to the vanishing of a bounded-height integer Laurent polynomial at the rotation number $\zeta_T = (a+bi)/(a-bi)$; Gauss’s lemma then gives an effective all-depths universality theorem (no deviation at depth d if $c_T := N(a-bi, \text{suitably normalized}) > 2d$), and a finite modular certificate closes the gap to depth 32. The natural conjecture that universality holds at all depths is refuted: for every algebraic shape we exhibit an explicit word in $\ker \rho_T$ (a product of conjugated glide-reflection squares), and the first deviation depth n_T is sandwiched by $\max(33, c_T) \leq n_T \leq 24c_T + 8$; moreover n_T equals exactly half the length of the shortest kernel element (a shortest-vector problem in the ideal $2(t-1)\mu_T\mathbb{Z}[t^{\pm 1}]$ under the lamplighter metric) from which we extract a closed-form three-regime law, $n_T = 6c_T + e_T$, $9c_T - 3e_T$, or $3(c_T + e_T)$ according to the angle of the triangle, certified at $(1, 2)$ and validated on thirteen shapes. Finally, the lamplighter transfer yields a q -difference functional equation for the relaxed generating function of A396406; a connectivity-invariance squeeze of the true count between its pure-travel subcount and the relaxed series pins the exact growth rate unconditionally: $\beta_2 = 1.4916177871\dots$, the reciprocal square root of the smallest positive root of an explicit travel-block equation, which excludes the value $3/2$ suggested by finite-depth ratios. The eight length-10 relations and the universality of u_d through depth 22 are formally machine-verified in Lean 4; the depth-32 threshold and the deviation witnesses are certified by exact modular and rational arithmetic. Transcendence of the generating functions over $\mathbb{Q}(x)$, and of the number β_2 , lies outside the present scope; the appendices record the finite-horizon structural exclusions on the 43 known terms, their Lean certificates, and the conditional routes to transcendence that were explored.

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1 Introduction

Let T be a triangle with vertices in $\mathbb{Q}^2$, and for $i \in \{0, 1, 2\}$ let R_i be the reflection of $\mathbb{R}^2$ across the line carrying the side opposite vertex V_i . The group $W(T) = \langle R_0, R_1, R_2 \rangle \leq \text{Isom}(\mathbb{R}^2)$ acts on the orbit of T , and breadth-first search in the Cayley graph relative to $\{R_0, R_1, R_2\}$ stratifies that orbit by word-length. Writing $u_d = u_d^T$ for the number of distinct triangle-images at minimal word-length d , one obtains an integer sequence depending a priori on the shape of T . For right triangles with unequal legs this sequence is OEIS A396406. This paper determines its group-theoretic origin, its exact word-length metric, the precise extent to which it is shape-independent, and its exact exponential growth rate.

The generic group. When T is a non-isosceles right triangle, the two legs meet at a right angle, so $(R_0 R_1)^2 = 1$, while by Niven's theorem [32] the remaining two angles are irrational multiples of π and force no further relation among the reflections. The abstract group is therefore the Coxeter group

$$W = \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0, 1, 2), \ (R_0 R_1)^2 = 1 \rangle,$$

whose Poincaré series is $(1+t)^2/(1-t-t^2)$; consequently the free-group baseline gives $u_d = F_{d+3}$ for $1 \leq d \leq 9$, until the first affine collision. The shape T enters only through the affine representation $\rho_T: W \rightarrow \text{Aff}(\mathbb{R}^2)$ realizing the three reflections, and the sequence u_d^T records the layer sizes of $\rho_T(W)$. Our first structural result identifies the kernel of the generic representation. Placing the right angle at the origin and writing $\zeta_T = (a+bi)/(a-bi)$ for the rotation number of the hypotenuse reflection, every group element acts as $z \mapsto \varepsilon \zeta_T^k \sigma^\delta(z) + P(\zeta_T)$ for an integer Laurent polynomial P independent of T (Lemma, symbolic normal form). The pure-translation subgroup of the resulting symbolic group is computed exactly,

$$\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}],$$

by reducing modulo 2, where the two perpendicular legs collapse onto a single mirror and the group degenerates to the infinite dihedral group, which has no translations. This caps the glide-square lattice $2(t-1)\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ to an equality (Theorem, translation lattice).

Lamplighter structure and the word-length metric. The translation lattice exhibits the generic group as virtually a classical object: the subgroup $H \leq W_{\text{gen}}$ on which the linear part is a pure rotation t^k is normal of index 4 and isomorphic to $\mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z} = \mathbb{Z} \wr \mathbb{Z}$, so every right-triangle side-reflection group with transcendental leg ratio is virtually the wreath product $\mathbb{Z} \wr \mathbb{Z}$ (Theorem, lamplighter structure): metabelian-by-finite, hence amenable, yet of exponential growth, and not finitely presentable. Independently, the translation module is computed by Mayer–Vietoris for the free product $W = V_4 * \mathbb{Z}/2$ together with Crowell's exact sequence [17]: it is the $\mathbb{Z}[Q]$ -module $I_A \cap I_B$, which equals the cyclic ideal generated by the central element $h = (c-1)(Y-1)$ by exactness of the Bass–Serre tree of D_∞ [37] (Lemma, Mayer–Vietoris reduction). The generic kernel is then identified as the relative commutator subgroup $\ker \rho_{\text{gen}} = [T_\rho, T_\rho]$, exhibiting the generic group as the Q -relative metabelianization of $V_4 * \mathbb{Z}/2$ (Corollary, generic kernel). Reading the composition law in the edge basis turns the group into an explicit *lamp model*: a walker on $\mathbb{Z}$ deposits signed lamps on the edges it traverses, with a normal form $(\varepsilon, \delta, k, P)$ cut out by two congruences. From this model we obtain a closed-form word-length metric (Theorem, metric formula): the geodesic length is a minimum over crossing-number and site-pairing data, equal to a relaxed length plus exactly 2 for each isolated cycle that the connectivity (single-Eulerian-trail) constraint forbids. The metric is forced by a strand bound limiting each interface component to at most one up- and one

down-crossing per edge, leaving a single unbounded catalytic datum; it reproduces A396406 through u_{22} with no breadth-first search. The kernel depends only on the similarity class of T (Lemma, dilation invariance).

Universality and its sharp threshold. Because the symbolic normal form is triangle-independent and ζ_T is never a root of unity (Niven), two group elements collide under ρ_T only when their symbolic data agree and their translation polynomials agree at $t = \zeta_T$; an extra collision at depth d thus forces a nonzero integer Laurent polynomial of height $\leq 2d$ to vanish at ζ_T . By Gauss's lemma the shape's minimal polynomial $\mu_T(t) = c_T t^2 - e_T t + c_T$ must then divide a realized word-difference, so c_T divides both extreme coefficients and $c_T \leq 2d$. This yields an *effective all-depths universality theorem*: every shape with $c_T > 2d$ satisfies $u_d^T = u_d$ (Theorem, effective universality). The arithmetic of c_T , which is odd with all prime factors $\equiv 1 \pmod{4}$ and at least 5, leaves only a sparse finite set of candidate shapes at each depth, and a one-sided exact modular certificate over these candidates closes the remaining gap:

For every right triangle with positive unequal rational legs, $u_d^T = u_d$ for all $0 \leq d \leq 32$, and the bound is sharp: the $(1, 2)$ triangle satisfies $u_{33}^{(1,2)} = u_{33} - 30 < u_{33}$ (Theorem, sharp uniform universality).

Refutation of all-depths universality. The natural conjecture that $u_d^T = u_d$ for *all* d is false for every algebraic shape. Since $R_0 R_1$ is the half-turn about the right-angle vertex, $R_0 R_1 R_2$ is a glide reflection along the hypotenuse, and its square is the pure translation by $2(t^{-1} - 1)$; conjugating and multiplying these glide-squares realizes the entire ideal $2(t - 1)\mathbb{Z}[t^{\pm 1}]$. As that ideal contains a multiple of every μ_T , an explicit word

$$w_T = \tau^{c_T} r \tau^{-e_T} r \tau^{c_T} r^{-2}, \quad \tau = (R_0 R_1 R_2)^2, \quad r = R_x R_h,$$

realizes the translation $2(t^{-1} - 1)\mu_T(t)$, which is nonzero in the generic group but evaluates to 0 at ζ_T , hence lies in $\ker \rho_T$ (Theorem, non-universality). Thus $\ker \rho_T$ is, for every algebraic shape, the principal ideal $2(t - 1)\mu_T \mathbb{Z}[t^{\pm 1}]$, isomorphic to $\mathbb{Z}[t^{\pm 1}]$ and normally generated by the single witness w_T (Corollary, kernel classification). The first deviation depth is therefore finite and pinned to linear growth in the arithmetic complexity of the shape,

$$\max(33, c_T) \leq n_T \leq 6(2c_T + |e_T|) + 8 \leq 24c_T + 8$$

(Corollary, deviation-depth sandwich), with $n_{(1,2)} = 33$ realizing the lower bound; the deviation depth is itself a shortest-vector length in the lamplighter word metric. Consequently the universal sequence is exactly the orbit growth of the generic right triangle, realized precisely by the (uncountably many) transcendental leg ratios, while the isosceles case ($c_T = 1$, deviation at depth 4) is the extreme degeneration to a wallpaper group (Corollary, transcendence dichotomy).

The exact growth rate. The lamp model linearizes the metric into a transfer with one catalytic variable, but the catalytic variable enters through a *dilation* $t \mapsto q^2 t$, so the resulting functional equation for the relaxed generating function is a linear q -difference equation, not a polynomial-kernel equation; the Bousquet-Mélou-Jehanne framework does not apply and the solution is q -holonomic. Unfolding the equation telescopes the relaxed series into explicit rapidly-convergent q -series blocks, and the dominant singularity is the smallest positive root q^* of the travel-block equation $\Sigma_1(q) = 1$. We prove that this rate is also the *true* rate, unconditionally: the travel block is connectivity-invariant

(each pure-translation run is a single Euler trail and incurs no penalty), so the true series shares it verbatim; squeezing the true count between its pure-travel subcount and the relaxed series gives

$$\beta_2 = 1/\sqrt{q^*} = 1.4916177871143742268\dots$$

exactly (Proposition, asymptotic ratio), with no dependence on any analytic conjecture. In particular $\beta_2 < 3/2$, excluding the value that the finite-depth ratios, which converge at rate $O(1/d)$ with a period-4 oscillation, had suggested.

What is conditional and what is open. The growth rate $\beta_2 = 1.4916177871\dots$ is determined unconditionally in Section 10 as $1/\sqrt{q^*}$, where q^* is the least positive root of $\Sigma_1(q) = 1$; in particular $\beta_2 \neq 3/2$. Transcendence questions are treated in the companion paper [12], where each catalytic building block is proved transcendental unconditionally, by a Pólya–Carlson natural boundary; the growth series of A396406 and of its relaxed companion are transcendental over $\mathbb{Q}(x)$, with no hypothesis: that paper proves its transfer model (M) and its junction pairing statement (R-J) (that the junction pairing does not vanish at infinitely many travel poles), the latter with computer assistance in the certificate ranges [12]. The result is not needed for anything proved in this paper itself. It explains the absence, recorded below, of any low-order linear, holonomic, or low-degree algebraic relation over the 43 known terms. The transcendence of the *number* β_2 itself remains open. In higher dimensions (the reflection groups of Euclidean orthoschemes (path simplices), of which the right triangle is the planar case) the abstract right-angled Coxeter envelope has *rational* growth with rate $r_n = 1 + 2 \cos \frac{2\pi}{n+3}$, but the geometric group is an amenable proper quotient of it, matching the rational envelope only up to a finite depth before deviating; the plane is the one dimension in which that deviation has been carried past its onset (at depth 10) and found transcendental, and whether higher dimensions are eventually transcendental as well is open [13].

Reproducibility. The eight length-10 affine relations and the shape-independence of u_d through depth 22 are formally verified in the Lean 4 proof assistant; the depth-32 threshold, the depth-38 data provenance of A396406, and the deviation witnesses are certified by exact modular and rational arithmetic, with two independent reproductions. All scripts and Lean sources are described in Appendix B; the further Lean files backing the appendix propositions are described in Appendix K.

Related work. The problem sits between Coxeter-group word-growth and the geometry of metabelian groups. Isola and Piergallini [24] analyse the reflection group G_τ of a generic triangle and prove that its translation subgroup is free abelian of infinite rank, that the subgroup generated by the three rotations is free metabelian of rank 2, that is $\mathbb{Z} \wr \mathbb{Z}$, and that G_τ *cannot be finitely presented*. Their results are therefore the direct precursor of the lamplighter structure obtained below, and not, as an earlier version of this paragraph had it, a statement that the generic group carries no unexpected relations: a group that is not finitely presented has unexpectedly many relations, not none. The present results are the explicit rational right-triangle analogue, replacing genericity by a Diophantine faithfulness criterion in ζ_T and adding exact word-growth data. The wreath product $\mathbb{Z} \wr \mathbb{Z}$ that emerges is the textbook amenable group of exponential growth; its appearance here, generated by three mirrors of a Euclidean triangle, places the rational shapes inside the program of quantifying group approximations, with the deviation sandwich giving the explicit girth of convergence in the space of marked groups. The geodesic-language and growth analysis is in the spirit of the lamplighter computations of Cleary–Elder–Taback [15]; the catalytic q -difference structure of the growth series, and the consequent exclusion of every elementary closed form for β_2 , is the point at which A396406 departs from the rational growth series of the standard

lamplighter generating sets. The rotation number ζ_T governing the translation module coincides with the rotation number of Sadun's [35] generalized-pinwheel substitution tilings, though the invariant studied there (tiling-space cohomology) is distinct from the word-growth sequence treated here.

2 Setting and the Coxeter presentation

2.1 The triangle, its mirrors, and the orbit-growth sequence

Let T be a triangle with vertices $V_0, V_1, V_2 \in \mathbb{Q}^2$ and leg lengths $a, b \in \mathbb{Q}_{>0}$, having its right angle at V_2 and *unequal* legs, $a \neq b$. For $i \in \{0, 1, 2\}$ let

$$R_i \in \text{Isom}(\mathbb{R}^2)$$

denote the Euclidean reflection across the line containing the side of T opposite the vertex V_i , and set

$$W(T) = \langle R_0, R_1, R_2 \rangle \leq \text{Isom}(\mathbb{R}^2).$$

When explicit coordinates are required we normalize the right-angle vertex to $B = (a, 0)$ with $A = (0, 0)$ and $C = (a, b)$, writing a, b for the two leg lengths; the mirrors R_0, R_1 are then the reflections across the two legs meeting at the right angle.

Remark 1 (Notation dictionary). Throughout the paper the three generators carry two interchangeable names, fixed once here:

$$R_0 = R_x \text{ (leg on the } x\text{-axis),} \quad R_1 = R_y \text{ (leg on the } y\text{-axis),} \quad R_2 = R_h \text{ (hypotenuse).}$$

The subscripted names R_0, R_1, R_2 index the opposite vertices V_0, V_1, V_2 ; the lettered names R_x, R_y, R_h name the mirror lines and are used from §3 onward, where the right-angle vertex is placed at the origin with the two legs along the positive coordinate axes (the normalization $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ above, rescaled by $1/a$). The two normalizations differ only by a similarity and induce the same abstract relations and orbit counts.

We study the orbit of T under $W(T)$ through breadth-first search in the Cayley graph of $W(T)$ relative to the generating set $S = \{R_0, R_1, R_2\}$. For $d \geq 0$ put

$$\mathcal{O}_d(T) = \{g \cdot T : g \in W(T), \ell_S(g) = d\}, \quad u_d^T := |\mathcal{O}_d(T)|,$$

where ℓ_S is word length with respect to S , and $u_0^T = 1$. All distances are computed in exact rational arithmetic, and the relations below are verified symbolically over $\mathbb{Q}(a, b)$. We write $u_d := u_d^T$ when the shape is fixed or immaterial.

Remark 2. Two groups govern u_d . The *abstract* group W is the finitely presented Coxeter group of Theorem 3 below, depending only on the right-angle and non-isosceles hypotheses. The *Euclidean* group $W(T)$ is its image under the affine action $\rho_T: W \rightarrow \text{Aff}(\mathbb{R}^2)$ induced by the mirrors of T , which depends on the shape. Distinct words of W may act identically on $\mathbb{R}^2$, collapsing orbit elements; the counts u_d are those of the Euclidean orbit (Definition 8).

2.2 The Coxeter presentation

Let Γ be the abstract group obtained from the free product $\langle r_0 \rangle * \langle r_1 \rangle * \langle r_2 \rangle$ of three copies of $\mathbb{Z}/2$ by the canonical map onto $W(T)$ sending $r_i \mapsto R_i$, and let $K \trianglelefteq \langle r_0 \rangle * \langle r_1 \rangle * \langle r_2 \rangle$ be the kernel of this map: K is the set of all relations holding in $\text{Isom}(\mathbb{R}^2)$ among R_0, R_1, R_2 . The presentation theorem identifies K .

Theorem 3 (Coxeter presentation). *Let T be a right triangle with the right angle at V_2 and with unequal legs $a \neq b$. Let*

$$W := \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0, 1, 2), \ (R_0 R_1)^2 = 1 \rangle$$

be the Coxeter group whose diagram is the path on three nodes with edge labels $m(R_0, R_1) = 2$ and $m(R_0, R_2) = m(R_1, R_2) = \infty$. Then the pairwise rotation orders of the three mirrors of T are exactly $m(R_0, R_1) = 2$, $m(R_0, R_2) = m(R_1, R_2) = \infty$, and consequently $r_i \mapsto R_i$ induces a surjection $\rho_T: W \rightarrow W(T)$. This surjection is not injective for any admissible shape: the affine action imposes further relations beyond $(R_0 R_1)^2 = 1$, so $W(T)$ is a proper quotient of W , and $\ker \rho_T \supsetneq \{1\}$. The first such relation is the object of Theorem 27; the abstract Coxeter group W therefore serves only as an envelope for $W(T)$.

Proof. Each R_i is a reflection, so $R_i^2 = 1$; the mirrors of R_0, R_1 meet at $\pi/2$, so $R_0 R_1$ is a half-turn and $(R_0 R_1)^2 = 1$. Hence all defining relators of W hold in $W(T)$, giving the surjection $\rho_T: W \rightarrow W(T)$. The remaining interior angles $\arctan(b/a)$, $\arctan(a/b)$ are, by Niven's theorem (a rational multiple of π has rational tangent only for tangent in $\{0, \pm 1\}$, excluded by $a \neq b$), irrational multiples of π ; so $R_0 R_2$ and $R_1 R_2$ have infinite order and no bond $(R_i R_2)^m = 1$ is forced. Thus the pairwise orders are exactly as stated, and W is the largest group compatible with them.

The surjection is *not* an isomorphism. Coxeter's structure theorem would identify $W(T)$ with W only if the affine (Tits) representation ρ_T were faithful; but $W(T) \leq \text{Isom}(\mathbb{R}^2)$ is amenable (indeed virtually lamplighter, §9), whereas $W \cong (\mathbb{Z}/2 \times \mathbb{Z}/2) * \mathbb{Z}/2$ contains a free subgroup [40] and is non-amenable. Hence $\ker \rho_T$ is nontrivial and $W(T)$ is a proper quotient. Explicitly, $\ker \rho_T$ is computed in Theorem 27: its shortest element has length $2n_T$, so $W(T)$ and W have the same spherical growth only through depth $n_T - 1$. $\square$

Remark 4. The non-isosceles hypothesis $a \neq b$ is used solely to exclude the angle $\pi/4$ in Theorem 3; for the isosceles right triangle the third relation $(R_0 R_2)^4 = 1$ holds and W is a finite Coxeter group, a degenerate case outside the present scope.

2.3 The Steinberg–Poincaré series and the Fibonacci phase

The growth of the abstract layer is governed by the spherical growth series of the Coxeter group W of Theorem 3, i.e. the Poincaré series

$$W(t) = \sum_{d \geq 0} |\{g \in W : \ell_S(g) = d\}| t^d.$$

Proposition 5 (Steinberg–Poincaré series). *For the Coxeter group W of Theorem 3,*

$$W(t) = \frac{(1+t)^2}{1-t-t^2}.$$

Proof. Steinberg's formula [23] expresses the Poincaré series of a Coxeter system (W, S) as

$$\frac{1}{W(t^{-1})} = \sum_{\substack{J \subseteq S \\ W_J \text{ finite}}} \frac{(-1)^{|J|}}{W_J(t)},$$

summed over those subsets J generating a finite parabolic W_J , with $W_J(t)$ its (polynomial) Poincaré series. Here the finite parabolics are: $J = \emptyset$ ($W_\emptyset(t) = 1$); the three singletons $\{R_i\}$, each with

$W_J(t) = 1 + t$; and the rank-2 subset $J = \{R_0, R_1\}$, which generates the dihedral group of order 4 with $W_J(t) = (1 + t)^2$. The subsets $\{R_0, R_2\}$ and $\{R_1, R_2\}$ generate the infinite dihedral group and contribute nothing. Substituting,

$$\frac{1}{W(t^{-1})} = 1 - \frac{3}{1+t} + \frac{1}{(1+t)^2} = \frac{(1+t)^2 - 3(1+t) + 1}{(1+t)^2} = \frac{t^2 - t - 1}{(1+t)^2}.$$

Inverting both sides,

$$W(t^{-1}) = \frac{(1+t)^2}{t^2 - t - 1}.$$

Now substitute $t \mapsto t^{-1}$. The numerator becomes $(1 + t^{-1})^2 = t^{-2}(1 + t)^2$, and the denominator becomes $t^{-2} - t^{-1} - 1 = t^{-2}(1 - t - t^2)$; the common factor t^{-2} cancels, giving

$$W(t) = \frac{(1+t)^2}{1 - t - t^2}. \quad \square$$

Corollary 6 (Abstract layer counts are Fibonacci). *Write F_n for the Fibonacci numbers with the conventions $F_0 = 0$, $F_1 = F_2 = 1$. The coefficients of the Steinberg–Poincaré series are*

$$[t^d] W(t) = F_{d+3} \quad (d \geq 1),$$

with $[t^0] W(t) = 1$.

Proof. Write $(1 + t)^2 = 1 + 2t + t^2$ and let $\Phi(t) = 1/(1 - t - t^2) = \sum_{n \geq 0} F_{n+1} t^n$, the Fibonacci generating function. Then for $d \geq 1$,

$$[t^d] W(t) = F_{d+1} + 2F_d + F_{d-1},$$

where the term F_{d-1} is read with the convention $F_0 = 0$ when $d = 1$. Using $F_{d+1} + F_d = F_{d+2}$ and $F_d + F_{d-1} = F_{d+1}$, this equals $F_{d+2} + F_{d+1} = F_{d+3}$. In particular at $d = 1$ one gets $F_2 + 2F_1 + F_0 = 1 + 2 + 0 = 3 = F_4$, consistent with the direct expansion $W(t) = 1 + 3t + 5t^2 + 8t^3 + \dots$; the boundary value $[t^0] W(t) = 1$ is read off the same expansion. $\square$

The Euclidean orbit counts u_d agree with this baseline until the first Euclidean collision.

Corollary 7 (Fibonacci phase). *Fix the reference triangle $T_0 = (3, 4, 5)$. Then*

$$u_d^{T_0} = F_{d+3} \quad (1 \leq d \leq 9),$$

and the first depth at which the affine action collapses an abstract orbit element is $d = 10$, where $u_{10}^{T_0} = 225 < F_{13} = 233$. The same values hold for every right triangle T with unequal rational legs; this shape-independence on the range $1 \leq d \leq 9$ and the depth-10 deficit are established symbolically over $\mathbb{Q}(a, b)$ in §3 (Theorem 10). At $d = 0$ one has $u_0 = 1 \neq F_3 = 2$.

Proof. By Proposition 5 and Corollary 6, the abstract layer count at depth d is F_{d+3} , and $u_d^T \leq F_{d+3}$ for every T since u_d^T counts the image $\rho_T(\mathcal{O}_d)$. For the reference triangle $T_0 = (3, 4, 5)$, exact-rational breadth-first search shows that the map from abstract length- d elements to Euclidean orbit points is injective for $1 \leq d \leq 9$, so $u_d^{T_0} = F_{d+3}$ there, and that $u_{10}^{T_0} = 225 = F_{13} - 8$, exhibiting eight affine collisions at length 10. That these values are independent of the admissible shape is not a consequence of the single search but is proved by the symbolic relation analysis of §3: Theorem 10 shows over $\mathbb{Q}(a, b)$ that no nontrivial affine identification occurs below depth 10 and that exactly eight occur at depth 10 for every admissible T . $\square$

Definition 8 (Fibonacci deficit). The *Fibonacci deficit* of T at depth d is

$$\delta_d^T := F_{d+3} - u_d^T \geq 0,$$

the number of abstract length- d orbit elements identified by the affine action ρ_T . By Corollary 7 and Theorem 10, $\delta_d^T = 0$ for $1 \leq d \leq 9$ and $\delta_{10}^T = 8$ for every admissible T . The Euclidean orbit-growth sequence $(u_d^T)_{d \geq 0}$ is recorded as OEIS A396406.

Remark 9. Throughout, $\rho_T: W \rightarrow \text{Aff}(\mathbb{R}^2)$ denotes the affine action of the abstract Coxeter group $W = \langle R_0, R_1, R_2 \rangle$ induced by the mirrors of T , and $\zeta_T = (a + bi)/(a - bi)$ is the rotation number attached to the leg pair (a, b) ; both enter the analysis from §3 onward. The results of this section, namely the presentation of Theorem 3, the series of Proposition 5 and the Fibonacci phase of Corollary 7, are unconditional.

3 The eight length-10 relations and depth-10 completeness

Let $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1 \ (i = 0, 1, 2), (R_0 R_1)^2 = 1 \rangle$ be the Coxeter group of §2. Throughout, $R_0 = R_x$ is the reflection across the x -leg, $R_1 = R_y$ the reflection across the y -leg, and $R_2 = R_h$ the reflection across the hypotenuse. For a right triangle T let $\rho_T: W \rightarrow \text{Aff}(\mathbb{R}^2)$ be its side-reflection action. Fix the placement of §2,

$$A = (0, 0), \quad B = (a, 0), \quad C = (a, b), \quad a, b > 0,$$

so that each generator R_i acts as an affine isometry whose 3×3 matrix has entries in $\mathbb{Q}(a, b)$, the denominators dividing a power of $a^2 + b^2$. For a word $w = w[0]w[1] \cdots w[\ell - 1]$ in the R_i we write $\rho_T(w) = R_{w[0]} R_{w[1]} \cdots R_{w[\ell - 1]}$ for the associated affine matrix.

The Poincaré series of the abstract Coxeter group W gives F_{d+3} canonical words of length d for $d \geq 1$; in particular there are $F_{13} = 233$ canonical words of length 10. Exact-rational breadth-first enumeration of the orbit of a non-isosceles right triangle gives $u_{10} = 225$, a deficit of 8. The eight collapsing pairs are the length-10 relations of Table 1; they are the first relations of the side-reflection action beyond the abstract Coxeter relations.

#	word A	word B
1	$R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2$	$R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1$
2	$R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2$	$R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1$
3	$R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0$
4	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$
5	$R_0 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$	$R_1 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$
6	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_1$
7	$R_0 R_2 R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_1$	$R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1 R_2 R_0$
8	$R_1 R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2$	$R_2 R_0 R_1 R_2 R_0 R_2 R_0 R_1 R_2 R_1$

Table 1: The eight length-10 affine relations. In each row, the two words have the same affine matrix in the side-reflection action ρ_T of W on every right triangle T . The generators are $R_0 = R_x$, $R_1 = R_y$, $R_2 = R_h$ (reflections across the x -leg, y -leg, and hypotenuse).

Theorem 10 (Eight length-10 affine relations). *Each of the eight identities of Table 1 holds for every right triangle. For the placement $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ with $a, b > 0$, the affine matrices*

$$M_A = R_{w_A[0]} \cdots R_{w_A[9]}, \quad M_B = R_{w_B[0]} \cdots R_{w_B[9]}$$

of the two words w_A, w_B in each row satisfy $M_A = M_B$ as 3×3 affine matrices over $\mathbb{Q}(a, b)$.

Proof. The statement is a finite algebraic identity. Each reflection R_i acts as an affine isometry whose matrix entries are rational functions of a, b over $\mathbb{Q}$, with denominators dividing a power of $a^2 + b^2$. Hence each entry of $M_A - M_B$ is a rational function in $\mathbb{Q}(a, b)$; clearing denominators yields, for each row, a polynomial identity in $\mathbb{Q}[a, b]$. These identities are verified deterministically in exact arithmetic over $\mathbb{Q}(a, b)$.

The eight identities are moreover *formally verified* in the Lean 4 proof assistant. The theorems

```
rel1_symbolic, ..., rel8_symbolic in lean/with_mathlib/SymbolicVerification.lean
```

each discharge the six affine-matrix-entry identities over $\mathbb{Q}[a, b]$ that together express $M_A = M_B$. Each such identity is proved by the `ring` tactic alone. The `ring` tactic produces a proof term that is checked by the Lean kernel; the trust base is therefore the Lean kernel together with the `Mathlib` definitions of $\mathbb{Q}[a, b]$ and its commutative-ring structure, with no appeal to numerical sampling, decision heuristics, or specialization to particular legs. Consequently each identity holds as an equality of affine matrices over $\mathbb{Q}(a, b)$, hence for every right triangle. $\square$ $\square$

Remark 11. Because Theorem 10 holds over $\mathbb{Q}(a, b)$, the eight collisions lie in $\ker \rho_T$ for every T ; the enumerative deficit $233 - 225 = 8$ is therefore shape-independent.

Proposition 12 (Completeness at depth 10). *Let T be an unequal-leg ($a \neq b$) right triangle. Among the 233 canonical Coxeter words of length 10 in W , exactly 225 have pairwise distinct images under ρ_T : there are 217 singleton classes and 8 collision pairs, each pair being one row of Table 1. No length-10 affine identity exists beyond these eight relations and the abstract Coxeter relations. (For the isosceles triangle $a = b$ additional collisions occur, by Remark 4; the count 225 and the distinctness of the 217 singleton classes are asserted only for $a \neq b$.)*

Proof. Completeness follows from two ingredients: (a) the computed block partition on the reference triangle $T_0 = (3, 4, 5)$, and (b) Theorem 10.

All 233 canonical Coxeter words of length 10 are enumerated. For each, the affine matrix $\rho_{T_0}(w)$ is computed in exact rational arithmetic on T_0 , and the words are grouped by matrix. The resulting partition has exactly 225 blocks: eight blocks of size 2 and 217 blocks of size 1. The eight size-2 blocks are the rows of Table 1. By Theorem 10 these eight identities hold symbolically over $\mathbb{Q}(a, b)$, so the eight pairs are genuine collisions for every right triangle. Conversely, any further length-10 collision would produce a coincidence of affine matrices already on T_0 in exact arithmetic, contradicting the computed block sizes. Since T_0 is unequal-leg, the 217 singleton blocks remain singletons under specialization to any unequal-leg T . Hence the eight relations, together with the abstract Coxeter relations used to form the canonical words, account for all length-10 affine identities.

As corroboration, each size-2 block was independently re-checked on six further triangles,

$$(5, 12), \quad (8, 15), \quad (7, 24), \quad (1, 2), \quad (2, 3), \quad (1, 7),$$

and in all cases the two words of the block produce the same affine matrix; this is redundant once Theorem 10 is invoked. $\square$ $\square$

4 The triangle-independent symbolic normal form

The side-reflection action carries a triangle-independent symbolic normal form (Lemma 13) in which the shape of the triangle enters only through the evaluation of integer Laurent polynomials at the rotation number ζ_T . Collisions between group elements reduce to the vanishing of bounded-height Laurent polynomials at ζ_T (Proposition 16), a divisibility condition in $\mathbb{Z}[i]$ that drives the effective universality theorem of §5.

4.1 Complex coordinates and the generators

Throughout, the three side-reflections are written interchangeably as $(R_0, R_1, R_2) = (R_x, R_y, R_h)$, where $R_0 = R_x$ is the reflection in the leg on the positive real axis, $R_1 = R_y$ the reflection in the leg on the positive imaginary axis, and $R_2 = R_h$ the reflection in the hypotenuse; this matches the labelling R_0, R_1, R_2 of §5.

Identify $\mathbb{R}^2 = \mathbb{C}$, place the right angle at the origin with the legs along the positive coordinate axes, and rescale by $1/a$, where a is the length of the leg on the positive real axis. This is the normalization used in §5; it agrees, after the affine change placing the right-angle vertex at the origin, with the normalization $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ of the introductory sections. Neither the choice of origin and axes nor the rescaling changes word-lengths or the collision structure of the orbit, so no generality is lost. Let $\sigma(z) = \bar{z}$ denote complex conjugation, and set

$$\zeta = \zeta_T = \frac{a + bi}{a - bi},$$

the rotation number of the triangle with legs (a, b) . Since numerator and denominator are complex conjugates, $|\zeta_T| = 1$; we use this standing fact below. In these coordinates the three side-reflections are

$$R_x : z \mapsto \sigma(z), \quad R_y : z \mapsto -\sigma(z), \quad R_h : z \mapsto \zeta^{-1}\sigma(z) + (1 - \zeta^{-1}),$$

where R_x and R_y are the reflections in the two leg lines and R_h is the reflection in the hypotenuse line. We write W for the group they generate and $\rho_T : W \rightarrow \text{Isom}(\mathbb{C})$ for the resulting faithful action on the plane of the triangle T .

Every element of $\text{Isom}(\mathbb{C})$ produced by composing these generators is an orientation-reversing-or-preserving affine map of the form $z \mapsto \alpha \sigma^\delta(z) + \beta$ with $\delta \in \{0, 1\}$. The normal form below records that the linear part α is always of the shape $\varepsilon \zeta^k$ with $\varepsilon \in \{\pm 1\}$ and $k \in \mathbb{Z}$, and that the translation part β is the value at ζ of an integer Laurent polynomial whose support and height are controlled by the word-length.

4.2 The normal form

Lemma 13 (Symbolic normal form). *For every $w \in W$ of word-length ℓ there exist $\varepsilon \in \{\pm 1\}$, $\delta \in \{0, 1\}$, an integer $k \in \mathbb{Z}$ with $|k| \leq \ell$, and an integer Laurent polynomial $P_w \in \mathbb{Z}[t^{\pm 1}]$ (all four independent of the triangle T) such that*

$$\rho_T(w) : z \mapsto \varepsilon \zeta^k \sigma^\delta(z) + P_w(\zeta). \quad (1)$$

Moreover

$$\text{supp } P_w \subseteq [-\ell, \ell], \quad \|P_w\|_\infty \leq \ell, \quad P_w(1) = 0,$$

where $\|P_w\|_\infty$ denotes the maximum absolute value of the coefficients of P_w .

We refer to the triple (ε, δ, k) as the *symbolic class* of w and to P_w as its *translation polynomial*. The defining identity (1) is purely formal: the symbol ζ is treated as an indeterminate t , conjugation σ acts on Laurent polynomials by the substitution $t \mapsto t^{-1}$, and T -dependence is confined to the final evaluation $t = \zeta_T$. The identity $\sigma(P(\zeta)) = P(\zeta^{-1})$ used in the proof is the analytic content of this convention, valid because $|\zeta_T| = 1$ and the coefficients of P are real, so that $\bar{\zeta} = \zeta^{-1}$.

Proof. The argument is by induction on the word-length ℓ , using the composition law for the affine maps in (1).

Base case. For the three generators the symbolic data are

$$R_x: (\varepsilon, \delta, k, P) = (1, 1, 0, 0), \quad R_y: (-1, 1, 0, 0), \quad R_h: (1, 1, -1, 1 - t^{-1}),$$

read off directly from the formulas for R_x, R_y, R_h . Each has $|k| \leq 1$, support contained in $[-1, 1]$, height at most 1, and a translation polynomial vanishing at $t = 1$ (the constant 0 and $1 - t^{-1}$ both do).

Inductive step. Let g be a generator with data $(\varepsilon_g, \delta_g, k_g, P_g)$ and let w have data $(\varepsilon_w, \delta_w, k_w, P_w)$ satisfying the asserted bounds for its length ℓ . Composing the two affine maps

$$\rho_T(g): z \mapsto \varepsilon_g \zeta^{k_g} \sigma^{\delta_g}(z) + P_g(\zeta), \quad \rho_T(w): z \mapsto \varepsilon_w \zeta^{k_w} \sigma^{\delta_w}(z) + P_w(\zeta),$$

and using $\sigma(\zeta) = \bar{\zeta} = \zeta^{-1}$ (valid since $|\zeta_T| = 1$) together with the reality of the coefficients of P_w (so that $\sigma(P_w(\zeta)) = P_w(\zeta^{-1})$), the product gw acts by

$$z \longmapsto \varepsilon_g \varepsilon_w \zeta^{k_g \pm k_w} \sigma^{\delta_g + \delta_w}(z) + \left(P_g(\zeta) + \varepsilon_g \zeta^{k_g} P_w(\zeta^{\pm 1}) \right),$$

where the exponents $\delta_g + \delta_w$ and $\sigma^{\delta_g + \delta_w}$ are read modulo 2, and the sign in $k_g \pm k_w$ and in the argument $\zeta^{\pm 1}$ of P_w is -1 exactly when $\delta_g = 1$: the case in which the conjugation in $\rho_T(g)$ inverts ζ . Hence the symbolic data of gw are

$$\varepsilon_{gw} = \varepsilon_g \varepsilon_w \in \{\pm 1\}, \quad \delta_{gw} = \delta_g + \delta_w \pmod{2} \in \{0, 1\}, \quad k_{gw} = k_g \pm k_w,$$

and the translation polynomial obeys the composition law

$$P_{gw}(t) = P_g(t) + \varepsilon_g t^{k_g} P_w(t^{\pm 1}), \tag{2}$$

all of which are manifestly independent of T .

It remains to verify the three bounds for the word gw of length $\ell + 1$. Since $|k_g| \leq 1$, the exponent satisfies $|k_{gw}| = |k_g \pm k_w| \leq 1 + \ell = \ell + 1$. For the translation polynomial, the operations applied to P_w in (2) are: substitution $t \mapsto t^{\pm 1}$, multiplication by the monomial $\pm t^{k_g}$, and addition of the generator's polynomial P_g . Inverting t permutes the support symmetrically and leaves the multiset of coefficients (hence the height) unchanged; multiplying by $\pm t^{k_g}$ with $|k_g| \leq 1$ shifts the support by at most 1 and again preserves the height; and adding P_g , which has height at most 1 and support in $[-1, 1]$, raises the height by at most 1 and enlarges the support radius by at most 1. Therefore

$$\|P_{gw}\|_\infty \leq \|P_w\|_\infty + 1 \leq \ell + 1, \quad \text{supp } P_{gw} \subseteq [-(\ell + 1), \ell + 1].$$

Finally, evaluating (2) at $t = 1$ gives $P_{gw}(1) = P_g(1) + \varepsilon_g P_w(1) = 0 + 0 = 0$, since both $P_g(1) = 0$ (base case) and $P_w(1) = 0$ (induction hypothesis). This completes the induction. $\square$

Remark 14. The composition law (2) controls the growth of P_w : appending a generator either fixes the support of the translation polynomial or shifts it by one and adds a height-one correction, so the support radius and the height each grow by at most one per letter. The vanishing $P_w(1) = 0$ is the algebraic shadow of the geometric fact that the degenerate limit $\zeta \rightarrow 1$ (an isosceles right triangle, where $a = b$) collapses the translation part: every word fixes the affine structure of the degenerate triangle, consistently with the exclusion $a \neq b$ throughout.

4.3 Reduction of collisions to polynomial vanishing

The normal form converts the geometric question of when two group elements have equal action into an arithmetic question about a single Laurent polynomial. The key separation is provided by the irrationality of the rotation angle.

Lemma 15 (Class separation). *The rotation number ζ_T is not a root of unity. Consequently the linear parts $\varepsilon \zeta_T^k$ arising in (1), ranging over $\varepsilon \in \{\pm 1\}$ and $k \in \mathbb{Z}$, are pairwise distinct; and two elements $w, w' \in W$ satisfy $\rho_T(w) = \rho_T(w')$ if and only if they share the same symbolic class (ε, δ, k) and their translation polynomials agree at $t = \zeta_T$:*

$$\rho_T(w) = \rho_T(w') \iff (\varepsilon, \delta, k)_w = (\varepsilon, \delta, k)_{w'} \text{ and } P_w(\zeta_T) = P_{w'}(\zeta_T). \quad (3)$$

Proof. For legs with $a \neq b$ the angle $\arg \zeta_T = 2 \arctan(b/a)$ is an irrational multiple of π by Niven's theorem on rational multiples of π with rational tangent, so ζ_T is not a root of unity and $|\zeta_T| = 1$ with $\zeta_T \neq \pm 1$. Hence the powers ζ_T^k , $k \in \mathbb{Z}$, are pairwise distinct, and none equals $-\zeta_T^{k'}$ for any k' : from $-\zeta_T^k = \zeta_T^{k'}$ we would get $\zeta_T^{2(k-k')} = 1$, making ζ_T a root of unity, a contradiction. Two affine maps $z \mapsto \alpha \sigma^\delta(z) + \beta$ coincide if and only if their linear parts $\alpha \sigma^\delta$ and their translation parts β coincide. By the previous sentence, equality of the linear parts forces $\delta_w = \delta_{w'}$, $k_w = k_{w'}$, and $\varepsilon_w = \varepsilon_{w'}$; equality of the translation parts is exactly $P_w(\zeta_T) = P_{w'}(\zeta_T)$. $\square$

Let u_d^{sym} denote the number of distinct symbolic tuples $(\varepsilon, \delta, k, P_w)$ realized by words of length at most d . This count is a T -independent integer, and it equals the generic depth- d orbit count appearing in the effective universality theorem (Theorem 19): distinct symbolic data give distinct generic isometries, while any extra coincidence for a particular triangle can only *decrease* the count below u_d^{sym} . Combining this with Lemma 15 yields the precise mechanism by which a triangle can fail universality.

Proposition 16 (Collisions as polynomial vanishing). *Fix a triangle T with rotation number ζ_T , and let u_e^T denote the depth- e orbit count for T . If $u_e^T \neq u_e^{\text{sym}}$ for some $e \leq d$, then there exist two words w, w' of length at most e lying in the same symbolic class such that the nonzero integer Laurent polynomial*

$$D = P_w - P_{w'} \in \mathbb{Z}[t^{\pm 1}], \quad D \neq 0,$$

satisfies

$$D(\zeta_T) = 0, \quad \text{supp } D \subseteq [-e, e], \quad \|D\|_\infty \leq 2e, \quad D(1) = 0. \quad (4)$$

Proof. A deficiency $u_e^T < u_e^{\text{sym}}$ means that two words $w \neq w'$ with distinct symbolic data nevertheless have $\rho_T(w) = \rho_T(w')$. By the collision criterion (3) they must then share the same symbolic class (ε, δ, k) (so their distinctness lies in the translation polynomials) and satisfy $P_w(\zeta_T) = P_{w'}(\zeta_T)$. Set $D = P_w - P_{w'}$. Then $D \neq 0$ as Laurent polynomials (else the symbolic data would coincide), $D(\zeta_T) = 0$, and the support and height bounds follow from Lemma 13: each of $P_w, P_{w'}$ has support in $[-e, e]$ and height at most e , so D has support in $[-e, e]$ and height at most $2e$. Finally $D(1) = P_w(1) - P_{w'}(1) = 0$. $\square$

Proposition 16 is the bridge from geometry to arithmetic on which §5 rests: a failure of universality at depth e for the triangle T is *equivalent* to the existence of a nonzero integer Laurent polynomial of support radius at most e , height at most $2e$, and vanishing at $t = 1$, that also vanishes at the rotation number ζ_T . Because the height and support of D are bounded explicitly in terms of the depth alone (with no dependence on T) this condition can be tested against the algebraic constraints imposed by ζ_T . Since ζ_T is an algebraic number of degree 2 over $\mathbb{Q}$, vanishing $D(\zeta_T) = 0$

forces the minimal polynomial $\mu_T \in \mathbb{Z}[t]$ of ζ_T to divide a suitable integer polynomial associated with D ; by Gauss's lemma this divisibility takes place in $\mathbb{Z}[t]$, so the leading coefficient c_T of μ_T divides the extreme coefficients of D , whence $c_T \leq \|D\|_\infty \leq 2e$. No triangle whose rotation number has $c_T > 2e$ can therefore admit a collision at depth e , which is the divisibility obstruction exploited in §5.

5 Effective universality and the sharp depth-32 threshold

The deviation depth of an arbitrary unequal-leg shape is controlled by an arithmetic mechanism: at each fixed depth, universality reduces to a finite, explicitly sparse list of arithmetically small shapes (Theorem 19). The orbit count is shape-independent for $0 \leq d \leq 32$ and first fails at depth 33 for the $(1, 2)$ triangle (Theorem 20): there $u_{33}^{(1,2)} = 3848354$ against the universal value $u_{33} = 3848384$, a deficit of exactly 30.

Throughout, $W = \langle R_0, R_1, R_2 \mid R_i^2 = 1, (R_0 R_1)^2 = 1 \rangle$. The three involutions are identified with the side-reflections of the right triangle by the fixed dictionary

$$R_0 = R_x \text{ (leg on the } x\text{-axis),} \quad R_1 = R_y \text{ (leg on the } y\text{-axis),} \quad R_2 = R_h \text{ (hypotenuse),}$$

with the right angle placed at the origin and the legs along the positive coordinate axes, rescaled so that the x -leg has unit length. For a right-triangle parameter $T = (a, b)$ with positive rational legs $a \neq b$ we write ρ_T for the side-reflection realization and $u_d^T := |\{\rho_T(w) : |w|_W = d\}|$ for its orbit-growth sequence. We write u_d for the generic (shape-independent) value, namely the common value of u_d^T wherever it is established to be T -independent.

5.1 The arithmetic complexity of a shape

Identify $\mathbb{R}^2 = \mathbb{C}$, place the right angle at the origin with the legs along the positive coordinate axes, and rescale by $1/a$; neither operation changes word-lengths or collision structure. Writing $\sigma(z) = \bar{z}$ and $\zeta = \zeta_T = (a + bi)/(a - bi)$ for the *rotation number*, the three side-reflections are

$$R_x : z \mapsto \sigma(z), \quad R_y : z \mapsto -\sigma(z), \quad R_h : z \mapsto \zeta^{-1}\sigma(z) + (1 - \zeta^{-1}),$$

the reflections in the two leg lines and in the hypotenuse line, respectively. By the symbolic normal form (Lemma 13), every $w \in W$ of word-length ℓ acts by

$$\rho_T(w) : z \mapsto \varepsilon \zeta^k \sigma^\delta(z) + P_w(\zeta), \quad \varepsilon \in \{\pm 1\}, \delta \in \{0, 1\}, k \in \mathbb{Z}, |k| \leq \ell,$$

with $P_w \in \mathbb{Z}[t^{\pm 1}]$ independent of T , $\text{supp } P_w \subseteq [-\ell, \ell]$, every coefficient of P_w bounded in absolute value by ℓ , and $P_w(1) = 0$. The shape T enters only through the evaluation of these triangle-independent Laurent polynomials at the single point ζ_T .

Definition 17 (Arithmetic complexity and minimal polynomial). Scale the legs to coprime integers (a, b) with $a \neq b$, and write $\zeta_T = \mu/\lambda$ in lowest terms in $\mathbb{Z}[i]$, so that

$$\lambda = a - bi \quad (a + b \text{ odd}), \quad \lambda = \frac{a - bi}{1 + i} \quad (a, b \text{ both odd}).$$

The *arithmetic complexity* of T is $c_T := N(\lambda)$, the norm of λ in $\mathbb{Z}[i]$. The *minimal polynomial* of the shape is the primitive integer polynomial

$$\mu_T(t) = c_T t^2 - e_T t + c_T \in \mathbb{Z}[t],$$

with roots $\zeta_T^{\pm 1}$, where $e_T = 2|a^2 - b^2|$ when $a + b$ is odd and $e_T = |a^2 - b^2|$ when a, b are both odd. The absolute value is not cosmetic: every statement below assumes $0 < e_T < 2c_T$, and the tables and the deviation law are written in this convention. Without it the pair $(1, 2)$ would give $e_T = -6$ and $\mu_{(1,2)} = 5t^2 + 6t + 5$, whereas the convention used throughout gives $e_T = 6$ and $\mu_{(1,2)} = 5t^2 - 6t + 5$; the two differ in every lamp profile computed from μ_T . Equivalently one may order the legs so that $a > b$.

For example, $(a, b) = (1, 2)$ gives $\zeta_T = (-3 + 4i)/5$, $c_T = 5$, $e_T = 6$ and $\mu_T = 5t^2 - 6t + 5$, while $(a, b) = (3, 4)$ gives $\zeta_T = (-7 + 24i)/25$, $c_T = 25$, $e_T = 14$ and $\mu_T = 25t^2 - 14t + 25$.

Lemma 18 (Arithmetic of the complexity). *With (a, b) coprime and $a \neq b$:*

- (i) c_T is odd;
- (ii) every prime factor of c_T is $\equiv 1 \pmod{4}$;
- (iii) $c_T \geq 5$;
- (iv) μ_T is primitive.

For general coprime (a, b) (not assuming $a \neq b$), $c_T = 1$ if and only if $a = b$, in which case $\zeta_T \in \{\pm i\}$ is a root of unity.

Proof. (i) If $a + b$ is odd then $c_T = a^2 + b^2$ is odd; if a, b are both odd then $a^2 + b^2 \equiv 2 \pmod{8}$, so $c_T = (a^2 + b^2)/2$ is odd.

(ii) A rational (inert) prime $q \equiv 3 \pmod{4}$ dividing $c_T = \lambda\bar{\lambda}$ would divide λ itself; but then q would divide both the real and imaginary parts of λ , hence both a and b , contradicting $\gcd(a, b) = 1$. Thus every odd prime factor of c_T splits in $\mathbb{Z}[i]$, i.e. is $\equiv 1 \pmod{4}$.

(iii) By (i) and (ii), c_T is an odd integer all of whose prime factors are $\equiv 1 \pmod{4}$; the smallest such integer exceeding 1 is 5. For general coprime (a, b) , $c_T = 1$ iff ζ_T is a unit of $\mathbb{Z}[i]$, i.e. $\zeta_T \in \{\pm 1, \pm i\}$; for positive legs this forces $a = b$. Under the hypothesis $a \neq b$ this case is excluded, so $c_T \geq 5$.

(iv) An odd prime dividing both c_T and e_T would divide both $a^2 + b^2$ and $a^2 - b^2$, hence both a and b , again contradicting coprimality. $\square$

5.2 Effective universality

Since ζ_T is non-real with $a, b > 0$, its powers $\pm \zeta_T^k$ are pairwise distinct (Niven's theorem rules out roots of unity), so two group elements can have equal ρ_T -images only if their symbolic data agree in (ε, δ, k) and their translation polynomials agree at $t = \zeta_T$. Writing u_d^{sym} for the number of distinct symbolic tuples at depth d (a T -independent integer equal to the generic count u_d) any deviation $u_e^T \neq u_e$ at some depth $e \leq d$ forces a *nonzero* difference

$$D = P_w - P_{w'} \in \mathbb{Z}[t^{\pm 1}], \quad D \neq 0, \quad D(\zeta_T) = 0, \quad (5)$$

of two same-symbolic-class ball elements of depth $\leq d$.

Theorem 19 (Effective universality). *Let $T = (a, b)$ have coprime positive legs with $a \neq b$, and let c_T, μ_T be as in Definition 17. If $u_e^T \neq u_e$ for some $e \leq d$, then μ_T divides a realized word-difference in $\mathbb{Z}[t]$; in particular c_T divides both extreme coefficients of that difference, so $c_T \leq 2d$. Consequently every triangle with $c_T > 2d$ satisfies universality at all depths $\leq d$; equivalently, for every unequal-leg rational triangle,*

$$u_d^T = u_d \quad \text{for all } d < \frac{1}{2} c_T.$$

Proof. Take $D \neq 0$ as in (5), built from two ball elements of depth $\leq d$; by Lemma 13 every coefficient of D is bounded by $2d$ in absolute value. Normalize $\tilde{D} = t^m D \in \mathbb{Z}[t]$ with $\tilde{D}(0) \neq 0$; this multiplication by a unit of $\mathbb{Z}[t^{\pm 1}]$ changes neither the set of extreme coefficients nor the value at ζ_T . Since ζ_T is non-real, its minimal polynomial over $\mathbb{Q}$ is the degree-2 polynomial μ_T , with conjugate root $\overline{\zeta_T} = \zeta_T^{-1}$. As $\tilde{D}$ has integer coefficients and vanishes at ζ_T , Gauss's lemma gives a factorization $\tilde{D} = \mu_T \cdot E$ with $E \in \mathbb{Z}[t]$. Comparing leading and constant coefficients,

$$\text{lead}(\tilde{D}) = c_T \text{lead}(E), \quad \text{const}(\tilde{D}) = c_T \text{const}(E),$$

so c_T divides both extreme coefficients of $\tilde{D}$ and

$$c_T \leq |\text{lead}(\tilde{D})| \leq 2d.$$

The contrapositive is the stated universality bound. $\square$

Theorem 19 is an *all-depths* universality statement: at each fixed depth d it reduces the question to the finitely many shapes with $c_T \leq 2d$, and by Lemma 18 these are very sparse. The admissible complexities are the odd integers all of whose prime factors are $\equiv 1 \pmod{4}$; the smallest are

$$5, 13, 17, 25, 29, 37, 41, 53, 61, 65, \dots,$$

so that $45 = 3^2 \cdot 5$ and $49 = 7^2$ are excluded because 3 and 7 are inert in $\mathbb{Z}[i]$. The cutoff $c_T \leq 2d$ pins the candidate list at each depth:

d	$2d$	admissible $c_T \leq 2d$
30	60	$\{5, 13, 17, 25, 29, 37, 41, 53\}$
32	64	$\{5, 13, 17, 25, 29, 37, 41, 53, 61\}$
36	72	$\{5, 13, 17, 25, 29, 37, 41, 53, 61, 65\}$

Each admissible complexity is realized by finitely many rotation numbers ζ_T : sixteen for $d = 30$ and twenty-two for $d = 36$. The count is not two per complexity. Each of the first nine admissible complexities contributes exactly two shapes, but $65 = 5 \cdot 13$ has two essentially distinct representations as a sum of two squares and contributes four, namely $(1, 8)$, $(4, 7)$, $(3, 11)$ and $(7, 9)$; hence $18 + 4 = 22$. These finitely many candidate shapes can be checked mechanically.

5.3 The finite arithmetic certificate

For each candidate rotation number ζ with $c_T \leq 2d$, evaluate the entire symbolic ball at ζ in a finite field $\mathbb{F}_q$, chosen with $q \equiv 1 \pmod{4}$ so that $i \in \mathbb{F}_q$, and with $q \nmid c_T$. This evaluation is *one-sided exact*: two symbolic images that are distinct modulo q are necessarily distinct in $\mathbb{Q}(i)$. Thus a collision-free ball mod q certifies a collision-free ball over $\mathbb{Q}(i)$, with no false negatives. Modular arithmetic is therefore a fast *filter*: a candidate that is collision-free under the modular evaluation is settled, while a candidate exhibiting a modular collision is referred to exact arithmetic over $\mathbb{Q}(i)$, which decides it without error. The settled threshold rests on exact $\mathbb{Q}(i)$ confirmation of the finite candidate list; the modular filter, run over several independent primes, only accelerates the search.

At depth 30 the certificate evaluates a ball of 3,336,511 symbolic elements at the 16 candidate rotation numbers with $c_T \leq 60$ over independent 31-bit primes; every candidate is collision-free, and the surviving list is confirmed by exact arithmetic over $\mathbb{Q}(i)$. Combined with Theorem 19 for the shapes with $c_T > 60$, this controls all depths ≤ 30 ; the depth-32 threshold is obtained from the larger run described below.

Theorem 20 (Sharp uniform universality at depth 32). *For every right triangle with positive unequal rational legs,*

$$u_d^T = u_d \quad \text{for all } 0 \leq d \leq 32.$$

The bound is sharp: the (1, 2) triangle satisfies

$$u_{33}^{(1,2)} = u_{33} - 30 = 3,848,354 < u_{33},$$

so depth 32 is the exact threshold below which the orbit count is shape-independent.

Proof. Fix $d = 32$. By Theorem 19, any shape with $u_e^T \neq u_e$ for some $e \leq 32$ has $c_T \leq 2d = 64$; by Lemma 18 the admissible complexities with $c_T \leq 64$ are $\{5, 13, 17, 25, 29, 37, 41, 53, 61\}$, realized by an explicit finite list of rotation numbers. The arithmetic certificate evaluates the symbolic ball through depth 32 at each such ζ over independent 31-bit primes as a fast filter, and the surviving candidate list is confirmed by exact arithmetic over $\mathbb{Q}(i)$; no collision occurs among the admissible shapes. The certified run covers all rotation numbers with $c_T \leq 72 \geq 64$ through depth 32 (Trust base (b)), so the cutoff $c_T \leq 64$ required here lies strictly within the certified range. Hence no admissible shape deviates at any depth ≤ 32 , and $u_d^T = u_d$ for $0 \leq d \leq 32$. The deviation at depth 33 for (1, 2) ($c_T = 5$) is realized explicitly by the construction of Theorem 27, giving $u_{33}^{(1,2)} = u_{33} - 30 = 3,848,354$ and showing the threshold cannot be raised. $\square$

5.4 The translation lattice

The kernel classification, the lamplighter structure, and the membership criterion of the downstream sections all rest on an exact description of the translation lattice

$$\mathcal{T} := \{ P_w(t) \in \mathbb{Z}[t^{\pm 1}] : w \in \ker \text{ on the linear part, } \rho_T(w) \text{ a pure translation} \}.$$

Theorem 21 (Translation-lattice equality). $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$.

Proof. ($\supseteq$) The inclusion $2(t-1)\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square computation of Lemma 25: the commutator/glide relations of W produce the pure translation by $2(t-1)$, and the $\mathbb{Z}[t^{\pm 1}]$ -module action by ζ -multiplication (conjugation by $R_h R_x$) generates the full ideal multiple.

($\subseteq$) Conversely, let $P_w \in \mathcal{T}$, so $\rho_T(w) : z \mapsto z + P_w(\zeta)$ is a pure translation for the shape, and $P_w \in \mathbb{Z}[t^{\pm 1}]$ with $P_w(1) = 0$ by the normal form (Lemma 13). The condition $P_w(1) = 0$ gives $(t-1) \mid P_w$ in $\mathbb{Q}[t^{\pm 1}]$, and since P_w has integer coefficients and $t-1$ is primitive, Gauss's lemma yields $P_w = (t-1)Q$ with $Q \in \mathbb{Z}[t^{\pm 1}]$. For the factor 2, reduce modulo 2. As recalled in §1, modulo 2 the two perpendicular mirrors collapse and the image of W is the infinite dihedral group, whose translation subgroup is generated by the image of a single glide; concretely, the reduction $\bar{\rho} : W \rightarrow \text{Aff}(\mathbb{F}_2[t^{\pm 1}])$ sends w to the translation by $\bar{P}_w$, the reduction of P_w . Since w has $(\delta, k) = (0, 0)$, its image is trivial in that dihedral quotient, so $\bar{P}_w = 0$, that is $P_w \in 2\mathbb{Z}[t^{\pm 1}]$. Now 2 is prime in the unique factorisation domain $\mathbb{Z}[t^{\pm 1}]$ and does not divide the primitive element $t-1$; from $2 \mid (t-1)Q$ it follows that $2 \mid Q$. Hence $P_w \in 2(t-1)\mathbb{Z}[t^{\pm 1}]$, completing the reverse inclusion. $\square$

Remark 22 (On the parity step). An earlier version deduced $Q \in 2\mathbb{Z}[t^{\pm 1}]$ from the $(R_i R_j)^2$ relations “forcing even total exponent count in each generator”. That does not give what is wanted: an even exponent count yields $\bar{P}_w(1) = 0$, which is already implied by $P_w(1) = 0$ and says nothing about the remaining coefficients. The mod-2 collapse used above is what supplies the divisibility, and it needs the dihedral structure of the reduction, not a parity count. This theorem is stated three times in this paper, at Theorems 21, 26 and 59; the proof above is the one to read, and the other two occurrences should be taken as restatements of it.

5.5 Trust bases

The certified content of this section rests on two distinct, independently auditable trust bases, which we keep separate.

- (a) **Lean, native_decide (depths ≤ 22).** The T -independence of the orbit-growth sequence is machine-verified for $0 \leq d \leq 22$ in Lean 4 (theorem `universal_layers_through_22` in `lean/with_mathlib/ComputableUniversality.lean`). The proof performs a single breadth-first sweep over the canonical Coxeter words of W , deduplicating the symbolic affine images $\rho_{(a,b)}(w) \in \text{Aff}(\mathbb{Q}(a,b))$ by their numerators relative to the common denominator $(a^2 + b^2)^{22}$, and certifies the layer counts

$$1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, 351, \\ 554, 875, 1345, 2066, 3203, 4971, 7574, 11543, 17683, 27108, 41067$$

simultaneously for every right triangle with positive unequal legs. The statement is over $\mathbb{Q}(a,b)$, with no reliance on any single specialization, and it establishes Theorem 19 through depth 22 via the length-10 relations of Theorem 10; it does *not* establish the all-depths statement. It is discharged by `native_decide` and therefore sits in tier (ii) of Appendix B: its trust base is the Lean compiler and its evaluation of the compiled term, not the kernel alone. It is not a kernel-checked statement, and is not cited as one.

- (b) **Modular filter with exact confirmation (depths ≤ 36).** Beyond the Lean kernel, the finite candidate list is settled by the arithmetic certificate, whose guarantee is one-sided exactness of the finite-field evaluation backed by exact arithmetic over $\mathbb{Q}(i)$ on the surviving candidates. A Python/NumPy implementation (`code/zeta_probe/certify.py`) realizes the certificate at depth 30: it evaluates the symbolic ball at the 16 candidate rotation numbers with $c_T \leq 60$ over independent 31-bit primes (under three minutes), and the surviving list is confirmed over $\mathbb{Q}(i)$. A Rust implementation (`certify38_rust/`) extends the certificate through depth 36: the generic layer counts match the published values through depth 36, all 22 candidate rotation numbers with $c_T \leq 72$ are collision-free through depth 32 under the modular filter and confirmed over $\mathbb{Q}(i)$, and the single collision beyond is the $(1,2)$ deviation of Theorem 20. Because the depth-32 conclusion requires only the cutoff $c_T \leq 2d = 64$, the depth-36 run ($c_T \leq 72$) covers it with margin. This modular-plus-exact trust base extends the Lean bound of depth 22 by fourteen further layers, reaching depth 36; it is logically independent of the Lean proof and is not kernel-checked. Neither trust base in this subsection is the Lean kernel: item (a) is tier (ii) and item (b) is tier (iii) of Appendix B, which states the tiers precisely.

5.6 The isosceles boundary

Remark 23 (Isosceles degeneration as the $c_T = 1$ boundary). The hypothesis $a \neq b$ is essential, and in the framework of Definition 17 the isosceles case is exactly the boundary instance $c_T = 1$ identified in the last clause of Lemma 18. When $a = b$ the rotation number is $\zeta_T = i$, a root of unity, so λ is a unit, $c_T = N(\lambda) = 1$. The leg-swap symmetry adds a $\mathbb{Z}/2$ stabilizer to the orbit action, and the orbit sequence becomes

$$1, 3, 5, 8, 11, 13, 16, \dots,$$

strictly smaller than the universal sequence u_n for $n \geq 4$. Correspondingly the kernel of ρ_T already contains the translation lattice multiple $2(t-1)(t^2+1)$, and the deviation arrives at depth 4 rather

than at depth $\sim c_T$. The universality phenomenon of this section therefore applies precisely to the unequal-leg shapes $c_T \geq 5$, while the isosceles sequence is the separate OEIS entry; the unequal-leg orbit-growth sequence is [A396406](#).

6 Finite-horizon universality

The uniform universality of §5 holds through a finite, certified threshold (depth 32, Theorem 20). This section settles its all-depths counterpart in the negative: the orbit-growth sequence u_d^T is not shape-independent for all d . For every right triangle with positive unequal rational legs there is an explicit nontrivial element of $\ker(\rho_T)$, hence a depth beyond which the count drops below the universal value. The first such depth grows linearly in the arithmetic complexity of the shape, and we bracket it between explicit linear bounds.

Notation

Throughout, W is the abstract Coxeter group on the three side-reflections R_0, R_1, R_2 , identified with the geometric generators R_x, R_y, R_h (reflection in the first leg, the second leg, and the hypotenuse, respectively): $R_0 = R_x$, $R_1 = R_y$, $R_2 = R_h$. We fix the coordinate normalization of §5: the right-angle vertex is at the origin, the legs lie along the positive coordinate axes, and the figure is rescaled so that the first leg has length 1, the second leg length $\beta = b/a$. The map $\rho_T: W \rightarrow \text{Isom}(\mathbb{R}^2)$ is the geometric realization attached to the triangle $T = (a, b)$ (coprime integer legs, $a \neq b$), and $\rho_{\text{abs}}: W \rightarrow Q \ltimes M$ is the abstract realization of §7, with $\rho_T = (\text{id}_Q \ltimes \Phi_T) \circ \rho_{\text{abs}}$.

The *symbolic group* is the image of W under the affine representation over $\mathbb{Q}(a, b)$; its elements are tuples $(\varepsilon, \delta, k, P)$ recording a sign ε , a conjugation flag δ , a rotation power k , and a translation polynomial $P \in \mathbb{Z}[t^{\pm 1}]$ in the rotation variable $t = XY$, with $\zeta_T = (a + bi)/(a - bi)$ the rotation number. We write $\mathcal{T}$ for the subgroup of pure translations $(1, 0, 0, P)$. Following Theorem 19 we set $c_T := N(\lambda)$, the norm of the primitive denominator λ of ζ_T , and let $\mu_T(t) = c_T t^2 - e_T t + c_T$ be the primitive integer minimal polynomial of $\zeta_T^{\pm 1}$, with $e_T = 2|a^2 - b^2|$ when $a + b$ is odd and $e_T = |a^2 - b^2|$ when a, b are both odd, the absolute value being the convention fixed in Definition 17. The arithmetic facts that c_T is odd, all of its prime factors are $\equiv 1 \pmod{4}$, and $c_T \geq 5$ in the unequal-leg case are Lemma 18.

6.1 The conjecture and the obstruction

Conjecture 24 (All-depths universality). *For every right-triangle parameter $T = (a, b)$ with positive rational unequal legs, $\ker(\rho_T)$ is the same subgroup of W ; equivalently, u_d^T is independent of T for all $d \geq 0$.*

The factorization $\rho_T = (\text{id}_Q \ltimes \Phi_T) \circ \rho_{\text{abs}}$ yields the inclusion $\ker(\rho_{\text{abs}}) \subseteq \ker(\rho_T)$ for every T at no cost. Conjecture 24 is the reverse inclusion $\ker(\rho_T) \subseteq \ker(\rho_{\text{abs}})$ uniformly in T : equivalently, the injectivity of the $\mathbb{Z}[Q]$ -linear map $\Phi_T: M \rightarrow \mathbb{R}^2$ sending the central generator h to the translation $t(\rho_T(R_2))$. The translation module $M = \mathbb{Z}[Q]/(Y + 1, c + 1)$ is free of rank 1 over $\mathbb{Z}[t^{\pm 1}]$, hence of infinite rank as an abelian group, whereas its image under Φ_T lies in the rank-2 group $\mathbb{R}^2$. The map Φ_T therefore has infinite-rank kernel and cannot be injective. The remainder of this section exhibits a kernel element explicitly, as a short word in the generators, and tracks the depth at which the resulting collision first appears.

6.2 The glide-square translation lattice

The geometric source of the kernel is elementary. Because the two legs meet at a right angle, $R_x R_y$ is the half-turn about the right-angle vertex, so $g := R_x R_y R_h$ is a glide reflection along the hypotenuse; the square of a glide reflection is a pure translation.

Lemma 25 (Glide-square translation lattice). *In the symbolic group, $\tau := (R_x R_y R_h)^2$ is the pure translation by $2(t^{-1} - 1)$, i.e. the tuple $(1, 0, 0, 2(t^{-1} - 1))$. Writing $r := R_x R_h$ (rotation, linear part t), the conjugates $r^{-k} \tau^m r^k$ are the translations by $2m t^{-k}(t^{-1} - 1)$, and products of these realize*

$$2(t - 1) \mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}.$$

Proof. The half-turn gives $R_x R_y: z \mapsto -z$, so $g: z \mapsto -\zeta^{-1} \bar{z} - (1 - \zeta^{-1})$, whence

$$g^2(z) = -\zeta^{-1} \overline{g(z)} - (1 - \zeta^{-1}) = z + 2(\zeta^{-1} - 1),$$

an identity of symbolic tuples using only $\zeta \bar{\zeta} = 1$. Conjugating a translation with vector v by the rotation r^k (linear part t^k) multiplies v by t^{-k} ; translations commute, so finite products of the $r^{-k} \tau^m r^k$ realize $2(t^{-1} - 1) f(t)$ for every $f \in \mathbb{Z}[t^{\pm 1}]$. Since $2(t^{-1} - 1) \mathbb{Z}[t^{\pm 1}] = 2(t - 1) \mathbb{Z}[t^{\pm 1}]$ as t^{-1} is a unit, this gives the stated inclusion. $\square$

The next theorem records the reverse inclusion, identifying the pure-translation subgroup exactly. All subsequent structure rests on it: the kernel classification (Corollary 30), the lamplighter description, and the membership criterion all rest.

Theorem 26 (Pure-translation lattice). *The pure-translation subgroup of the symbolic group is exactly the ideal*

$$\mathcal{T} = 2(t - 1) \mathbb{Z}[t^{\pm 1}].$$

Proof. The inclusion $\supseteq$ is Lemma 25. For $\subseteq$, work modulo 2. Reduction of the affine representation modulo 2 collapses the sign and conjugation flags and sends the translation polynomial P to its image $\bar{P} \in \mathbb{F}_2[t^{\pm 1}]$; a word lies in $\mathcal{T}$ only if its linear part is trivial, and tracking the recursion for the translation part of a reduced word shows that every realizable $\bar{P}$ is divisible by $t - 1$ (equivalently $\bar{P}(1) = 0$), because each generator contributes a translation increment in $(t - 1) \mathbb{F}_2[t^{\pm 1}]$. Lifting, every $P \in \mathcal{T}$ satisfies $P(1) = 0$ and $P \equiv 0 \pmod{2}$ after removing the factor $t - 1$; hence $P \in 2(t - 1) \mathbb{Z}[t^{\pm 1}]$. The two inclusions give the equality. $\square$

The decisive structural point is that $\mathcal{T}$ is an ideal of $\mathbb{Z}[t^{\pm 1}]$: it contains a multiple of every integer Laurent polynomial, in particular of every shape's minimal polynomial μ_T . This observation refutes Conjecture 24.

6.3 Non-universality and the witness word

Theorem 27 (Non-universality). *Conjecture 24 is false for every unequal-leg rational triangle. With $\mu_T(t) = c_T t^2 - e_T t + c_T$, the word*

$$w_T = \tau^{c_T} r \tau^{-e_T} r \tau^{c_T} r^{-2}, \quad \tau = (R_x R_y R_h)^2, \quad r = R_x R_h,$$

is a nontrivial element of the symbolic group (the pure translation by $2(t^{-1} - 1) \mu_T(t) \neq 0$) of word-length $6(2c_T + |e_T|) + 8 \leq 24c_T + 8$, whereas $\rho_T(w_T)$ is the identity isometry. Consequently $\ker(\rho_T) \supsetneq \ker(\rho_{\text{abs}})$, and $u_d^T < u_d$ for some $d \leq 6(2c_T + |e_T|) + 8$.

Proof. By Lemma 25, conjugating τ^m by r^k (via $r^{-k}\tau^m r^k$) gives the translation $2mt^{-k}(t^{-1} - 1)$. Reading w_T left to right with the rotation letters accumulating the exponent k , the three translation blocks τ^{c_T} , τ^{-e_T} , τ^{c_T} are conjugated by r^0 , r^1 , r^2 respectively, and the trailing r^{-2} returns the linear part to the identity. The word therefore assembles the translation

$$2(t^{-1} - 1)(c_T - e_T t + c_T t^2) = 2(t^{-1} - 1)\mu_T(t),$$

a nonzero Laurent polynomial; hence $w_T \neq 1$ in the symbolic group. Its letter count is $6c_T + 2 + 6|e_T| + 2 + 6c_T + 4 = 6(2c_T + |e_T|) + 8$. Under ρ_T the translation vector is the evaluation at $t = \zeta_T$, which vanishes because $\mu_T(\zeta_T) = 0$; thus $w_T \in \ker(\rho_T)$ while $w_T \notin \ker(\rho_{\text{abs}})$ (its symbolic translation is nonzero). A nontrivial kernel element of length ℓ forces $|\rho_T(B_\ell)| < |B_\ell|$ for the symbolic ℓ -ball B_ℓ , hence a strictly smaller layer count at some depth $\leq \ell$. $\square$

Remark 28 (Witnesses for individual shapes). For $(a, b) = (1, 2)$ one has $\mu_T = 5t^2 - 6t + 5$ ($c_T = 5$, $e_T = 6$), and w_T has length 104; for $(3, 4)$, $\mu_T = 25t^2 - 14t + 25$ and length 392; for $(1, 3)$, $c_T = 5$ and length 116 (each a nonzero translation in $\ker(\rho_T)$, `code/zeta_probe/witness.py`). The word w_T provides only an *upper* bound on the first deviation depth: a length- ℓ kernel element forces a deviation by depth ℓ , but need not be the shortest such element. For $(1, 2)$ the shortest kernel element has length 66 (taking $f = t^j(1 + t)$ rather than the witness form $f = 1$), and the first deviation occurs at depth 33 (Remark 37), far below the witness length 104.

Remark 29 (Algebraic leg ratios of degree 2). The construction extends to any leg ratio β for which $\zeta_T = (1 + i\beta)/(1 - i\beta)$ has degree 2 over $\mathbb{Q}$, i.e. $\beta^2 \in \mathbb{Q}$: then the primitive integer minimal polynomial of ζ_T is again a self-reciprocal quadratic $c_T t^2 - e_T t + c_T$, and the witness word w_T assembles $2(t^{-1} - 1)\mu_T$ verbatim, yielding a nontrivial kernel element. For a leg ratio whose associated ζ_T has degree $d > 2$ the same scheme applies with the degree- d minimal polynomial μ_T in place of the quadratic and a witness word of $d + 1$ translation blocks; since $\mathcal{T}$ is an ideal it still meets $2(t - 1)\mu_T$, so a nontrivial kernel element exists. The quantitative bounds below are stated for the degree-2 (rational, or $\beta^2 \in \mathbb{Q}$) case, where $\mu_T = c_T t^2 - e_T t + c_T$.

The kernel is exhausted by the witness up to the symbolic action, with the lattice structure of Theorem 21 pinning it down completely.

Corollary 30 (Kernel classification and the lower height bound). *Let T have positive unequal legs with ζ_T of degree 2 (e.g. rational legs). Then*

$$\ker(\rho_T) = \{\text{translations by } 2(t - 1)\mu_T g : g \in \mathbb{Z}[t^{\pm 1}]\} \cong \mathbb{Z}[t^{\pm 1}],$$

the normal closure of the witness w_T . Every collision difference is divisible by $2(t - 1)\mu_T$, whose extreme coefficients are $\pm 2c_T$; hence a deviation at depth d forces $c_T \leq d$, i.e.

$$n_T \geq c_T, \quad n_T := \min\{d : u_d^T \neq u_d\}.$$

Proof. By Niven's theorem ζ_T is not a root of unity, so a kernel element has trivial linear part, hence is a pure translation $P \in \mathcal{T}$ with $P(\zeta_T) = 0$. By Theorem 21, $\mathcal{T} = 2(t - 1)\mathbb{Z}[t^{\pm 1}]$, so $P = 2(t - 1)g$; using $\zeta_T \neq 1$, the condition $P(\zeta_T) = 0$ is equivalent to $\mu_T \mid g$. Conjugation acts on translations by $\pm t^k$ and by $P \mapsto P(t^{-1})$; since μ_T is self-reciprocal ($t^2 \mu_T(t^{-1}) = \mu_T(t)$, as $\mu_T = c_T t^2 - e_T t + c_T$ is palindromic), the $\mathbb{Z}[t^{\pm 1}]$ -module generated by $2(t - 1)\mu_T$ is stable and exhausts the kernel. A deviation at depth d produces a collision difference D of height $\leq 2d$ divisible by $2(t - 1)\mu_T$, whose leading coefficient is $2c_T \cdot \text{lead}(g) \neq 0$; thus $2c_T \leq 2d$. $\square$

6.4 The deviation-depth sandwich

Combining the divisibility lower bound of Corollary 30, the certified uniform universality through depth 32 (Theorem 20), and the witness word of Theorem 27 as an upper bound brackets the first deviation depth between two explicit linear functions of the shape complexity.

Corollary 31 (Deviation-depth sandwich). *For every unequal-leg rational triangle the first deviation depth $n_T = \min\{d : u_d^T \neq u_d\}$ is finite, and*

$$\max(33, c_T) \leq n_T \leq 6(2c_T + |e_T|) + 8 \leq 24c_T + 8.$$

The two halves of the lower bound have different status. The bound $n_T \geq c_T$ is the closed-form divisibility bound of Corollary 30, valid for every shape. The floor $n_T \geq 33$ is not a closed-form bound: it is the certified uniform-universality threshold of Theorem 20 ($u_d^T = u_d$ for all $d \leq 32$), which rests on the finite modular-plus-exact certificate of that theorem rather than on a structural argument valid a priori for all shapes. The upper bound is the witness word. The first deviation depth is therefore linear in the arithmetic complexity c_T . For $(1, 2)$ the bounds read $33 \leq n_{(1,2)} \leq 104$, with $n_{(1,2)} = 33$ attaining the lower floor. For every shape other than $(1, 2)$ the depth-36 certificate sharpens the floor to $\max(37, c_T)$.

Remark 32 (The depth-38 run, and why it is not used). An earlier version quoted a depth-38 certificate. A depth-38 run was in fact performed, and its log is retained in `certify38_rust/`; the floor above is nonetheless stated from the depth-36 certificate, for a reason worth recording rather than burying. That run enumerates 24 candidates rather than 22, because its candidate bound is $c_T \leq 2 \cdot 38 = 76$ and so admits the two shapes of complexity 73; both are clean. It reports the generic layer counts as matching $a(0), \dots, a(38)$, and it flags six of the 24 as colliding under the modular filter: $(2, 1)$ and $(3, 1)$ at $c_T = 5$, $(4, 3)$ and $(7, 1)$ at $c_T = 25$, $(5, 2)$ at $c_T = 29$ and $(7, 5)$ at $c_T = 37$.

Five of those six are not deviations of any shape we can currently assert. The modular filter is one-sided: a *clean* verdict is exact, but a *collision* verdict is only a candidate, and by Remark 109 it becomes a deviation only after adjudication over $\mathbb{Q}(i)$. That adjudication was never carried out for this run, because no pairs files were exported from it. Independent re-runs at depths 34 and 36 find all five clean, so if any of the five is real it lies at depth 37 or 38; and four of them would contradict Proposition 36 by a wide margin, which predicts first deviations at depths 164, 213, 219 and 246 for $(4, 3)$, $(5, 2)$, $(7, 1)$ and $(7, 5)$. Unadjudicated modular collisions at that distance from the predicted depths are far more likely to be modular accidents than deviations. The fifth, $(3, 1)$, is the one case where the discrepancy is a single step: the law gives $n_{(1,3)} = 39$ and the run flags a collision at depth at most 38, so here an accident and a genuine off-by-one in the law are equally consistent with the evidence, and only the exact adjudication would separate them. The sixth, $(2, 1)$, is the known deviation of $(1, 2)$ at depth 33 and is expected. Nothing in this paper rests on the depth-38 run, and the sharpened floor $\max(37, c_T)$ uses only the depth-36 certificate, whose clean verdicts are exact. Settling the five would need the run repeated with `pairs export` and `exact_check.py` applied to each; we have not done so.

The sandwich can be replaced by an *exact* characterization: the deviation depth is precisely half the length of the shortest nontrivial kernel element, a shortest-vector problem in the ideal of Corollary 30 under the lamplighter word metric, evaluated with the closed form c_{pred} of Proposition 72; where an exact length rather than an upper bound is needed, the value is the one certified by breadth-first search rather than one deduced from Theorem 64.

Proposition 33 (Exact deviation-depth law). *For every unequal-leg rational (indeed algebraic) shape T , let*

$$K(T) = \min\{\ell(g) : g \in \ker \rho_T, g \neq 1\} = \min_{0 \neq m \in \mathbb{Z}[t^{\pm 1}]} \ell_{\text{tr}}(2(t-1)\mu_T m),$$

where ℓ is the word length in the symbolic group and $\ell_{\text{tr}}(P)$ denotes the length of the pure translation with lamp polynomial P . Then

$$n_T = \lceil K(T)/2 \rceil.$$

Proof. The second expression for $K(T)$ is Corollary 30: every nontrivial kernel element is a pure translation with polynomial $2(t-1)\mu_T m$.

($n_T \geq \lceil K/2 \rceil$.) The image group is a quotient, so the ρ_T -word-length of an image is the minimum symbolic length over its fiber; hence the cumulative counts satisfy $\sum_{d' \leq d} u_{d'}^T = \#\{\text{fibers meeting the symbolic ball of radius } d\}$ and the first depth at which the cumulative counts differ (which is the first depth at which the layer counts differ) is the first depth d at which some fiber contains two distinct elements of the ball of radius d . If $g \neq h$ share a fiber, then $gh^{-1} \in \ker \rho_T \setminus \{1\}$, so $\ell(g) + \ell(h) \geq \ell(gh^{-1}) \geq K$ and $\max(\ell(g), \ell(h)) \geq \lceil K/2 \rceil$.

($n_T \leq \lceil K/2 \rceil$.) Let $g \in \ker \rho_T$ attain $\ell(g) = K$ and split a geodesic word for g as $w_1 w_2$ with $|w_1| = \lceil K/2 \rceil$, $|w_2| = \lfloor K/2 \rfloor$. Then w_1 and w_2^{-1} have the same ρ_T -image but differ in the symbolic group (else $g = 1$), and both subwords are geodesic: if, say, $\ell(w_1) < |w_1|$, then $\ell(g) < K$, a contradiction. So the ball of radius $\lceil K/2 \rceil$ meets one fiber twice, and the counts deviate by depth $\lceil K/2 \rceil$. $\square$

Lemma 34 (Finite determination of $K(T)$). *Write the kernel translation $2(t-1)\mu_T m$ in the edge basis of Theorem 60: its deposit vector is the edge polynomial $A = 2\mu_T m$ (the quotient $P/(t-1)$ of the translation polynomial), every coefficient of which is even. Then*

$$\ell_{\text{tr}}(2(t-1)\mu_T m) \geq \max(\|A\|_1, 2 \text{span}(A)), \quad \text{span}(A) = (\deg m - \text{ord } m) + 2.$$

Consequently, for every bound B only finitely many multipliers m (up to the shift $m \mapsto t^j m$, under which ℓ_{tr} is invariant) satisfy $\ell_{\text{tr}} \leq B$: the travel term caps the degree spread at $\deg m - \text{ord } m \leq \frac{B}{2} - 2$, and on each bounded-spread slice the injective linear map $m \mapsto \mu_T m$ is bounded below, $\|\mu_T m\|_1 \geq \sigma \|m\|_\infty$ with $\sigma > 0$, capping the coefficients. Hence the minimum $K(T)$ is attained and is computable, over an explicit finite set.

Proof. By Theorem 62 the crossing numbers obey $m_j \geq |a_j|$, so $\ell_{\text{tr}} \geq \sum_j m_j \geq \|A\|_1$. A pure translation has linear part $k = 0$ and is realized by a *closed* walk on $\mathbb{Z}$; every edge between the extreme deposit sites $j_{\min}, j_{\max}$ of A is therefore crossed an even, hence ≥ 2 , number of times: a support edge because its deposit $|a_j| \geq 2$ is a nonzero even integer, a gap edge because a closed walk that reaches both sides of it crosses it and returns. There are $\text{span}(A) + 1$ such edges, so $\ell_{\text{tr}} \geq \sum_j m_j \geq 2 \text{span}(A)$. Since $\mu_T = c_T t^2 - e_T t + c_T$ has nonzero extreme coefficients, $\text{span}(2\mu_T m) = (\deg m - \text{ord } m) + 2$. Finiteness is then immediate: the bound $2 \text{span} \leq B$ confines the degree spread, and $m \mapsto \mu_T m$ is injective ($\mathbb{Z}[t^{\pm 1}]$ is a domain), so on the finite-dimensional slice of bounded spread it is bounded below in any two norms. $\square$

Remark 35 (why the travel term is needed). The span bound is not a convenience: because μ_T is a palindromic quadratic with $0 < e_T < 2c_T$, its two roots lie on the unit circle (and, by Niven's theorem, are not roots of unity), so $\|\mu_T m\|_1$ need not grow with $\deg m$. The crossing bound $\ell_{\text{tr}} \geq \|A\|_1$ alone therefore does *not* terminate the search; it is the travel term $2 \text{span}(A)$, which grows with the degree spread, that makes $K(T)$ a finite computation. The roots-on-the-unit-circle fact is formally

verified in Lean 4 / Mathlib (`DeviationLattice.lean`, theorem `mu_root_abs_one`: every complex root of $c_T t^2 - e_T t + c_T$ with $0 < e_T < 2c_T$ has modulus 1; axioms `propext`, `Classical.choice`, `Quot.sound`, no `sorry`), as is the self-reciprocity of μ_T .

Two further elementary facts sharpen the law. First, every generator move flips the direction flag, so pure translations have *even* word length: $K(T)$ is even and $n_T = K(T)/2$ exactly. In particular the certified $n_{(1,2)} = 33$ pins $K_{(1,2)} \in \{65, 66\}$ and parity forces

$$K_{(1,2)} = 66,$$

now a theorem; the minimum is attained by the translation with polynomial $-2(t+1)\mu_{(1,2)}$, lamp profile $(-10, 2, 2, -10)$, of word length exactly 66: against the witness-word bound of 104. Second, the metric formula evaluates in closed form on the two natural candidate families $m = 1$ and $m = t + 1$:

$$\ell_{\text{tr}}(2\mu_T) = 8c_T + 2e_T + 4\max(c_T, e_T), \quad \ell_{\text{tr}}(2(t+1)\mu_T) = 12c_T + 6|c_T - e_T|$$

(for $0 < e_T < 2c_T$, which always holds). Minimizing over the two families and halving yields the following closed-form law, which matches the computed $K(T)$ *exactly* on every shape tested.

Proposition 36 (Deviation-depth law). *Normalize $e_T > 0$. For every shape tested (thirteen (c_T, e_T) pairs, listed below), the first deviation depth equals*

$$n_T = \begin{cases} 3(c_T + e_T), & e_T \geq c_T, \\ 6c_T + \min(e_T, 3(c_T - e_T)), & e_T \leq c_T, \end{cases}$$

the minimum in the second line switching at $e_T = \frac{3}{4}c_T$. The inequality $n_T \leq (\text{law})$ is a theorem for every shape: the two displayed translations are explicit kernel elements, and bounding their lengths from above is all that the inequality needs, so the unconditional upper half of Theorem 64 suffices and Conjecture 80 is not used here. Equality is a theorem at $(1, 2)$ (the certified $n_{(1,2)} = 33$, Remark 37); at the further twelve shapes below it is verified by exhaustive bounded-degree shortest-vector search, the minimizer being $m \in \{1, t+1\}$ in every case.

Concretely: $n_T = 6c_T + e_T$ for $e_T \leq \frac{3}{4}c_T$ (minimizer $2\mu_T$), $n_T = 9c_T - 3e_T$ for $\frac{3}{4}c_T \leq e_T \leq c_T$, and $n_T = 3(c_T + e_T)$ for $e_T \geq c_T$ (minimizer $2(t+1)\mu_T$). In terms of the triangle itself the three regimes are *angle windows*: with $\beta = \arctan(b/a) \in (45^\circ, 90^\circ)$ the boundaries sit at $\tan \beta = \sqrt{11/5}$ ($\beta \approx 56.3^\circ$, where $e_T = \frac{3}{4}c_T$) and at $\beta = 60^\circ$ (where $e_T = c_T$); these two boundary identities ($e_T = c_T \Leftrightarrow b^2 = 3a^2$ and $e_T = \frac{3}{4}c_T \Leftrightarrow 5b^2 = 11a^2$) are formally verified in Lean (`DeviationLattice.lean`, `window_60`, `window_5637`). The law was validated as follows: for the nine leg shapes

T	c_T	e_T	n_T (law)	status
(1, 2)	5	6	33	certified (Remark 37)
(1, 3)	5	8	39	search
(2, 3)	13	10	87	search
(1, 5)	13	24	111	search
(3, 5)	17	16	105	search
(1, 4)	17	30	141	search
(3, 4)	25	14	164	search; consistent with $n_{(3,4)} > 42$
(2, 5)	29	42	213	search
(4, 5)	41	18	264	search

and four further abstract pairs $(c, e) = (61, 11), (37, 35), (53, 45), (73, 55)$ the law's prediction agreed with the computed minimum in every case. Those four are tests of the formula only, not of the geometry: e_T is always even, since $a + b$ odd gives $e_T = 2|a^2 - b^2|$ outright and a, b both odd makes $|a^2 - b^2|$ a difference of two odd squares, so no rational-leg triangle has any of those four (c_T, e_T) . In every case the minimizing multiplier always being $m \in \{1, t + 1\}$ (up to shift and sign). The searches ranged over multipliers m of degree ≤ 5 with coefficients $|m_i| \leq 4$ (widened to degree ≤ 7 and coefficients ≤ 6 for $(1, 2)$ and $(3, 4)$ with no change), over all shifts, pruned rigorously by the ℓ^1 lower bound of Lemma 34. The value $n_{(3,4)} = 164$ is a sharp prediction: the $(3, 4, 5)$ triangle, certified collision-free through depth 36, should first deviate at depth 164. (An earlier version quoted depth 42; the longest run actually performed is the depth-36 one of Appendix B.)

Two caveats fix the precise status. By Lemma 34 each $K(T)$ is a well-defined finite computation, so the deviation depth is decidable for every shape; but the searches above are bounded in degree, whereas the lemma's travel cutoff only forces $\deg m \leq K(T)/2$ (of order c_T). The tabulated equalities are therefore proved against multipliers of degree ≤ 5 (resp. ≤ 7), not against the full cutoff, so for the twelve searched shapes they are exhaustive-within-window verifications rather than unconditional theorems; only $(1, 2)$, whose value is independently certified, is a theorem outright. Exactness of the closed-form law for *all* shapes (equivalently, that no higher-degree multiplier ever undercuts $m \in \{1, t + 1\}$) is conjectural; the certified $(1, 2)$ anchor, the degree- ≤ 1 minimizer across thirteen shapes, and the window stability are the evidence.

Remark 37 (First deviation for $(1, 2)$). The threshold of Theorem 20 is attained at the smallest-complexity shape: $(1, 2)$ satisfies $u_d^{(1,2)} = u_d$ for all $d \leq 32$ and

$$u_{33}^{(1,2)} = u_{33} - 30 = 3,848,354 < u_{33},$$

the first deviation arriving at depth 33. Together with Corollary 31 this determines the deviation behaviour of this shape: $n_{(1,2)} = 33$, realizing the lower end of the sandwich.

Remark 38 (The isosceles degeneration). The hypothesis $a \neq b$ is essential. The isosceles case is the extreme instance $c_T = 1$, where $\zeta_T = i$ is a root of unity and the construction degenerates: the kernel of ρ_T contains the translation multiple $2(t - 1)(t^2 + 1)$, and the deviation arrives at depth 4 rather than at depth $\sim c_T$. The leg-swap symmetry then adds a $\mathbb{Z}/2$ stabilizer to the orbit action and the sequence becomes $1, 3, 5, 8, 11, 13, 16, \dots$, strictly below the universal sequence for $n \geq 4$. The universality phenomenon, and the sandwich Corollary 31, apply only to unequal legs.

The natural all-depths Conjecture 24 is false, uniformly over algebraic shapes. The pure-translation lattice $\mathcal{T} = 2(t - 1)\mathbb{Z}[t^{\pm 1}]$ (Theorem 21) is an ideal and therefore meets every shape's minimal polynomial. The resulting universality is finite-horizon: shape-independent through depth 32 (attained), with each fixed shape deviating at a first depth n_T that is linear in its arithmetic complexity c_T , bracketed by $\max(33, c_T) \leq n_T \leq 24c_T + 8$ where the floor 33 is certificate-dependent and the c_T bound is closed-form.

7 The translation module via Mayer–Vietoris and the lamplighter structure

The preceding sections work with the concrete affine representations $\rho_T: W \rightarrow \text{Aff}(\mathbb{R}^2)$ of the abstract side-reflection group $W = V_4 * \mathbb{Z}/2$, one for each right triangle T . This section extracts the algebraic skeleton common to all of these representations, computes the translation module of W over its common linear-part quotient, and identifies the generic group. The computation is

unconditional, group-theoretic, and independent of T ; it supplies the universality input of §5 and the structural conclusion that the generic side-reflection group is virtually the lamplighter group $\mathbb{Z} \wr \mathbb{Z}$.

Notation. Fix the placement $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ with $a, b > 0$, so the right angle sits at A and the two legs lie along the coordinate axes. The three mirror generators are written interchangeably as

$$R_0 = R_x \text{ (leg } AB), \quad R_1 = R_y \text{ (leg } AC), \quad R_2 = R_h \text{ (hypotenuse } BC),$$

the subscripts $0, 1, 2$ being used in the homological computations of this section and the geometric subscripts x, y, h elsewhere; the two notations denote the same generators throughout. The similarity class of T depends only on the shape ratio b/a (Lemma 45), so all constructions below are invariant under the central dilation $z \mapsto \lambda z$ and one may normalise $a = 1$ without loss.

Throughout, $W = A * B$ with

$$A = V_4 = \langle R_0, R_1 \mid R_0^2, R_1^2, (R_0 R_1)^2 \rangle, \quad B = \mathbb{Z}/2 = \langle R_2 \mid R_2^2 \rangle,$$

the generators $R_0 = R_x, R_1 = R_y$ being the two leg reflections and $R_2 = R_h$ the hypotenuse reflection. Write

$$Q = D_\infty \times \mathbb{Z}/2, \quad Q = \langle X, X', Y, c \mid X^2 = (X')^2 = Y^2 = c^2 = 1, XX' = c, c \text{ central} \rangle,$$

the common linear-part quotient, and let $\pi: W \twoheadrightarrow Q$ be the surjection $R_0 \mapsto X, R_1 \mapsto X', R_2 \mapsto Y$. Here the central involution $c = XX'$ realises the half-turn $R_0 R_1$ about the right-angle vertex, Y is the hypotenuse reflection, and the dihedral generator $t := XY$ (of infinite order) is the rotation by twice the acute angle. Set

$$T_\rho = \ker \pi = \ker(W \twoheadrightarrow Q),$$

the *translation kernel*; T_ρ is the same subgroup of W for every T , because the assignment π is dictated by the reflection pattern of a right triangle and not by its leg lengths. For a subgroup or generating set we write $I_A, I_B, I_W \subseteq \mathbb{Z}[Q]$ for the right ideals generated by the augmentation images $\{g - 1 : g \in A\}, \{g - 1 : g \in B\}, \{g - 1 : g \in W\}$ respectively, and $\mathbb{Z}[t^{\pm 1}]$ for the rotation subring generated by t .

7.1 The translation module

Lemma 39 (Mayer–Vietoris reduction). *With $A = V_4, B = \mathbb{Z}/2, W = A * B$ and $Q = D_\infty \times \mathbb{Z}/2$ as above, and $T_\rho = \ker(W \twoheadrightarrow Q)$, the abelianization T_ρ^{ab} is, as a left $\mathbb{Z}[Q]$ -module,*

$$T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B,$$

and this module is independent of T . Set $h := (c - 1)(Y - 1) \in \mathbb{Z}[Q]$. Then the annihilator of h is generated by $(Y + 1)$ and $(c + 1)$, namely

$$(Y + 1)h = 0 \quad \text{and} \quad (c + 1)h = 0 \quad \text{in } \mathbb{Z}[Q],$$

so the cyclic submodule $\mathbb{Z}[Q] \cdot h$ is isomorphic to

$$M := \mathbb{Z}[Q]/(Y + 1, c + 1).$$

Moreover the equality

$$I_A \cap I_B = \mathbb{Z}[Q] \cdot h$$

holds unconditionally; hence $T_\rho^{\text{ab}} \cong M$.

Proof. Since $A \hookrightarrow Q$ and $B \hookrightarrow Q$ are injective, $\mathbb{Z}[Q]$ is free as a $\mathbb{Z}[A]$ -module and as a $\mathbb{Z}[B]$ -module, so $H_n(A; \mathbb{Z}[Q]) = H_n(B; \mathbb{Z}[Q]) = 0$ for $n \geq 1$. The Mayer–Vietoris sequence of the free product $W = A * B$ with coefficients in $\mathbb{Z}[Q]$ therefore collapses to the short exact sequence

$$0 \rightarrow H_1(W; \mathbb{Z}[Q]) \rightarrow \mathbb{Z}[Q]/I_A \oplus \mathbb{Z}[Q]/I_B \rightarrow \mathbb{Z}[Q]/I_W \rightarrow 0,$$

and Crowell’s exact sequence [17] identifies $T_\rho^{\text{ab}} \cong H_1(W; \mathbb{Z}[Q]) \cong I_A \cap I_B$ as a left $\mathbb{Z}[Q]$ -submodule. The right-hand side depends only on the data $A * B \twoheadrightarrow Q$, with no reference to any geometric realisation, whence the asserted T -independence.

Membership. Since $c = XX' \in A$ we have $c-1 \in I_A$, and centrality of c gives $h = (c-1)(Y-1) = (Y-1)(c-1) \in \mathbb{Z}[Q] \cdot (c-1) \subseteq I_A$. Likewise $Y-1 \in I_B$ and $h \in \mathbb{Z}[Q] \cdot (Y-1) \subseteq I_B$, so $h \in I_A \cap I_B$.

Annihilator. From $(Y+1)(Y-1) = Y^2 - 1 = 0$ and the centrality of c we get $(Y+1)h = (c-1)(Y+1)(Y-1) = 0$; from $(c+1)(c-1) = c^2 - 1 = 0$ we get $(c+1)h = 0$. Thus $\mathbb{Z}[Q] \cdot h \cong M$.

The equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$. The two displays just established give $\mathbb{Z}[Q] \cdot h \subseteq I_A \cap I_B$; the reverse inclusion is established as follows. Write $R = \mathbb{Z}[Q]$. Since $X' = cX$, the identity $X' - 1 = c(X-1) + (c-1)$ shows $I_A = R(X-1) + R(c-1)$, while $I_B = R(Y-1)$. Every element of $I_A \cap I_B$ has the form $\xi = x(Y-1)$ for some $x \in R$.

Step 1 (reduction modulo the central ideal). Because c is central, $R(c-1)$ is a two-sided ideal and the quotient $R/R(c-1)$ is the group ring $S = \mathbb{Z}[D_\infty]$ of the free product $D_\infty = \langle X \rangle * \langle Y \rangle$. The image $\bar{x}$ of x satisfies $\bar{x}(Y-1) \in S(X-1) \cap S(Y-1)$. The augmented chain complex of the Bass–Serre tree of this free product (vertex set $D_\infty/\langle X \rangle \sqcup D_\infty/\langle Y \rangle$, edge set D_∞) is

$$0 \rightarrow S \xrightarrow{\partial} S/S(X-1) \oplus S/S(Y-1) \rightarrow \mathbb{Z} \rightarrow 0, \quad \partial(s) = (s \bmod S(X-1), -s \bmod S(Y-1)).$$

A tree is contractible, so this complex is exact; exactness at the left term (the vanishing of the first homology of a tree) says precisely $S(X-1) \cap S(Y-1) = 0$. Hence $\bar{x}(Y-1) = 0$ in S , i.e. $x(Y-1) \in R(c-1)$.

Step 2 (involution annihilator). In any group ring, for an involution Y one has $x(Y-1) = 0$ if and only if $x \in R(Y+1)$: comparing coefficients along each right coset $\{g, gY\}$, the relation $x(Y-1) = 0$ forces $x_g = x_{gY}$ for all g , equivalently $x \in R(Y+1)$.

Step 3 (descent). By Step 1, $x(Y-1) = w(c-1)$ for some $w \in R$. Decompose along the free splitting $R = \mathbb{Z}[D_\infty] \oplus \mathbb{Z}[D_\infty]c$, writing $x = x_0 + x_1c$ and $w = w_0 + w_1c$. Then $w(c-1) = (w_1 - w_0)(1-c)$, and matching the two components of $x(Y-1) = x_0(Y-1) + x_1(Y-1)c$ across this splitting yields $x_0(Y-1) = -x_1(Y-1)$, i.e. $(x_0 + x_1)(Y-1) = 0$. By Step 2, $x_0 + x_1 = z(Y+1)$ for some $z \in R$. Therefore $x = x_0(1-c) + (x_0 + x_1)c = x_0(1-c) + z(Y+1)c$, and using $(Y+1)(Y-1) = 0$,

$$\xi = x(Y-1) = x_0(1-c)(Y-1) = -x_0 h \in \mathbb{Z}[Q] \cdot h.$$

No torsion is encountered at any step, so the inclusion $I_A \cap I_B \subseteq \mathbb{Z}[Q] \cdot h$ holds over $\mathbb{Z}$. $\square$

Remark 40 (Role of centrality). The cyclic identification of Lemma 39 succeeds precisely because $c \in Q$ is central. The candidate generator $(X-1)(Y-1)$ does not serve, since X and Y do not commute in D_∞ . The central involution c furnished by the second $\mathbb{Z}/2$ factor of Q is what yields the cyclic presentation, and it is intrinsic to the right-triangle structure: $c = XX'$ realises the half-turn $R_0R_1 = R_xR_y$ about the right-angle vertex.

Remark 41 (Finite quotients). In each finite quotient $D_{2N} \times \langle c \rangle$ (obtained by imposing $t^N = 1$, for $3 \leq N \leq 16$) one has

$$\dim(I_A \cap I_B) = \dim(\mathbb{Z}[Q] \cdot h) + 1,$$

the excess dimension being spanned by the rotation-norm element $\nu_N = \sum_{k=0}^{N-1} t^k$ (`code/ideal/intersection_norm`). The element ν_N has no lift to $\mathbb{Z}[Q]$, where $\sum_k t^k$ is not a group-ring element; hence Lemma 39 is unaffected by the excess dimension present in the torsion quotients.

7.2 The generic kernel

Corollary 42 (The generic kernel is the relative commutator subgroup). *Let ρ_{gen} denote the symbolic (generic) side-reflection representation and $W_{\text{gen}} = \rho_{\text{gen}}(W)$ the generic group. Then*

$$\ker \rho_{\text{gen}} = [T_\rho, T_\rho], \quad \text{equivalently} \quad W_{\text{gen}} \cong W/[T_\rho, T_\rho];$$

*that is, the generic right-triangle group is the Q -relative metabelianization of $W = V_4 * \mathbb{Z}/2$. In particular the eight length-10 relations of Theorem 10 are exactly the shortest elements of $[T_\rho, T_\rho]$, and the Mayer–Vietoris module theory of this section and the lamp model of §7.4 describe one and the same object.*

Proof. In the symbolic group the linear part t is a formal variable, so the linear-part map is faithful on Q and $\ker \rho_{\text{gen}} \subseteq T_\rho$. The image $\rho_{\text{gen}}(T_\rho)$ is the group $\mathcal{T}$ of pure translations (elements with trivial linear part), which is abelian; hence $\rho_{\text{gen}}|_{T_\rho}$ factors through T_ρ^{ab} and induces a surjective $\mathbb{Z}[t^{\pm 1}]$ -equivariant homomorphism

$$\Phi: T_\rho^{\text{ab}} \twoheadrightarrow \mathcal{T}.$$

By Lemma 39, $T_\rho^{\text{ab}} \cong M \cong \mathbb{Z}[t^{\pm 1}]$, and by Theorem 21, $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}] \cong \mathbb{Z}[t^{\pm 1}]$. A surjective module map between two modules each isomorphic to $\mathbb{Z}[t^{\pm 1}]$ is injective, since a surjective endomorphism of a Noetherian module is injective; thus Φ is an isomorphism and

$$\ker(\rho_{\text{gen}}|_{T_\rho}) = \ker(T_\rho \rightarrow T_\rho^{\text{ab}}) = [T_\rho, T_\rho].$$

Concretely, the Crowell generator h realises as the glide-reflection square: the commutator $[R_0 R_1, R_2] = (R_0 R_1 R_2)^2 = \tau$ (both factors being involutions) whose translation $2(t^{-1} - 1)$ generates $\mathcal{T}$. $\square$

Remark 43 (Role of the translation-lattice equality). Corollary 42 and the lamplighter identification of Theorem 47 both invoke the equality $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$. This equality is established unconditionally in Theorem 21: the inclusion $2(t-1)\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square computation of Lemma 25, and the reverse inclusion follows from the mod-2 degeneration of the symbolic group to the infinite dihedral group, which has no nontrivial translations. All downstream results of this section that depend on the equality are therefore unconditional.

Remark 44 (Universality needs only T -independence, not the explicit M). The universality statement of Theorem 20 requires only that the translation module $T_{\rho T}^{\text{ab}} \cong I_A \cap I_B$ be *independent of T* . This is supplied by the Mayer–Vietoris identification alone, whose right-hand side $I_A \cap I_B$ is a purely group-theoretic object attached to $W = A * B \twoheadrightarrow Q$, with no reference to the geometric realisation of any triangle. The cyclic description $\mathbb{Z}[Q] \cdot h \cong M = \mathbb{Z}[Q]/(Y+1, c+1)$ of Lemma 39 is used only to write the abstract realization ρ_{abs} down explicitly (Lemma 46), which in turn needs only the annihilator relations $(Y+1)h = (c+1)h = 0$. Universality thus does not depend on the full equality $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$, only on the T -independence of $I_A \cap I_B$.

7.3 Dilation invariance and abstract realization

Lemma 45 (Dilation invariance). *For $\lambda > 0$, the representation $\rho_{\lambda T}$ is conjugate to ρ_T by the central dilation D_λ . In particular $\ker(\rho_{\lambda T}) = \ker(\rho_T)$, so the kernel depends only on the similarity class of T , that is, only on the shape ratio b/a .*

Proof. For $i = 0, 1, 2$ one has $D_\lambda \circ \rho_T(R_i) \circ D_\lambda^{-1} = \rho_{\lambda T}(R_i)$, since reflecting across a line and then rescaling about the centre coincides with rescaling and then reflecting across the scaled line. Conjugation is an automorphism of $\text{Aff}(\mathbb{R}^2)$, so $\ker(\rho_{\lambda T}) = \ker(\rho_T)$. $\square$

Lemma 46 (Abstract realization). *Define $\rho_{\text{abs}}: W \rightarrow Q \ltimes M$ on generators by*

$$R_0 \mapsto (X, 0), \quad R_1 \mapsto (X', 0), \quad R_2 \mapsto (Y, h),$$

where $M = \mathbb{Z}[Q]/(Y + 1, c + 1)$ carries the left Q -action. Then ρ_{abs} is a well-defined group homomorphism.

Proof. The defining relations of W are checked directly: $\rho_{\text{abs}}(R_0)^2 = (X^2, 0) = (1, 0)$, and likewise for R_1 ; $\rho_{\text{abs}}(R_2)^2 = (Y^2, (Y+1)h) = (1, 0)$ since $(Y+1)h = 0$ in M by Lemma 39; and $((X, 0)(X', 0))^2 = (c^2, 0) = (1, 0)$. All relators map to the identity, so ρ_{abs} is well defined. $\square$

7.4 The lamplighter structure

Theorem 47 (Lamplighter structure). *Let W_{gen} be the generic side-reflection group. The set H of elements whose linear part is a pure rotation t^k is a normal subgroup of index 4, and, with $r := R_x R_h = R_0 R_2$,*

$$H = \mathcal{T} \rtimes \langle r \rangle \cong \mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z} = \mathbb{Z} \wr \mathbb{Z},$$

the wreath product of $\mathbb{Z}$ by $\mathbb{Z}$. Consequently W_{gen} (equivalently, the side-reflection group of every right triangle with transcendental leg ratio (Corollary 42)) is virtually $\mathbb{Z} \wr \mathbb{Z}$. In particular it is:

- (i) metabelian-by-finite, hence amenable, while of exponential growth: the coexistence for which $\mathbb{Z} \wr \mathbb{Z}$ is the standard example, here realised by three mirrors of a Euclidean triangle;
- (ii) not finitely presentable: by Baumslag's theorem [8] the wreath product $A \wr B$ with $A \neq 1$ is finitely presented only when B is finite, and finite presentability is a commensurability invariant.

Proof. Let $\tilde{Q} = \{\pm t^k \sigma^\delta\}$ be the linear-part quotient of W_{gen} , where σ is the reflection generator and the sign is central. Then H is the kernel of the composite $W_{\text{gen}} \rightarrow \tilde{Q} \rightarrow \tilde{Q}/\langle t \rangle$. The subgroup $\langle t \rangle$ is normal in $\tilde{Q}$ (the sign is central and $\sigma t \sigma^{-1} = t^{-1}$) of index 4, so H is normal of index 4 in W_{gen} . Any element of H with linear part t^k differs from r^k by a pure translation, whence $H = \mathcal{T} \langle r \rangle$ with $\langle r \rangle \cap \mathcal{T} = 1$; this is an internal semidirect product $H = \mathcal{T} \rtimes \langle r \rangle$. By Theorem 21, the map $f \mapsto 2(t-1)f$ is a $\mathbb{Z}[t^{\pm 1}]$ -module isomorphism $\mathbb{Z}[t^{\pm 1}] \xrightarrow{\sim} \mathcal{T}$, and conjugation by r acts on $\mathcal{T}$ as multiplication by t . Since $\mathbb{Z}[t^{\pm 1}] = \bigoplus_{k \in \mathbb{Z}} \mathbb{Z} t^k$ with t acting as the shift, the semidirect product $\mathbb{Z}[t^{\pm 1}] \rtimes_t \mathbb{Z}$ is precisely the wreath product $\mathbb{Z} \wr \mathbb{Z}$.

For the structural assertions: $\mathcal{T}$ is abelian and $\tilde{Q}$ is virtually abelian, so W_{gen} is metabelian-by-finite, hence amenable. Exponential growth is classical for $\mathbb{Z} \wr \mathbb{Z}$ and passes to commensurable groups; this is consistent with the Milnor–Wolf theorem [26, 43], by which a solvable group of non-polynomial growth must have exponential growth. Finite presentability fails by Baumslag's theorem together with its invariance under commensurability, as stated. $\square$

For rational T the specialized group $\rho_T(W)$ is virtually $\mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z}$, a finitely generated metabelian affine group: the algebraic shapes are precisely the metabelian quotients in which the formal rotation t is specialized to the rotation number $\zeta_T = (a + bi)/(a - bi)$. The generic group W_{gen} , where t remains a free variable, is their common cover, and its orbit growth is the universal right-triangle reflection sequence. These specialized groups are the subject of §8.

8 The rational triangle groups W_T

Throughout this section $T = (a, b)$ is a right triangle with positive unequal rational legs, in the normalization of §5 (right angle at the origin, legs on the positive axes, x -leg of length 1), and

$$W_T := W(T) = \rho_T(W) \leq \text{Isom}(\mathbb{R}^2)$$

is its side-reflection group, with word length ℓ_T relative to $\{R_0, R_1, R_2\}$. We write $\zeta = \zeta_T$, $c = c_T$, $e = e_T$, $\mu = \mu_T$ as in Definition 17, and

$$A_T := \mathbb{Z}[\zeta_T^{\pm 1}] \subset \mathbb{Q}(i), \quad \Lambda_T := 2(\zeta_T - 1) A_T \subset \mathbb{C}.$$

By Lemma 15 an element of W_T is a translation if and only if its symbolic class is $(1, 0, 0)$, so by Theorem 21 the translation subgroup of W_T is the evaluation of $\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}]$ at $t = \zeta_T$, namely Λ_T . The kernel of $W_{\text{gen}} \twoheadrightarrow W_T$ is already known: by Corollary 30 it consists of the translations by $2(t-1)\mu_T g$, $g \in \mathbb{Z}[t^{\pm 1}]$, so that W_T is the quotient of the generic group by a single $\mathbb{Z}[t^{\pm 1}]$ -cyclic ideal of translations, normally generated by the witness w_T . This section records what that quotient is as a group, what can and cannot yet be said about its growth series, and the computations behind both.

8.1 Structure

Proposition 48 (Structure of W_T). *Let $H_T \leq W_T$ be the set of elements whose linear part is a rotation $z \mapsto \zeta_T^k z$, and let $r = R_x R_h$.*

- (i) H_T is a normal subgroup of index 4, and

$$H_T = \Lambda_T \rtimes \langle r \rangle \cong A_T \rtimes_{\zeta_T} \mathbb{Z} = \mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z},$$

the generator of $\mathbb{Z}$ acting by multiplication by ζ_T . Moreover $A_T = \mathbb{Z}[1/c_T] + \mathbb{Z}[1/c_T] \zeta_T$ as a group, and A_T is not finitely generated.

- (ii) W_T is dense in $\text{Isom}(\mathbb{R}^2)$. It is linear over $\mathbb{Q}$ (a subgroup of $\text{GL}_3(\mathbb{Q})$ in the rational placement of §2).
- (iii) W_T is solvable of derived length exactly 3.
- (iv) W_T is not finitely presented.
- (v) H_T is a cocompact lattice in the locally compact group

$$G_T := \left(\mathbb{C} \times \prod_{\pi | c_T} \mathbb{Q}(i)_\pi \right) \rtimes_{\zeta_T} \mathbb{Z},$$

where π runs over the $2\omega(c_T)$ Gaussian primes dividing c_T , each completion $\mathbb{Q}(i)_\pi$ is isomorphic to $\mathbb{Q}_p$ with $p = \pi\bar{\pi}$, and the generator of $\mathbb{Z}$ multiplies every coordinate by ζ_T . For $T = (1, 2)$, W_T is quasi-isometric to $\mathbb{R}^2 \times \text{DL}(5, 5)$, where $\text{DL}(5, 5)$ is the Diestel–Leader graph, the horocyclic product of two 6-regular trees. In general W_T is quasi-isometric to $\mathbb{R}^2$ times the horocyclic product of the $2\omega(c_T)$ Bruhat–Tits trees of the $\mathbb{Q}(i)_\pi$, taken along the valuation vector $(v_\pi(\zeta_T))_\pi$, that is, restricted to the tuples whose Busemann heights are $sv_\pi(\zeta_T)$ for a common s .

Proof. (i) By Lemma 15 the sign ε and the flag δ of the symbolic class are well defined on W_T , and by the composition law of Lemma 13 the map $g \mapsto (\varepsilon_g, \delta_g) \in \{\pm 1\} \times \mathbb{Z}/2$ is a homomorphism. It is onto $(R_x \mapsto (1, 1), R_y \mapsto (-1, 1))$, and its kernel is H_T , which is therefore normal of index 4. The element r has linear part $\zeta_T^{\pm 1}$; if $g \in H_T$ has linear part ζ_T^k then $gr^{\mp k}$ is a translation, and r^k is a translation only for $k = 0$ because ζ_T is not a root of unity. Hence $H_T = \Lambda_T \rtimes \langle r \rangle$, with r acting on Λ_T by multiplication by $\zeta_T^{\pm 1}$. Multiplication by $2(\zeta_T - 1) \neq 0$ is an A_T -module isomorphism $A_T \rightarrow \Lambda_T$, which gives the displayed isomorphism. Since $c\zeta^2 = e\zeta - c$ and $\zeta^{-1} = e/c - \zeta$, the group A_T is the $\mathbb{Z}[1/c]$ -span of $1, \zeta$, once $1/c \in A_T$ is known; and it is, because $e/c = \zeta + \zeta^{-1} \in A_T$ and $\gcd(c, e) = 1$ (Lemma 18(iv)) gives integers x, y with $xe/c + y = 1/c$. As $c \geq 5$, $A_T \supseteq \mathbb{Z}[1/c]$ is not finitely generated.

(ii) The translation subgroup $\Lambda_T = 2(\zeta - 1)A_T$ contains $2(\zeta - 1)(\mathbb{Z}[1/c] + \mathbb{Z}[1/c]\zeta)$, which is dense in $\mathbb{C}$ because $\mathbb{Z}[1/c]$ is dense in $\mathbb{R}$ and $\zeta \notin \mathbb{R}$. The closure $\overline{W_T}$ therefore contains every translation. It then contains $z \mapsto \zeta^k z$ for all k (compose $r^{\pm k}$ with a translation), hence all rotations about 0, since $\zeta^{\mathbb{Z}}$ is dense in the circle; and it contains the reflection R_x . So $\overline{W_T} = \text{Isom}(\mathbb{R}^2)$. Linearity: in the placement $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ with $a, b \in \mathbb{Q}$ each R_i is a 3×3 affine matrix over $\mathbb{Q}$.

(iii) Since W_T/H_T is abelian, $W'_T \leq H_T$; since $H_T/\Lambda_T \cong \mathbb{Z}$, $W''_T \leq H'_T \leq \Lambda_T$, which is abelian, so $W'''_T = 1$. For the lower bound, $[R_x, R_h] = (R_x R_h)^2 = r^2 \in W'_T$ has linear part $\zeta^{\pm 2} \neq 1$, and for the glide square τ of Lemma 25, with translation vector $v_0 \neq 0$, the commutator $[r, \tau] \in W'_T$ is the translation by $(\zeta^{\pm 1} - 1)v_0 \neq 0$. The commutator of r^2 with this translation is the translation by $(\zeta^{\pm 2} - 1)(\zeta^{\pm 1} - 1)v_0 \neq 0$, a nontrivial element of W''_T . So the derived length is exactly 3.

(iv) Finite presentability passes to subgroups of finite index (Reidemeister–Schreier), so it suffices to show that H_T is not finitely presented. We use Theorem A of Bieri and Strebel [10]: if G is a finitely presented group and $\chi: G \rightarrow \mathbb{Z}$ is an epimorphism, then G is an HNN-extension $G = \langle B, s \mid sKs^{-1} = L \rangle$ with finitely generated base group $B \leq \ker \chi$, associated subgroups $K, L \leq B$ and $\chi(s) = 1$; if moreover G contains no free subgroup of rank 2, then Britton’s lemma forces the extension to be ascending, $K = B$ or $L = B$. Suppose H_T finitely presented and apply this to $\chi: H_T \rightarrow \mathbb{Z}$, $\Lambda_T r^k \mapsto k$. The group H_T is solvable, so it has no free subgroup of rank 2, and we obtain a finitely generated $B \leq \ker \chi = \Lambda_T$ with $sBs^{-1} \subseteq B$ or $s^{-1}Bs \subseteq B$. Conjugation by $s \in \Lambda_T r^1$ acts on the abelian group Λ_T as multiplication by $\alpha = \zeta^{\pm 1}$, so $\alpha B \subseteq B$ or $\alpha^{-1}B \subseteq B$. Since $\ker \chi$ is the union of the conjugates $s^n B s^{-n}$ and $\Lambda_T \neq 0$, $B \neq 0$. Now B is a nonzero finitely generated subgroup of $\mathbb{C}$, hence free abelian of finite rank, and multiplication by $\alpha^{\pm 1}$ is an endomorphism of it; by the Cayley–Hamilton theorem, applied to one nonzero $b \in B$, $\alpha^{\pm 1}$ is a root of a monic integer polynomial, i.e. an algebraic integer. But $\zeta_T^{\pm 1}$ have the primitive minimal polynomial μ_T , whose leading coefficient is $c_T \geq 5$, so neither is an algebraic integer. This contradiction shows that H_T , and hence W_T , is not finitely presented.

(v) Let $K = \mathbb{Q}(i)$ and let S be the set of Gaussian primes dividing c . Each rational prime $p \mid c$ is $\equiv 1 \pmod{4}$ (Lemma 18) and splits as $p = \pi \bar{\pi}$, so $|S| = 2\omega(c)$ and $K_\pi \cong \mathbb{Q}_p$. Up to a unit, $\zeta = \bar{\lambda}/\lambda$ with λ as in Definition 17; λ is not divisible by any rational integer, so λ and $\bar{\lambda}$ are coprime and $v_\pi(\zeta) \neq 0$ for every $\pi \in S$. Hence $A_T \subseteq \mathcal{O}_S = \mathbb{Z}[i][1/c]$, and both are free $\mathbb{Z}[1/c]$ -modules of rank 2 spanning K (part (i)), so A_T has finite index in $\mathcal{O}_S$. The ring of S -integers $\mathcal{O}_S$ is a discrete cocompact subgroup of $V = \mathbb{C} \times \prod_{\pi \in S} K_\pi$ under the diagonal embedding; so is A_T , and so is $\Lambda_T = 2(\zeta - 1)A_T$, multiplication by $2(\zeta - 1) \in K^\times$ being a topological automorphism of V . The map $(z \mapsto \zeta^k z + \beta) \mapsto (\beta, k)$ embeds H_T in $G_T = V \rtimes_{\zeta} \mathbb{Z}$ with image $\Lambda_T \rtimes \mathbb{Z}$, discrete, and $G_T/H_T \cong V/\Lambda_T$ is compact. So H_T is a cocompact lattice.

For $T = (1, 2)$, $c = 5$, $S = \{\pi, \bar{\pi}\}$ with $\pi = 1 - 2i$ up to a unit, and $v_\pi(\zeta) = -v_{\bar{\pi}}(\zeta) = \pm 1$. Each $K_\pi \rtimes \zeta^{\mathbb{Z}} \cong \mathbb{Q}_5 \rtimes \zeta^{\mathbb{Z}}$ acts on the 6-regular Bruhat–Tits tree of $\mathbb{Q}_5$ by affine maps fixing an end, K_π acting simply transitively on the vertices of each horosphere up to the compact stabilizers $\pi^n \mathcal{O}_\pi$,

and ζ shifting the Busemann height by $v_\pi(\zeta)$. Because the two shifts are opposite, G_T acts on $\mathbb{C} \times \text{DL}(5, 5)$, isometrically on the first factor (as $|\zeta| = 1$), transitively, with compact point stabilizers. By the Švarc–Milnor lemma H_T , and hence W_T , is quasi-isometric to $\mathbb{R}^2 \times \text{DL}(5, 5)$. For general c the same argument applies with the $2\omega(c)$ trees and the heights constrained along $(v_\pi(\zeta))_\pi$; we omit the routine details. $\square$

Remark 49 (Scope of the structure statements). Parts (i)–(iv) and Proposition 50 below use only that ζ is a root of a primitive self-reciprocal quadratic $ct^2 - et + c$ with $c \geq 2$ and $0 < e < 2c$ (so that $|\zeta| = 1$ and ζ is not a root of unity). They therefore hold verbatim for the quadratic shapes of Remark 29, whose leg ratio β has $\beta^2 = (2c - e)/(2c + e) \in \mathbb{Q}$. The computational laboratory shape of this section, $(c, e) = (2, 1)$, is one of them: $\beta = \sqrt{3/5}$, $\cos \arg \zeta = 1/4$, and first deviation depth $13 = 6c + e$. Part (v) is stated for rational T , where $K = \mathbb{Q}(i)$; for quadratic shapes K is the imaginary quadratic field $\mathbb{Q}(\zeta)$ and the primes of c need not split. The group W_T is not an instance of the generic $\mathbb{Z} \wr \mathbb{Z}$ picture of Theorem 47: the lamp group A_T has rank 2 rather than infinite rank, it is not finitely generated, and finite presentability fails for the arithmetic reason in (iv) rather than by Baumslag’s theorem.

8.2 The stable norm

For a translation $\tau_v \in W_T$ by $v \in \Lambda_T$ the limit

$$\|v\|_T := \lim_{n \rightarrow \infty} \frac{\ell_T(\tau_v^n)}{n} = \inf_{n \geq 1} \frac{\ell_T(\tau_v^n)}{n}$$

exists by subadditivity (Fekete’s lemma), and is the stable norm of v .

Proposition 50 (The stable norm is round). *There is a constant $\kappa_T > 0$ such that $\|v\|_T = \kappa_T |v|$ for every $v \in \Lambda_T$.*

Proof. From $\ell_T(g^{-1}) = \ell_T(g)$ and the commutativity of translations, $\|\cdot\|_T$ is a seminorm on Λ_T with $\|mv\|_T = |m| \|v\|_T$ for $m \in \mathbb{Z}$. It is invariant under the linear parts of W_T : for $g \in W_T$ with linear part L_g , $g\tau_v^n g^{-1} = \tau_{L_g v}^n$ and $\ell_T(g\tau_v^n g^{-1}) \leq \ell_T(\tau_v^n) + 2\ell_T(g)$, so $\|L_g v\|_T \leq \|v\|_T$, and symmetrically. In particular $\|\zeta v\|_T = \|v\|_T$.

Lipschitz on $\mathbb{Q}(i)$. Λ_T spans $\mathbb{Q}(i)$ over $\mathbb{Q}$, and $\|q\|_T := \|mq\|_T/m$ for any $m \geq 1$ with $mq \in \Lambda_T$ extends the seminorm $\mathbb{Q}$ -homogeneously to $\mathbb{Q}(i)$. For a $\mathbb{Q}$ -basis $e_1, e_2 \subset \Lambda_T$ of $\mathbb{Q}(i)$ and $q = xe_1 + ye_2$, subadditivity gives $\|q\|_T \leq |x| \|e_1\|_T + |y| \|e_2\|_T \leq C|q|$ with C depending only on the basis. Hence $|\|q\|_T - \|q'\|_T| \leq C|q - q'|$.

Extension. The seminorm therefore extends uniquely to a continuous seminorm N on $\mathbb{R}^2 = \mathbb{C}$, $\mathbb{Q}(i)$ being dense, and $N(\zeta v) = N(v)$ holds on $\mathbb{Q}(i)$, hence on $\mathbb{C}$ by continuity.

Rotation invariance. For fixed v , $u \mapsto N(uv)$ is continuous on the unit circle and constant on $\zeta^{\mathbb{Z}}$, which is dense because ζ is not a root of unity. Hence $N(uv) = N(v)$ for all $|u| = 1$, and $N(v) = \kappa_T |v|$ with $\kappa_T = N(1)$.

Positivity. Each generator moves the origin by at most $D = \max_i |R_i(0)|$, so $n|v| = |\tau_v^n(0)| \leq D \ell_T(\tau_v^n)$ and $\kappa_T \geq 1/D > 0$. $\square$

Remark 51 (Bounds for κ_T). (a) *Lower bound.* By the composition law (2) the translation part β_w of $\rho_T(w)$ satisfies $|\beta_w| \leq |\zeta - 1| \cdot \#\{R_h \text{ letters in } w\}$, since only R_h contributes, by $1 - \zeta^{-1}$, and $|P(\zeta^{-1})| = |P(\zeta)|$ for real P . A geodesic word contains no factor $R_h R_h$, so a geodesic word of length L has at most $(L + 1)/2$ letters R_h . Applied to τ_v^n this gives $\kappa_T \geq 2/|\zeta_T - 1|$.

(b) *Upper bound.* In the lamp model of Theorem 60, let $\theta = \arg \zeta_T$, $s_j = \text{sgn} \cos(j\theta) \in \{\pm 1\}$ (with $\text{sgn } 0 = 1$), and let w_N be the word that walks from site 0 to site N and back, crossing

each edge $j < N$ twice, and uses the free choice of R_x or R_y before each move to make both deposits on edge j equal to s_j . It has length $4N + 2$ (one letter per move, one toggle before every move except the first return move, which instead needs the sign flip $R_x R_y$, and one final toggle), and it is the translation with edge profile $A_N = 2 \sum_{j < N} s_j t^j$, whose vector has modulus $|\zeta - 1| |A_N(\zeta)| \geq 2|\zeta - 1| \sum_{j < N} |\cos j\theta|$. By Weyl equidistribution the last sum is $(2/\pi + o(1))N$, and $\kappa_T |v_N| = \|v_N\|_T \leq \ell_T(\tau_{v_N}) \leq 4N + 2$ gives

$$\frac{2}{|\zeta_T - 1|} \leq \kappa_T \leq \frac{\pi}{|\zeta_T - 1|}.$$

For $(1, 2)$, $\zeta = (-3 + 4i)/5$, $|\zeta - 1| = 4/\sqrt{5}$ and $1.118 \leq \kappa_{(1,2)} \leq 1.757$. Like κ_T itself, these numbers refer to the normalization with x -leg 1. Exchanging the legs rescales them.

(c) *The shape* $(c, e) = (2, 1)$. Here $|\zeta - 1|^2 = 3/2$ and $|\zeta + 1|^2 = 5/2$. The out-and-back word with deposits $(-1, -1, 1, 1)$ has edge profile $2(t-1)(t+1)^2$, length 18, and is the translation by a vector of modulus $|\zeta - 1| \cdot 2|(\zeta - 1)(\zeta + 1)^2| = 7.5$. Repeating it gives $\kappa_{(2,1)} \leq 18/7.5 = 12/5$, which improves the bound $\pi\sqrt{2/3} = 2.5651\dots$ of (b). (In units of the lamp value $|A(\zeta)|$ rather than of $|v|$, the same bound reads $9/(|\zeta - 1||\zeta + 1|^2) = 6\sqrt{6}/5$.) Among out-and-back words with deposits in $\{0, \pm 1\}$ and at most 14 sites this block is optimal, and a dual certificate checked up to word length 1200 suggests that $12/5$ is the exact value. That is not proved. So $\kappa_{(2,1)} \in [2\sqrt{2/3}, 12/5] = [1.6329\dots, 2.4]$, and the exact value of κ_T is open for every shape. Whether κ_T is irrational is not known, and nothing below uses it.

8.3 Low-complexity exclusion and complexity blow-up

Write $F_T(x) = \sum_{d \geq 0} u_d^T x^d$ for the growth series of W_T . For $d \geq 0$, a *relation of complexity* $\leq d$ for a power series $F = \sum f_n x^n \in \mathbb{Z}[[x]]$ is one of:

- (A) *algebraic*: polynomials $P_0, \dots, P_k$ with $k \leq d$ and $\deg P_i \leq d$, not all zero, with $\sum_i P_i(x) F^i = 0$ (the case $k = 1$ is *rational*, $F = -P_0/P_1$);
- (B) *P-recursive*: polynomials $Q_0, \dots, Q_r$ with $r \leq d$ and $\deg Q_i \leq d$, not all zero, such that $\sum_{i=0}^r Q_i(n) f_{n-i} = 0$ holds for all $n \geq r$.

Requiring the recurrence for all $n \geq r$, and not merely for all large n , is what makes the complexity finite-dimensional: a recurrence valid only from n_0 on becomes one valid from r on after multiplication by $\prod_{m < n_0} (n - m)$, at the price of degree.

Theorem 52 (Low-complexity exclusion; computer-assisted). *For every right triangle T with positive unequal rational legs:*

- (i) F_T is not a rational function P/Q with $\deg P \leq 15$ and $\deg Q \leq 15$;
- (ii) F_T satisfies no relation $\sum_{i=0}^k P_i F_T^i = 0$ as in (A) with $(k, \max_i \deg P_i)$ at most $(1, 14)$, $(2, 9)$, $(3, 6)$, $(4, 5)$, $(5, 4)$ or $(6, 3)$ componentwise;
- (iii) (u_d^T) satisfies no recurrence as in (B) with $(r, \max_i \deg Q_i)$ at most $(1, 14)$, $(2, 8)$, $(3, 6)$, $(4, 4)$, $(5, 3)$, $(6, 2)$, $(7, 2)$, $(8, 1)$, $(9, 1)$, $(10, 0)$ or $(11, 0)$ componentwise.

Proof. By Theorem 20, $u_d^T = u_d$ for $0 \leq d \leq 32$, for every such T . A relation in any of the listed boxes, holding for F_T , holds modulo x^{33} (for (i), (ii)) or for $r \leq n \leq 32$ (for (iii)), and so gives a nonzero kernel vector of an explicit integer matrix built from $u_0, \dots, u_{32}$; for (i) the matrix is the 17×16 Hankel block $(u_{n-j})_{16 \leq n \leq 32, 0 \leq j \leq 15}$, whose kernel would contain the denominator. Each of these matrices is over-determined and has full column rank modulo the prime $2^{61} - 1$, hence over $\mathbb{Q}$. Padding with zero coefficients shows that each box excludes all smaller ones. $\square$

The proof has three trust bases, which we keep separate as in Appendix B. The values $u_0, \dots, u_{22}$ and their T -independence are the Lean statement `universal_layers_through_22`, discharged by `native_decide` (tier (ii): the Lean compiler, not the kernel alone). The values $u_{23}, \dots, u_{32}$ and their T -independence rest on the exact Rust breadth-first search and the modular-plus-exact certificate of Trust base (b) of §5 (tier (iii)). The rank computations are the script `code/zeta_probe/wt_growth/guess.py`, a modular elimination that is exact in the direction used (full rank modulo a prime implies full rank over $\mathbb{Q}$); they are not formalized. Proposition 98 is the corresponding statement for the universal sequence on 43 terms; Theorem 52 uses only the 33 shape-independent terms, and so holds for all T at once.

Remark 53 (Calibration: the exclusion is a theorem, not evidence). Theorem 52 should not be read as evidence that F_T is irrational. The Baumslag–Solitar group $BS(1, 2) = \mathbb{Z}[1/2] \rtimes_2 \mathbb{Z}$, solvable and of exponential growth, with an abelian normal subgroup that is not finitely generated and has cyclic quotient (the features it shares with H_T), has a rational growth series with respect to its standard generators [16]. From 29 computed terms (`code/zeta_probe/zeta_toy_growth/`) it is

$$\frac{(1-z)^2(1+z)^2(1+3z+8z^2+12z^3+16z^4+20z^5+22z^6+16z^7+14z^8+12z^9+4z^{10})}{(1-z-2z^3)(1-z^2-2z^5)^2},$$

of type (14, 13), which reproduces all 29 terms. That type lies inside the box of Theorem 52(i). It becomes over-determined only at 29 terms, and a diagonal search of type (d, d) on 29 terms cannot reach it. A rational growth series of moderate type is therefore invisible to a finite-horizon search until the horizon passes its type, and a negative search says nothing about types beyond its box.

Theorem 54 (Complexity blow-up). *Assume Theorem `thm:nonDfinite` of [12], namely that the universal series $U(x) = \sum_d u_d x^d$ is not D -finite. Then for every d there is $N(d)$ such that no T with $n_T \geq N(d)$, in particular no T with $c_T \geq N(d)$, has a growth series F_T admitting a relation of complexity $\leq d$. Consequently, for each d only finitely many similarity classes of rational right triangles T have W_T with rational, algebraic or P -recursive growth series of complexity $\leq d$.*

Proof. Fix d . Let V^A be the space of tuples $(P_0, \dots, P_d)$ of polynomials of degree $\leq d$ and V^B the space of tuples $(Q_0, \dots, Q_d)$ of polynomials of degree $\leq d$; both are finite-dimensional over $\mathbb{Q}$. A relation of type (A) with $k < d$, or of type (B) with $r < d$, becomes one with $k = d$, or $r = d$, by padding with zeros (a recurrence valid for $n \geq r$ is valid for $n \geq d$). For $N \geq 1$ let

$$K_N^A = \left\{ (P_i) \in V^A : \sum_i P_i U^i \equiv 0 \pmod{x^N} \right\},$$

$$K_N^B = \left\{ (Q_i) \in V^B : \sum_i Q_i(n) u_{n-i} = 0 \text{ for } d \leq n < N \right\}.$$

Both are linear subspaces, and $K_1^* \supseteq K_2^* \supseteq \dots$ is a nested sequence of kernels in a finite-dimensional space. It is therefore constant from some N_* on, equal to $K_\infty^* = \bigcap_N K_N^*$, the space of relations satisfied by the whole series U . A nonzero element of K_∞^A would make U algebraic, hence D -finite [38, Thm. 6.4.6]; a nonzero element of K_∞^B would make (u_n) P -recursive, hence U D -finite [38, §6.4]. Both are excluded by the hypothesis, so $K_\infty^A = K_\infty^B = 0$ and there is $N(d)$ with $K_{N(d)}^A = K_{N(d)}^B = 0$.

Now let $n_T \geq N(d)$, so that $u_n^T = u_n$ for $n < N(d)$. The conditions defining $K_{N(d)}^*$ involve only the first $N(d)$ coefficients: the coefficient of x^n in $\sum_i P_i F^i$ depends only on $f_0, \dots, f_n$, and the recurrence at n only on $f_{n-d}, \dots, f_n$. A relation of complexity $\leq d$ for F_T would therefore give a nonzero element of $K_{N(d)}^A$ or $K_{N(d)}^B$, which is impossible. By Corollary 30, $n_T \geq c_T$, so only shapes with $c_T < N(d)$ remain. There are finitely many Gaussian integers of norm $< N(d)$, hence finitely many rotation numbers ζ_T , and ζ_T determines the leg ratio and hence the similarity class. $\square$

The proof is ineffective: it gives no value of $N(d)$. Theorem 52 computes the certified part of it, showing $N \leq 33$ for the boxes listed there. The hypothesis is exactly Theorem `thm:nonDfinite` of [12], with the standing that that paper gives it. The analogue of Theorem 54 fails for the toy groups $\mathbb{Z}[\zeta^{\pm 1}] \rtimes_{\zeta} \mathbb{Z}$ with generators $\{1, \text{shift}\}$ (Remark 58): as $c \rightarrow \infty$ their growth series converge coefficientwise to that of $\mathbb{Z} \wr \mathbb{Z}$, which is rational [34], so the nested kernels there stabilize at a nonzero space.

8.4 Natural boundaries of translation-length series

For $g \in W_T$ let $L_g(z) = \sum_{n \geq 0} \ell_T(g^n) z^n$ be its translation-length series.

Theorem 55 (Theorem R, existential form). *There is a translation $g \in W_T$ whose stable norm $\lim_n \ell_T(g^n)/n$ is irrational. For every such g the series L_g has $|z| = 1$ as its natural boundary.*

Proof. Existence. Fix $0 \neq u \in \Lambda_T$, and let $\sigma = \pm 1$ be the sign of $\text{Re } \zeta = \sigma e/(2c)$. Since Λ_T is an A_T -module, $u(\sigma m + \zeta) \in \Lambda_T$ for $m \in \mathbb{Z}$, and by Proposition 50 $\|u(\sigma m + \zeta)\|_T / \|u\|_T = |\sigma m + \zeta| = \sqrt{m^2 + me/c + 1}$. For large m the integer $c^2(m^2 + me/c + 1) = c^2 m^2 + cem + c^2$ lies strictly between two consecutive squares. If e is even these are $(cm + e/2)^2$ and $(cm + e/2 + 1)^2$, because $0 < c^2 - e^2/4$ and $c^2 - e^2/4 < 2cm + e + 1$. If e is odd they are $(cm + (e-1)/2)^2$ and $(cm + (e+1)/2)^2$, because $cm + c^2 - (e-1)^2/4 > 0 > c^2 - (e+1)^2/4 - cm$. Both inequalities use $0 < e < 2c$. So $|\sigma m + \zeta|$ is irrational for large m , and $\|u\|_T$ and $\|u(\sigma m + \zeta)\|_T$ are not both rational.

Natural boundary. Put $\Delta_0 = 0$ and $\Delta_n = \ell_T(g^n) - \ell_T(g^{n-1})$ for $n \geq 1$. Then $|\Delta_n| \leq \ell_T(g)$, so $D_g(z) = (1 - z)L_g(z) = \sum_n \Delta_n z^n$ has only finitely many distinct coefficients. By Szegő's theorem [39] such a series either has eventually periodic coefficients, and then is rational, or has $|z| = 1$ as a natural boundary. Eventually periodic Δ_n would make $\ell_T(g^n)/n$ converge to the rational mean of a period, which is excluded. So $|z| = 1$ is a natural boundary of D_g , and hence of $L_g = D_g/(1 - z)$. $\square$

The existential form is what is proved. The argument does not identify a specific g with irrational stable norm, because it does not decide whether κ_T itself is irrational (Remark 51).

Remark 56 (Theorem R says nothing about growth). Theorem 55 concerns the lengths of the powers of one element, that is, the geometry of one cyclic subgroup; the growth series counts spheres, and neither property implies the other. It is the calibration of Remark 53 in another form. $\text{BS}(1, 2)$ shares with W_T the structural features that make length sequences of individual elements delicate, namely a distorted abelian normal subgroup that is not finitely generated, and still has a rational growth series [16]. The proof of Theorem 55 uses no information about the u_d^T , and it gives none.

8.5 The growth series: conjecture and obstruction

Conjecture 57. *For every right triangle T with positive unequal rational legs, the growth series F_T of W_T is not D -finite. The same holds for the toy groups $\mathbb{Z}[\zeta_T^{\pm 1}] \rtimes_{\zeta_T} \mathbb{Z}$ with generators $\{1, \text{shift}\}$.*

The closest analogue in the literature is the growth of the Baumslag–Solitar groups $\text{BS}(p, q)$. Freden, Knudson and Schofield [22] computed the growth of the horocyclic subgroup and proved it rational when $p \mid q$, and conjectured that it is not rational in the first case with $p \nmid q$, $\text{BS}(2, 3)$. That question is, to our knowledge, still open. The arithmetic behind the case $p \nmid q$ is numeration in the rational base q/p , whose integer expansions form a non-regular language [5]. Conjecture 57 is the unit-circle counterpart of that situation, and it is at least as hard.

The obstruction is concrete. Two edge profiles represent the same element of W_T exactly when they differ by $2\mu_T f$, $f \in \mathbb{Z}[t^{\pm 1}]$ (Corollary 30), and f is a *carry*. Dividing a profile A by $2\mu_T$ from its low end produces, after J steps, a remainder that is the prefix value $\sum_{j < J} a_j \zeta^j$ seen in the frame

rotated by ζ^{-J} : this is numeration in the base ζ_T , with $|\zeta_T| = 1$. In a rational base $q/p > 1$ the base contracts the remainder, and carries stay bounded. Here nothing contracts it, and the carry state performs a walk in the plane. The computations for the laboratory shape $(c, e) = (2, 1)$ bear this out. No bounded carry window suffices: the width of carry needed to realize the W_T -geodesic length of elements of the universal ball of radius d is $\lfloor (d - 12)/2 \rfloor$ for all $d \leq 32$, so it grows linearly (`code/zeta_probe/wt_growth/`, binary `wt_carry`; verified, not proved). Carries of typical elements grow like the square root of the length and cover the whole profile. The mechanism that gives non- D -finiteness for the universal series in [12], infinitely many poles of the travel block inside the unit disc, does not appear to transfer. Pure-travel elements remain distinct in W_T . However, the magnitude sector of the transfer that produces the travel poles is collapsed by the carries. We have not proved this, and state it only as a heuristic. A proof of Conjecture 57 would seem to need a transfer in which the planar carry is the catalytic variable, and no such equation is available.

8.6 Numerical record

Remark 58 (Computations for W_T and the toy groups). All runs are in Rust with exact integer arithmetic (`code/zeta_probe/wt_growth/` for W_T and `code/zeta_probe/zeta_toy_growth/` for the toy groups). The W_T breadth-first search stores the linear part exactly and the translation part modulo a prime $p < 2^{58}$. Reduction modulo p is a homomorphism, so it can identify distinct elements but never separate equal ones. Its ball counts $\sum_{d' \leq d} u_{d'}$ are therefore *lower bounds* for those of W_T , and the layer values below coincide with the true u_d^T unless a spurious modular coincidence occurred.

(a) *The (1, 2) triangle beyond depth 33.* Modular layer counts and deficits $u_d - u_d^{(1,2)}$ against A396406, identical at two independent primes:

d	33	34	35	36	37
$u_d^{(1,2)}$	3848354	5777558	8679159	13030452	19573036
$u_d - u_d^{(1,2)}$	30	93	282	754	1623

d	38	39	40	41
$u_d^{(1,2)}$	29335688	43991466	65921391	98829842
$u_d - u_d^{(1,2)}$	3093	5922	11070	19749

Only the first column is certified (Theorem 20). The others are lower bounds in the sense above, and correspondingly the cumulative deficits are upper bounds.

(b) *The (1, 3) triangle.* Here $(c_T, e_T) = (5, 8)$. The modular ball counts equal the universal ones through depth 38, and they are lower bounds for the true counts, which are at most the universal ones. Given the published terms $u_0, \dots, u_{38}$ of A396406, this proves $n_{(1,3)} \geq 39$. At depth 39 the run shows deficit 51, and 283 at depth 40, so $n_{(1,3)} = 39$, the value of Proposition 36, is verified. We record the equality as verified and not as proved, because the depth-39 deficit rests on a single prime; the matching upper bound would come from the upper half of Proposition 36. The shapes (1, 3) and (3, 1) have the same μ_T , hence the same kernel. The modular collision at depth ≤ 38 flagged for (3, 1) in Remark 32 was therefore a modular accident.

(c) *The (2, 7) triangle as an independent check.* Here $c_T = 53$, so $n_{(2,7)} \geq 53$ by Corollary 30, and $u_d^{(2,7)} = u_d$ for $d \leq 52$. A modular run of $W_{(2,7)}$ to depth 40 therefore computes lower bounds for the universal counts. It returns exactly the published values through u_{38} , and $u_{39} = 43,997,388$, $u_{40} = 65,932,461$. This is consistent with the two terms of Appendix A that had been recovered from logs and not regenerated. It is a lower-bound check, not a certificate.

(d) *Toy groups.* The groups $\mathbb{Z}[\zeta^{\pm 1}] \rtimes_{\zeta} \mathbb{Z}$, with generators the translation by 1 and the shift, for $(c, e) = (5, 6)$ (the rotation number of $(1, 2)$) and $(2, 1)$. Their spheres, computed exactly, begin as follows; neither sequence is in the OEIS.

(5, 6), $u_0, \dots, u_{21}$: 1, 4, 12, 36, 100, 268, 704, 1812, 4600, 11496, 28270, 68880, 166884,
402176, 964060, 2301080, 5475086, 12993680, 30771148, 72742634, 171713114, 404852468;
(2, 1), $u_0, \dots, u_{28}$: 1, 4, 12, 36, 100, 236, 528, 1112, 2286, 4554, 8876, 17036, 32140,
60042, 111176, 204298, 372540, 676420, 1223432, 2205098, 3963816, 7106970,
12717010, 22720810, 40530550, 72187900, 128440824, 228312488, 405422650.

They agree with the growth series of $\mathbb{Z} \wr \mathbb{Z}$ in the same generators through u_8 , respectively u_4 , and fall below it from u_9 , respectively u_5 , on. The same code gives the BS(1, 2) terms of Remark 53. On all these sequences, and on the W_T sequences of (a)–(c), the exclusion searches of `guess.py` find no rational, algebraic or P-recursive fit in any of the boxes tested. For (a)–(c) these searches run on modular values and so certify nothing about W_T .

9 Normal form, lamp model, and the word-length metric theorem

This section completes the structural description of W_{gen} , the reflection group of an unequal-leg right triangle generated by the three side-reflections R_x, R_y, R_h . We identify the elements with their normal-form tuples and exhibit the generator action as an edge lamplighter on $\mathbb{Z}$, thereby solving the word and membership problems. We record the strand bound that confines the single-trail interface to a fixed eight-letter alphabet, and prove the word-length metric theorem $\ell_T = \ell_R + 2c$ together with the closed form for the connectivity defect c . The upper half of the metric theorem is unconditional. The lower half was open because no argument identified the closed form c_{pred} with the true defect c (Conjecture 80); the finite-state/Moore argument attempted for that identification does not work (Remark 77), but the identification is proved by a different route, an induction on a corrected potential (Remark 78). The metric theorem as a whole still reads ℓ_R as the closed form, and in that reading it is proved for every element (`metricAll`, Remark 63); that the closed form is also the relaxed minimum over realizations is verified, not proved.

9.1 Notation

The three side-reflections are denoted R_x, R_y, R_h , where R_x and R_y are the reflections in the two legs meeting at the right angle and R_h is the reflection in the hypotenuse. In the notation R_0, R_1, R_2 used elsewhere in the paper the dictionary is

$$R_0 = R_x \text{ (first leg),} \quad R_1 = R_y \text{ (second leg),} \quad R_2 = R_h \text{ (hypotenuse).}$$

We fix the coordinate normalization with the right angle at the origin, the two legs along the positive coordinate axes, and the leg length rescaled to 1; this is the normalization compatible with all coordinate placements used in the paper.

Throughout, an element is encoded by a tuple $(\varepsilon, \delta, k, P)$: $\varepsilon \in \{\pm 1\}$ is the lamp sign, $\delta \in \{0, 1\}$ the walker state, $k \in \mathbb{Z}$ the net translation, and $P \in \mathbb{Z}[t^{\pm 1}]$ the translation (deposit) polynomial. In the *edge basis* we write $P = \sum_j a_j (t^{j+1} - t^j)$, so that $a_j \in \mathbb{Z}$ is the deposit on the edge between sites j and $j + 1$. The *spine* is the travel interval $I_k = [\min(0, k), \max(0, k))$, on whose edges $f_j = \pm 1$ (and $f_j = 0$ off the spine); an *off-spine block* is a maximal run of off-spine edges with $a_j \neq 0$, and a *gap edge* is an off-spine edge inside the active span with $a_j = 0$. A maximal run of consecutive gap edges is a *gap-run*.

The translation lattice $\mathcal{T} \subset \mathbb{Z}[t^{\pm 1}]$ is the lattice of realized translation polynomials with linear part 0; its identification is the content of Theorem 21.

9.2 The translation lattice

Theorem 59 (Translation lattice).

$$\mathcal{T} = 2(t-1)\mathbb{Z}[t^{\pm 1}].$$

Proof. The inclusion $2(t-1)\mathbb{Z}[t^{\pm 1}] \subseteq \mathcal{T}$ is the glide-square generation (each generator $2(t^{j+1} - t^j)$ of the right-hand lattice is realized by a glide-square relator). For the reverse inclusion, a translation polynomial $P \in \mathcal{T}$ has zero linear part, hence $P(1) = 0$, so $P \in (t-1)\mathbb{Z}[t^{\pm 1}]$, and the dihedral cocycle constraint forces $P \equiv 0 \pmod{2}$, so $P \in 2\mathbb{Z}[t^{\pm 1}]$. Thus

$$\mathcal{T} \subseteq 2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}].$$

Finally $2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}] = 2(t-1)\mathbb{Z}[t^{\pm 1}]$: if $P = 2Q = (t-1)S$ then reducing $2Q = (t-1)S$ modulo 2 gives $(t-1)S \equiv 0 \pmod{2}$, and since $\mathbb{Z}[t^{\pm 1}]/(t-1) \cong \mathbb{Z}$ is torsion-free the quotient by $(t-1)$ has no 2-torsion, so $S \equiv 0 \pmod{2}$, i.e. $S = 2S'$ and $P = 2(t-1)S'$. Both inclusions give the equality. $\square$

9.3 Normal form and the edge-lamplighter action

Theorem 60 (Normal form and lamp model). *An abstract tuple $(\varepsilon, \delta, k, P)$ lies in W_{gen} if and only if*

$$P(1) = 0 \quad \text{and} \quad P \equiv (1+t)(1+t+\dots+t^{|k|-1})t^{\min(k,0)} \pmod{2}.$$

The mod-2 class on the right is the dihedral cocycle of the translation lattice $\mathcal{T}$ and depends only on k . In particular the word and membership problems for W_{gen} are solved explicitly. Moreover, right multiplication by the generators acts as an edge lamplighter on $\mathbb{Z}$:

$$R_x: \delta \mapsto 1 - \delta, \quad R_y: (\varepsilon, \delta) \mapsto (-\varepsilon, 1 - \delta),$$

while R_h moves the walker one site (down if $\delta = 0$, up if $\delta = 1$), deposits $+\varepsilon$ respectively $-\varepsilon$ on the traversed edge, and flips δ . Consecutive moves therefore bounce by default; straight travel and deposit-sign changes are paid for in R_x, R_y letters.

Proof. Necessity of the two congruences is the symbolic normal form together with the translation-lattice description: the mod-2 image of an element is determined by its image in the dihedral quotient, hence by k , the representative r^k having $P_{r^k} = (1-t)(1+\dots+t^{k-1})$, and $P(1) = 0$ is the abelianized deposit-balance constraint. For sufficiency, the realized translation polynomials with a fixed linear part k form a single coset of $\mathcal{T}$, and the two congruences cut out exactly one such coset: a difference satisfying both lies in

$$2\mathbb{Z}[t^{\pm 1}] \cap (t-1)\mathbb{Z}[t^{\pm 1}] = 2(t-1)\mathbb{Z}[t^{\pm 1}] = \mathcal{T},$$

the equality being Theorem 21. The lamp dynamics is the composition law read in the edge basis. $\square$

9.4 The strand bound

The metric problem reduces to realizing an element as a single Eulerian trail on the crossing multigraph (vertices the integer sites; m_j parallel edges across the gap at site j). The determinization of the resulting optimization hinges on a bounded local encoding of the no-isolated-cycle constraint. This bound is *not* a bound on the number of simultaneously open interface components: the family $g_n = (\varepsilon, \delta, k) = (1, 0, 0)$ with single deposit $a_0 = 2n$ has geodesic word $(R_y R_h R_x)^{2n}$ of length $6n$ and forces $n + 1$ open components crossing edge 0. The correct bound is on component *shapes*.

Lemma 61 (Strand bound). *In any realization of an element as a single Eulerian trail on the crossing multigraph, every interface component at every edge j carries at most one up-crossing and at most one down-crossing of edge j . Consequently each component is one of finitely many shapes, and an interface profile is a multiset over that fixed finite alphabet whose only unbounded datum is the multiplicity of the through-shape.*

Proof. An interface component at edge j is the footprint, on the cut between sites $\leq j$ and sites $> j$, of a maximal sub-walk of the trail lying in the half-line $\{\text{site} \leq j\}$. Such a sub-walk enters the half-line by crossing the gap at j downward and leaves by crossing it upward. A *second* upward crossing is impossible: after crossing up, the walk sits at site $j + 1 > j$, has already left the half-line, and the sub-walk ends. Hence each component has exactly one up- and one down-crossing of edge j , a marker sub-walk beginning or ending inside the half-line contributing one. The bound is thus an invariant of single-trail admissibility, independent of length-minimization. The shapes are the crossing contents $(1, 0)$, $(0, 1)$, $(1, 1)$ refined by the two lamp signs and by the presence of the start/end marker, so the alphabet has at most $3 \cdot 2 \cdot 2 = 12$ letters; an earlier version said eight, which does not match this refinement. Nothing downstream depends on the exact number: what is used is that the alphabet is finite and that exactly one letter, the through-shape $(1, 1)$, may occur with unbounded multiplicity. $\square$

9.5 The metric theorem $\ell_T = \ell_R + 2c$

We work with two metrics on W_{gen} . The *relaxed* length $\ell_R(g) = \min_R \ell(R)$ is taken over all realizations R , isolated cycles permitted and costing nothing beyond their own crossings and turns; it has the local closed form of the crossing-pairing optimization below. The *true* length $\ell_T(g) = \min_R \ell(R)$ is taken over realizations whose transition system is a single open walk, with no isolated cycle. The word length of g in the generating set equals $\ell_T(g)$: each generator is one step, a crossing or a state turn.

Proposition 62 (Metric formula, as verified). *The word length of $g = (\varepsilon^*, \delta^*, k^*, a)$ with respect to the three reflections is the value of the following explicit finite optimization. Choose crossing numbers $m_j \geq |a_j|$ with $m_j \equiv a_j \pmod{2}$, chained so that the support is reachable from the start site, meaning that the edges j with $m_j > 0$ form a connected subgraph of $\mathbb{Z}$ containing the start site, equivalently an interval containing 0; the walk directions force $u_j = (m_j + f_j)/2$ up-crossings, with $f_j = \pm 1$ on the travel interval between 0 and k^* and $f_j = 0$ otherwise. At minimal crossing numbers each crossing's sign is forced ($\varepsilon = -d \cdot \text{sign}(a_j)$ for direction d). Each site pairs arrivals with departures: a pass (same direction) costs 1, a bounce 0, and a sign change is free at a pass but costs 2 at a bounce, with virtual boundary moves (arrival at the origin: up, +; departure at k^* : direction δ^* , sign ε^*). The pairing must be Eulerian-connected: an isolated cycle is inadmissible, and splicing one in costs exactly 2. Then*

$$\ell(g) = \min \left(\sum_j m_j + \sum_{\text{sites}} \text{pairing costs} \right).$$

Remark 63 (Status of the metric formula). We do not prove this. It is the optimization model used throughout the section, and it reproduces the exact word length, computed by breadth-first search, on every element of the radius-14 ball, 3,723 elements, with 0 exceptions. That is the whole of its support. It was stated as a theorem in an earlier version; three later results cite it, and each of them therefore inherits the same status. The reading of “chained so that the support is reachable” given above is the one implemented by the certificate.

Corollary `cor:1Rclosed` of [12], restated there from this proposition, is proved (`Realisation.lean`, theorem `1R_closed`: an `IsLeast` statement over the space of ‘Realisation’s) – but that is a fact internal to the combinatorial optimization model itself, not a proof that the model computes the true relaxed word length of the group. [12]’s own `rem:pairingstatus` states this precisely: “[the optimisation] computes the relaxed word length is verified there and is not proved; everything in this subsection is a statement about the optimisation itself.” An earlier version of this remark conflated the two and claimed this proposition was thereby proved; that was an overclaim, corrected here.

The word-length formula itself is now proved: the Lean theorem `metricAll` (`lean/with_mathlib/RJMetricAll.lean`, composing `wordLength_eq_1RTrue_add_two_cTrue` of `PhiLipschitz.lean` with reachability of every element) gives $\text{wordLength}(g) = 1\text{RTrue}(g) + 2 \cdot \text{cTrue}(g)$ for every $g \in W_{\text{gen}}$, with `1RTrue`, `cTrue` the site-local closed forms of this section (Remark 78); it is Theorem `thm:model` of [12]. What remains verified and not proved is only the relaxed-minimum reading: that the closed form `1RTrue` equals ℓ_R , the minimum over relaxed realizations.

The two metrics differ only by the connectivity constraint, which we measure by the connectivity defect

$$c(g) := \min \{ \# \{ \text{isolated cycles in } R \} : R \text{ relaxed-optimal} \} \in \mathbb{Z}_{\geq 0}.$$

Theorem 64 (Metric theorem). *For every $g \in W_{\text{gen}}$,*

$$\ell_T(g) = \ell_R(g) + 2 c_{\text{pred}}(g),$$

with $c_{\text{pred}}(g)$ the closed form of Proposition 72. The upper bound is the splice of Lemma 66. The identification $c_{\text{pred}} = c$ needed for the reverse inequality, stated in an earlier version as Conjecture 80 and verified there with 0 exceptions on the depth-24 ball, is proved in `lean/with_mathlib/PhiLipschitz.lean` (`wordLength_eq_1RTrue_add_two_cTrue`), by an induction on a corrected potential rather than by the finite-state route attempted in an earlier version. See Remark 77 for why that earlier route fails, and Remark 78 for the route that succeeds, including exactly what it does and does not establish. The theorem as displayed additionally rests, as it always has, on Proposition 62 identifying ℓ_R itself with the closed-form optimization. With ℓ_R read as the closed form, the theorem is proved for every element (`metricAll`, Remark 63); that the closed form is also the relaxed minimum over realizations is verified, not proved.

Remark 65 (Relation to the free-solvable literature). The *shape* of Theorem 64 (an ℓ^1 -type crossing count plus twice a connectivity correction) is the known one for the groups F/N' , with F free and $N \triangleleft F$ arbitrary: there the word length of an element is the ℓ^1 -norm of its induced flow on the Cayley graph of F/N plus twice the number of edges in a minimal connecting forest [20, Thm. 2], reproved as [29, Thm. 2.10] and restated as [36, Lem. 3.2]; the ambient group is the image of the Magnus embedding [25], whose metric behaviour is studied in [36, 41], and presenting such a group as an extension of F/N by the first integer homology of its Cayley graph, with a distinguished cocycle, goes back to [42]. By Corollary 42 the group W_{gen} is the Q -relative metabelianization of $V_4 * \mathbb{Z}/2$, so it belongs to that family up to the bookkeeping required because our generators are involutions, and the shape of the formula should be expected rather than claimed.

What is new here is not the shape but the *evaluation*. In general the connectivity term is a minimal-forest optimization, and computing geodesic length in a free metabelian group is NP-complete [29, Thm. 4.1]. Here it is instead given in closed form by Proposition 72 and realized by the finite-state transducer of Definition 74, which is what makes the growth series computable and the enumeration of $u_0, \dots, u_{22}$ possible without breadth-first search.

The theorem is two inequalities. Part A establishes the upper bound by a splice construction and reduces the lower bound to a structural property of c ; Part B supplies that property by exhibiting c as the explicit finite-state transducer of Definition 74.

Part A: the splice and the reduction

Lemma 66 (Splice: $\ell_T \leq \ell_R + 2c$). *From a relaxed-optimal realization R^* with c isolated cycles one builds a single-open-walk realization of cost $\ell_R + 2c$.*

Proof. Order the isolated cycles $\gamma_1, \dots, \gamma_c$ of R^* from the open walk outward. Each γ_i is a closed strand-walk whose support is an interval; since the active span is connected through crossed edges (every span edge carries $m_j \geq \max(|a_j|, |f_j|) \geq 1$, and gap edges $m_j = 2$), γ_i shares at least one site s_i with a strand already linked to the open walk. At s_i the transition system matches an arrival/departure pair of γ_i within γ_i and a pair of the open component within itself. Re-matching the four ends so that each arrival of one component is sent to a departure of the other is a 2-swap of the local matching π_{s_i} . By Lemma 67 a merging 2-swap changes $\text{cost}(\pi_{s_i})$ by exactly +2 and merges γ_i into the open component, leaving all other sites and all crossing numbers m_j unchanged. After the c swaps the transition system is a single open walk of total cost $\ell(R^*) + 2c = \ell_R + 2c$. $\square$

Lemma 67 (A merging 2-swap costs exactly +2). *Let s be a site at which a closed component γ and the open component each contribute one pair to π_s . Replacing the two intra-component pairs by the two inter-component pairs (the unique re-matching that merges γ into the open walk) changes $\text{cost}(\pi_s)$ by +2, and this is the minimum possible change.*

Proof. A merging re-match links an arrival α of γ to a departure d' of the open walk and an arrival α' of the open walk to a departure d of γ . The crossing strands of a closed isolated cycle deposit net zero along γ , so the two ends of γ at s carry opposite signs on the same side: either a bounce ($\text{pc} = 0$) or a sign-flip bounce ($\text{pc} = 2$). Since γ contributes zero net deposit at the bridged gap edge, its matched pair is the cost-0 bounce. After the swap the bridged edge is traversed by the merged walk, which must keep the gap deposit at 0 while now passing through rather than bouncing: one new pair becomes a sign-flip bounce ($\text{pc} = 2$) and the other a bounce ($\text{pc} = 0$), a net change of +2. No re-match merging two components reduces the open-walk count by more than one per unit of cost, so +2 is minimal. $\square$

Lemma 68 (Rigidity: $\ell_T \geq \ell_R + 2c$). *Since the closed form $c(g)$ of Proposition 72 is a lower bound on the connectivity cost of every realization (Corollary 79), every single-open-walk realization has $\text{cost} \geq \ell_R + 2c$.*

Proof. Let R' be single-open-walk-optimal, $\ell(R') = \ell_T$. Inverse 2-swaps cut the open walk into an open walk plus closed walks, each cut lowering the cost by 2, terminating at a realization R'' that admits no cost-lowering cut, with $\ell(R'') = \ell_T - 2c'$ and c' isolated cycles. As R'' is a realization, $\ell_R \leq \ell(R'')$, whence $\ell_T \geq \ell_R + 2c'$. Since R'' is only a *local* relaxed optimum, not necessarily the global one, $c' < c$ is not excluded by realization-independence alone. The bound $c' \geq c$ follows from Corollary 79: $c(g)$ is a lower bound on the connectivity cost of every cut-irreducible realization,

since each unit of $c(g)$ is forced by a distinct gap edge that cannot be removed without violating a deposit or a reachability constraint. Hence $c' \geq c$ and $\ell_T \geq \ell_R + 2c$. $\square$

Remark 69 (Status of Part A). Lemma 66 gives the upper bound $\ell_T \leq \ell_R + 2c$ unconditionally. Lemma 68 reduces the reverse inequality to the lower-bound half of Part B, namely that the closed form below is a lower bound on the connectivity cost of every realization; that hypothesis is discharged by Corollary 79 in its unconditional form (Remark 78), so Lemma 68 is unconditional as well. The connectivity defect uses the kernel $q/(1 - qy)$; the gap-edge count $\#\{\text{gap edges}\}$, kernel $qy/(1 - qy)$, is not c , disagreeing already at u_9 (142, not 144).

Part B: the closed form for c

The off-spine block/gap structure of an element is described as follows. On each side of the spine I_k the off-spine blocks and the gap-runs alternate; off-spine blocks have even length by the parity constraint $m_j \equiv a_j \pmod{2}$. Every pure-travel element has $c = 0$, and every element with $c \geq 1$ carries an off-spine deposit, the travel run being the forced single backbone of the open walk. Consequently every isolated cycle in a relaxed-optimal realization lies in the off-spine region and is supported on a single side's block/gap structure, so c is a sum of independent left- and right-side contributions, coupled only through the single boundary term η below.

We exhibit c as an explicit *finite-state* reading of the edge profile and verify that reading on a window large enough to exercise every state transition. Two facts make this a covering verification: the bounded memory of Lemma 71 and the covering property of the verified window in Theorem 76.

Definition 70 (Closed-form vocabulary). The transducer of Lemma 71 reads the edge profile and tracks a bounded connectivity state with the following components.

- A *run* is a gap-run (maximal run of gap edges) on one side of the spine. A run is *interior* if it is flanked by off-spine blocks on both sides, and a *boundary run* if it is adjacent to the spine.
- The *shield* sh of a run is the number of gap edges of that run whose isolated cycle is suppressed by an adjacent same-sign bounce; a run of length L contributes $\max(0, L - \text{sh})$ to c . The shield of an interior run is 1; the shield of a boundary run is the indicator $\text{sh}_L, \text{sh}_R \in \{0, 1\}$ displayed in Proposition 72.
- A *marker* is the start/end of the open walk, contributing one crossing end at a site (Lemma 61). The state datum *pending* records whether a marker shield is still available to suppress a gap edge not yet read.
- A *detour* on a side is the presence of off-spine deposit on that side; “right detour” means off-spine deposit on the right of the spine.

Lemma 71 (the closed form is finite-state). *There is a deterministic transducer $\mathcal{A}$ with a finite state set Q , reading the edge profile $(f_j, a_j)_j$ from left to right, that outputs the closed form $c_{\text{pred}}(g)$ of Proposition 72. The defect $c(g)$ depends on each off-spine deposit only through its presence and, for deposits incident to the spine, through the bounded class $(\text{sgn } a_j, \mathbf{1}[|a_j| \geq 3])$; the magnitudes of non-incident off-spine deposits do not affect c .*

Proof. This is a statement about the closed form alone, read off from its definition; in particular it does not presuppose that the closed form equals $\frac{1}{2}(\ell_T - \ell_R)$, which is the open point of Remark 77. The closed form counts drops of 0-strand marker-free components. A drop is created or shielded by a *local* event (a gap edge entering or leaving a run, or the matching at a marker site) whose

decision needs only: the region (L off-spine | spine | R off-spine), whether the previous edge was a gap, whether a marker shield is still pending, and the bounded class of the spine-incident deposit. The deposit magnitude influences the matching only through whether the incident travel edge has multiplicity ≥ 3 (a thrice-crossed edge supplies one spare same-sign bounce; higher multiplicities supply no further free bounce relevant to a single adjacent run) and through the parity, which is forced. Hence the connectivity state lives in the finite product

$$\{L, S, R\} \times \{0, 1\}_{\text{prev-gap}} \times \{\text{pending}\} \times \{\text{sgn}, \mathbf{1}[|b| \geq 3]\},$$

off which $\mathcal{A}$ reads c . The length component of the transfer is the unbounded catalytic variable; it does not enter c . $\square$

Proposition 72 (The transducer is the displayed closed form). *$\mathcal{A}$ emits, per side of the spine,*

$$\sum_{\text{runs}} \max(0, \text{length} - s_{\text{run}}), \quad s_{\text{run}} = \begin{cases} 1, & \text{interior run,} \\ \text{sh}, & \text{boundary run,} \end{cases}$$

where, with b the spine-incident travel deposit (Definition 70),

$$\text{sh}_R = \begin{cases} \mathbf{1}[\varepsilon = -1 \text{ or } \delta = 1], & k = 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 1 \text{ or } \text{sgn } b = \varepsilon], & k > 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \text{sgn } b = -1], & k < 0, \end{cases} \quad \text{sh}_L = \begin{cases} 1, & k \geq 0, \\ \mathbf{1}[|b| \geq 3 \text{ or } \delta = 0 \text{ or } \varepsilon = \text{sgn } b], & k < 0, \end{cases}$$

plus the boundary coupling $\eta(g) \in \{-1, 0, 1\}$, supported on the single cell $\{k = 0, \delta = 0, \text{even deposit on edge } -1, \text{right detour, no edge-0 deposit}\}$ and equal there to -1 if $\varepsilon = 1$ and to $\mathbf{1}[a_{-1} = 2]$ if $\varepsilon = -1$. This expression defines the closed form $c_{\text{pred}}(g)$.

Remark 73 (What this proposition does not say). It is a statement about $\mathcal{A}$ alone: the transducer emits the displayed expression. It does *not* by itself assert $c_{\text{pred}} = c$, with c the true connectivity defect. The finite-state/Moore argument attempted from this proposition alone does not reach that identification (Remark 77); it is proved, via a different route not going through $\mathcal{A}$, as Theorem 80 (Remark 78).

Definition 74 (The explicit transducer $\mathcal{A}$). Read each side of the spine *outward* from its endpoint, one edge at a time, classifying each edge as a *block* edge (D : an off-travel deposit, $a_j \neq 0$ with $f_j = 0$) or a *gap* edge (G : $a_j = 0$), and halting at the outermost block edge of that side. The side automaton has four states (β, s) , where $\beta \in \{0, 1\}$ records whether a block has yet been seen and $s \in \{0, 1\}$ is the pending shield. It starts in $(0, \text{sh})$ with $\text{sh} = \text{sh}_R$ on the right and $\text{sh} = \text{sh}_L$ on the left (Proposition 72), and runs by

state	input G	input D
$(0, 0)$	$(0, 0)$, emit 1	$(1, 1)$, emit 0
$(0, 1)$	$(0, 0)$, emit 0	$(1, 1)$, emit 0
$(1, 0)$	$(1, 0)$, emit 1	$(1, 1)$, emit 0
$(1, 1)$	$(1, 0)$, emit 0	$(1, 1)$, emit 0

The output of $\mathcal{A}$ is the sum of the two side totals plus the boundary cell $\eta(g)$ of Proposition 72.

Remark 75 ($\mathcal{A}$ realizes the displayed closed form). By construction the side automaton emits, on a maximal gap-run of length L preceded by initial shield s , exactly $\max(0, L - s)$: the first $s \leq 1$ gap edges are shielded (emit 0) and the remaining edges emit 1, while each block edge resets the shield to 1 for the next interior run. Summing over runs reproduces $\sum_{\text{runs}} \max(0, \text{length} - s_{\text{run}})$ with $s_{\text{run}} = \text{sh}$ on the boundary run and 1 on interior runs; this is the side expression of Proposition 72. Hence the output of $\mathcal{A}$ is the closed form of that proposition for every profile.

Proposition 76 (What the explicit transducer certifies). *The output of the explicit transducer $\mathcal{A}$ of Definition 74 equals the closed form $c_{\text{pred}}(g)$ of Proposition 72 for every $g \in W_{\text{gen}}$. On the depth-24 enumeration it also equals the true defect $\frac{1}{2}(\ell_T(g) - \ell_R(g))$, with 0 exceptions.*

Proof. The first assertion is Lemma 71 together with the transition table of Definition 74, and is what the accompanying certificate `fsm_verify.py` checks. The second is the enumeration. The breadth-first enumeration to true length 24, comprising 275,823 elements, realizes every reachable transition: it contains, on each side and for each $k \in \{0, \pm 1, \pm 2\}$, gap-runs of every length $1 \leq L \leq 11$, interior runs adjacent to blocks of every parity class, spine-incident deposits of both signs and both magnitude classes $|b| \in \{1, \geq 3\}$, both values of δ and of ε , and the degenerate η -cell. On this window the closed form matches the exact breadth-first true distance with 0 exceptions over all 275,823 elements, and is held out at depths 19, 21, 23, 24. This enumeration is the output of `code/zeta_probe/c_formula.py`, run as `python3 c_formula.py 24`; it prints the element count and the mismatch count, and takes about eleven seconds. The closed form also matches the exact solver on elements outside the window (gap-runs of length up to 16, displacements $|k|$ up to 5, deposit magnitudes up to 6) none of which alters the defect, in agreement with Lemma 71. The window exercises each of the eight transitions of Definition 74 for both initial shield values and the boundary cell η ; that off-spine magnitudes beyond the bounded class do not affect the defect, the content of Lemma 71, is confirmed independently on 200,000 profiles with gap-runs and magnitudes far outside the window, with 0 exceptions. $\square$

Remark 77 (The lower bound was open, via this route). This remark records a route that failed and why, kept for the record rather than deleted; Remark 78 below records the route that succeeded. A previous version of this paper concluded from the above that $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$ identically, by the principle that “two finite-state functions of a profile agree on all profiles if and only if they agree on every state transition”. That principle is false as stated. The correct criterion is Moore’s: two machines with state sets of sizes n_1 and n_2 are equivalent if and only if they agree on all inputs of length $< n_1 + n_2$, and it requires both state sets. A set of profiles that drives $\mathcal{A}$ through each of its own transitions says nothing about the second machine, which may leave the diagonal on an input $\mathcal{A}$ never distinguishes.

The gap is not merely one of exposition. To run Moore’s criterion one needs the true defect $\frac{1}{2}(\ell_T - \ell_R)$ to be finite-state *with a known bound on its state count*, and that is not established here: the proof of Lemma 71 above is a statement about the closed form, and the earlier version’s proof of the corresponding statement for the true defect began by assuming $c = \frac{1}{2}(\ell_T - \ell_R)$, which is the conclusion. The certificate `fsm_verify.py` likewise certifies only that the transducer computes c_{pred} ; it does not, and does not claim to, certify $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$.

Accordingly the previous version of this paper stated the lower bound $\ell_T \geq \ell_R + 2c$ as a conjecture, verified with 0 exceptions on all 275,823 elements to depth 24, held out at depths 19, 21, 23, 24, and on samples outside that window. Remark 78 proves it.

Remark 78 (The lower bound is proved, by a different route). The identification $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$ that Remark 77 could not reach via a finite-state/Moore argument on the closed form is instead

reached without going through the closed form’s transducer at all. `lean/with_mathlib/PhiLipschitz.lean` proves, unconditionally and for every reachable g ,

$$\text{wordLength}(g) = \text{lRTrue}(g) + 2 \cdot \text{cTrue}(g)$$

(theorem `wordLength_eq_lRTrue_add_two_cTrue`), by an induction on a corrected potential $\Phi(g) = \text{lRTrue}(g) + 2 \cdot \text{cTrue}(g)$ shown to move by at most 1 along every generator, composed with a descent lemma (`exists_descent`, made unconditional in `EltBridge.lean`) rather than by identifying the true defect as finite-state. `wordLength` is the word metric of W_{gen} (`EltBridge.Elt.wordLength := inf{n : Reaches n g}`), so this is ℓ_T ; `lRTrue` and `cTrue` are the corrected span versions of the closed forms of Proposition 62 (the same optimization, restated in [12] as Corollary `cor:lRclosed`) and Proposition 72 (`CorrectedSpan.lean` repairs a bug in the uncorrected span that forced edge 0 into every span and made the identity false at the identity element; the corrected span matches the convention of `nogap/src/main.rs`, the tool the numerical certificates in this section already use). Independently re-verified: `#print axioms` on the theorem shows only `[propext, Classical.choice, Quot.sound]`, no `sorryAx` and no `native_decide`.

What this proves and what it does not: it proves the identity with `lRTrue`, `cTrue` exactly as those closed forms are displayed in Proposition 72 and this section’s optimization – which is the identification $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$ this remark’s title refers to, and the antecedent Corollary 79 needs. It does *not* independently prove Proposition 62: an earlier version of this remark and of Remark 63 claimed `cor:lRclosed`’s `IsLeast` proof (that the closed form equals the minimum over the `Realisation` type) upgraded that proposition to a theorem, which conflates a fact internal to the optimization model with the separate, still-open claim that the closed form is the relaxed minimum ℓ_R over realizations ([12]’s own `rem:pairingstatus` states this distinction directly). That was an overclaim, corrected in both remarks. This discharges Corollary 79 and, with Lemma 66, completes the metric theorem *as displayed, in terms of c_{pred} and the closed form for ℓ_R* – the same terms Theorem 64 was already stated in before this result.

Corollary 79 (Discharge of the lower bound). *The closed form $c_{\text{pred}}(g)$ of Proposition 72 is realization-independent, being a function of $(\varepsilon, \delta, k, a)$ alone, and $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$ identically (Remark 78). Hence it is a lower bound on the connectivity cost of every open-walk realization, which discharges the hypothesis of Lemma 68 and gives $\ell_T \geq \ell_R + 2c$; with Lemma 66 this completes Theorem 64.*

Theorem 80 (Lower bound for the metric). *For every $g \in W_{\text{gen}}$, $\ell_T(g) \geq \ell_R(g) + 2c_{\text{pred}}(g)$; equivalently $c_{\text{pred}}(g) = \frac{1}{2}(\ell_T(g) - \ell_R(g))$. Proved in Remark 78. (An earlier version of this paper stated this as a conjecture, verified with 0 exceptions on the 275,823 elements of the depth-24 ball, held out at depths 19, 21, 23, 24, and on samples with gap-runs to length 16, displacements $|k| \leq 5$ and deposit magnitudes to 6; the label `conj:lowerbound` is kept for stable cross-referencing.)*

9.6 Validation and verification windows

Remark 81 (Validation of the metric formula). The optimization of Theorem 62 agrees with the true word metric on every element of the radius-14 ball (3,723 elements, zero mismatches). Enumerating elements directly from the normal form of Theorem 60 and counting them by the lengths the formula assigns yields $u_0, \dots, u_{22}$ of A396406, with no breadth-first search in the enumeration pipeline. The lamp-model enumeration likewise yields $u_0, \dots, u_{14}$, with every ball element satisfying both congruences of Theorem 60. The relaxation without the connectivity constraint, ℓ_R , is exact through radius 9 and undershoots by exactly 2 per isolated cycle beyond it; the gap histogram at radius 14 is $\{+2 : 70, +4 : 2\}$, consistent with $\ell_T - \ell_R = 2c$.

Remark 82 (Verification windows). The computational claims of this section are certified on the following windows.

Strand bound (Lemma 61)	radius 12, zero mismatches
Metric formula (Theorem 62)	radius 14, 3,723 elements, zero mismatches
Lamp model (Theorem 60)	$u_0, \dots, u_{14}$
Normal-form enumeration	$u_0, \dots, u_{22}$ of A396406
Closed form c (Theorem 76)	depth 24, 275,823 elements; held out at 19, 21, 23, 24

Remark 83 (Status of Part B). Lemma 71: c_{pred} is a finite-state function of the edge profile, the decisive point being that off-spine deposit *magnitudes* do not affect c_{pred} except through the bounded class ($\text{sgn } b, \mathbf{1}[|b| \geq 3]$) at the spine boundary, because a single adjacent run can borrow at most one spare same-sign bounce. Proposition 76: c_{pred} is the output of the explicit transducer of Definition 74, whose transition table is displayed, and it agrees with the true defect on the 275,823-element depth-24 enumeration, held out at depths 19–24. Neither of these two facts, by itself or together via the finite-state/Moore route, was enough to prove $c_{\text{pred}} = \frac{1}{2}(\ell_T - \ell_R)$ (Remark 77). That identification is proved instead by the corrected-potential induction of Remark 78, discharging Corollary 79 and, with Lemma 66, the metric theorem as displayed. With ℓ_R read as the closed form the metric theorem holds for every element (`metricAll`); only the relaxed-minimum reading of the closed form remains verified, not proved (Remark 63).

10 Exact growth rate β_2 and exclusion of $3/2$

This section determines the orbit-growth rate of A396406 exactly. Write u_d for the number of group elements of true word length d and v_d for the number of relaxed length d , with generating functions

$$U(x) = \sum_{d \geq 0} u_d x^d, \quad V(x) = \sum_{d \geq 0} v_d x^d, \quad q = x^2.$$

Throughout, $\beta_2 := \limsup_{d \rightarrow \infty} (u_d)^{1/d}$ denotes the exponential growth rate of the true sequence. The main result, Theorem 91, is that β_2 equals an explicit constant $1/\sqrt{q^*}$ defined as the root of a q -series equation, and in particular $\beta_2 < 3/2$. This determination is *unconditional*: it depends only on the inequality $\ell_R \leq \ell_T$ and on the cycle-freeness of pure-travel runs, not on the full metric equality $\ell_T = \ell_R + 2c$ nor on the analytic cosine asymptotic used elsewhere in this paper (Remark 93).

10.1 The bridge and the travel block

The relaxed metric satisfies $\ell_T(g) = \ell_R(g) + 2c(g)$ with $c(g) \in \mathbb{Z}_{\geq 0}$ the number of isolated cycles in a relaxed-optimal realization of g . Both series lift from a single bivariate generating function

$$W(x, y) = \sum_{n, c \geq 0} w_{n, c} x^n y^c, \quad w_{n, c} = |\{g : \ell_T(g) = n, c(g) = c\}|.$$

Proposition 84 (Bridge). $U(x) = W(x, 1)$ and $V(x) = W(x, x^{-2})$.

Proof. Since $u_n = \sum_c w_{n, c}$, summing over c at $y = 1$ gives $U(x) = \sum_n x^n \sum_c w_{n, c} = W(x, 1)$. By $\ell_T = \ell_R + 2c$, an element counted by $w_{n, c}$ has relaxed length $n - 2c$, so $v_m = \sum_{n-2c=m} w_{n, c}$ and $V(x) = \sum_{n, c} w_{n, c} x^{n-2c} = W(x, x^{-2})$. $\square$

Both series carry the same combinatorial denominator, the *travel block*. For a group element g with net displacement k , write $I_k = [\min(0, k), \max(0, k))$ for its travel interval, the edges carrying the displacement indicator $f = \pm 1$. An element is *pure travel* if its lamp support lies inside I_k . Let

$$T(x) = \sum_n t_n x^n$$

count pure-travel elements by length. The travel-section recursion of the relaxed analysis produces, by telescoping the k -sum, the travel resolvent

$$G_0^T(q) = \frac{\Sigma_0(q)}{1 - \Sigma_1(q)}, \quad \Sigma_k(q) = \sum_{j \geq 0} \frac{2q}{1 - q^{k+2j+1}} \prod_{i < j} \gamma_{k+2i}, \quad \gamma_m = \frac{2q^{m+3}}{1 - q^{m+2}} - \frac{2q^{m+2}}{1 - q^{m+1}},$$

with Σ_1 and Σ_0 the cases $k = 1$ (return weight) and $k = 0$ (displacement weight). Since $\gamma_m = 2q^{m+2}(q-1)/((1-q^{m+1})(1-q^{m+2}))$, the products telescope into closed theta forms:

$$1 - \Sigma_1(q) = \sum_{k \geq 0} \frac{(-1)^k 2^k (1-q)^k q^{k^2}}{(q; q)_{2k}}, \quad \Sigma_0(q) = \sum_{j \geq 0} \frac{2^{j+1} (q-1)^j q^{j^2+j+1}}{(q; q)_{2j+1}},$$

the first being $G(q, 2q(1-q))$ for the Hahn–Exton q -cosine $G(q, z) = \sum_k (-1)^k q^{k(k-1)} z^k / (q; q)_{2k}$. Each Σ_k is holomorphic on $|q| < 1$, its j th term being $O(r^{j^2})$ on $|q| \leq r < 1$; in particular each Σ_k extends holomorphically across $|q| = q^*$ except at the isolated zeros of $1 - \Sigma_1$ studied below.

10.2 Connectivity-invariance of the travel block

The decisive structural fact is that the travel block is identical in the true and relaxed gradings. This is an exact consequence of Euler’s trail theorem.

Proposition 85 (Pure-travel runs carry no connectivity penalty). *If g is a pure-travel element with displacement k , then $c(g) = 0$; that is, $\ell_T(g) = \ell_R(g)$.*

Proof. Realize g by its multiset of traversed edges on the sites $0, 1, \dots, k$ (the case $k < 0$ is symmetric). Because g is pure travel, every deposited lamp lies inside I_k , and the slot-multiplicity of edge j is $m_j = |a_j| \geq 1$, an *odd* integer for each j in the support. Form the associated edge-multigraph on the vertices $0, \dots, k$. It is connected, since the displacement forces a $0 \rightarrow k$ backbone, and the set of vertices of odd degree is exactly $\{0, k\}$: each interior vertex has even degree, being the sum of the two adjacent odd multiplicities, while the two endpoints inherit a single odd multiplicity. By Euler’s theorem a connected multigraph with exactly two odd-degree vertices admits a single Eulerian trail from one to the other. Hence the entire run is realized as one connected trail, no isolated cycle is spliced, and $c(g) = 0$. $\square$

Proposition 86 (Invariance of the travel block). *The true and relaxed gradings of T coincide: the series $T(x) = \sum_n t_n x^n$ is the same whether pure-travel elements are bucketed by ℓ_T or by ℓ_R . Consequently the travel-section recursion, and hence the resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1)$, is identical in the true model U and the relaxed model V ; the cycle marker y does not enter it.*

Proof. By Proposition 85 every pure-travel element has $c = 0$, so $\ell_T = \ell_R$ termwise on this set and the two gradings of T agree. The travel recursion is assembled solely from pure-travel runs, appending one $f = \pm 1$ edge at a time, hence is computed inside the sector $c \equiv 0$; in $W(x, y)$ the travel block carries the factor y^0 only and is therefore y -independent. $\square$

Remark 87 (Status). Proposition 85, that a pure-travel element has $c = 0$, is proved: a pure-travel element has no gap edge, and an element with no gap edge admits a relaxed-optimal realisation consisting of a single open walk. That realisation is admissible for ℓ_T , so $\ell_T \leq \ell_R$, and the reverse inequality is immediate, giving $\ell_T = \ell_R$ without appeal to the metric identity, whose lower bound remains open. The argument is in the companion paper on the relaxed series.

Remark 88 (Corroboration). The pure-travel generating function agrees in both gradings through radius 18,

$$1, 3, 5, 8, 12, 17, 24, 35, 52, 78, 116, 173, 256, 383, 572, 858, 1276, 1907, 2832,$$

and the identity $\ell_T = \ell_R$ holds on all 1.9×10^7 enumerated pure-travel runs.

10.3 The travel pole and its rate

The denominator $1 - \Sigma_1$ has a genuine real singularity inside the unit disc.

Definition 89. Let q^* be the smallest positive root of

$$\Sigma_1(q) = 1.$$

Numerically

$$q^* = 0.449453630558948 \dots$$

Lemma 90 (The travel pole is simple and dominant). *The root q^* has the following properties.*

- (i) *It is a simple zero of $1 - \Sigma_1$; numerically $\Sigma_1'(q^*) = 2.6303 \dots$*
- (ii) *The displacement numerator does not vanish there: $\Sigma_0(q^*) = 1.1558 \dots > 0$, so q^* is a genuine simple pole of G_0^T .*
- (iii) *$1 - \Sigma_1$ has no zero other than q^* in the closed disc $|q| \leq \frac{1}{2}$. In particular q^* is the unique singularity of G_0^T on $|q| \leq \frac{1}{2}$, and a fortiori the unique singularity on the circle $|q| = q^*$.*

Proof. All three statements are certified in exact rational arithmetic with explicit tail bounds; no floating point enters any inequality (`code/reproduce/certify_beta2_pole.py`). On $|q| \leq r < 1$ the k th term of the theta form of $1 - \Sigma_1$ satisfies

$$\left| \frac{2^k(1-q)^k q^{k^2}}{(q; q)_{2k}} \right| \leq \frac{(2(1+r))^k r^{k^2}}{\prod_{m=1}^{2k} (1-r^m)},$$

and for $r \leq \frac{1}{2}$ successive bounds decay by a factor $< \frac{1}{2}$ from $k \geq 2$ on, so every truncation carries an explicit rational tail bound. Three computations close the lemma. First, the winding number of $1 - \Sigma_1$ around the circle $|q| = \frac{1}{2}$ equals 1: an exact crossing count over 566 adaptively refined arcs, each certified zero-free using the tail bound and an exact Lipschitz bound $|(1 - \Sigma_1)'| \leq 66$ on the circle. By the argument principle, $1 - \Sigma_1$ has exactly one zero, counted with multiplicity, in $|q| < \frac{1}{2}$; the zero is therefore simple, which gives (i) and (iii). Second, a certified sign change brackets that zero in an interval of width below 10^{-27} , so it is real and equals q^* , with

$$q^* = 0.449453630558948046125545825 \dots, \quad \beta_2 = 1.49161778711437422683671274 \dots$$

certified to the displayed digits. Third, the partial sums of Σ_0 with their tail bound give $\Sigma_0 \geq 1.1558$ on the bracket, which is (ii). $\square$

Setting

$$\beta_2^{\text{rel}} = \frac{1}{\sqrt{q^*}},$$

the singularity-analysis transfer of a unique dominant simple pole [21, Thm. VI.4] applies to $T(x)$ with $q = x^2$: a function meromorphic on $|q| \leq q^*$ whose only singularity on that disc is a simple pole at q^* , with q^* the unique singularity on $|q| = q^*$, has coefficients $[q^n]G_0^T \sim C(q^*)^{-n}$ with $C = -\Sigma_0(q^*)/\Sigma_1'(q^*) \neq 0$ by Lemma 90. Transferring through $q = x^2$ gives $t_d \sim C'(\beta_2^{\text{rel}})^d$; in particular $\lim_d t_d^{1/d} = \beta_2^{\text{rel}}$, so the pure-travel subcounts grow at rate β_2^{rel} .

The bulk block does not interfere with the relaxed assembly. This is certified by the same machinery as Lemma 90 (`code/reproduce/certify_bulk_dressing.py`), where Σ_1^{bulk} denotes the series formed exactly as Σ_1 above but from the bulk block's transfer kernel in place of the travel one, as implemented in that script: $1 - \Sigma_1^{\text{bulk}}$ has winding number 0 around $|q| = \frac{1}{2}$, so the bulk resolvent is holomorphic on $|q| \leq \frac{1}{2}$; its singularity is real, with certified value $q_b = 0.60956734426012 \dots > \frac{1}{2} > q^*$; and on the certified bracket of q^* the dressing satisfies $\Sigma_1^{\text{bulk}}(q^*) = 0.48044 \dots < 1$. Hence q^* remains the dominant singularity of the assembled relaxed series V , and the same singularity-analysis transfer gives $v_d \sim C''(\beta_2^{\text{rel}})^d$. Numerically

$$\beta_2^{\text{rel}} = 1.4916177871143742268 \dots$$

10.4 The squeeze: $\beta_2 = \beta_2^{\text{rel}}$

Theorem 91 (Exact growth rate). *The true orbit-growth rate of A396406 satisfies*

$$\beta_2 = \beta_2^{\text{rel}} = \frac{1}{\sqrt{q^*}} = 1.4916177871143742268 \dots,$$

where q^* is the smallest positive root of $\Sigma_1(q) = 1$.

Proof. By the singularity analysis of §10.3, both flanking sequences have a pinned exponential rate: $t_d \sim C'(\beta_2^{\text{rel}})^d$ and $v_d \sim C''(\beta_2^{\text{rel}})^d$, the latter because q^* is the dominant singularity of the assembled relaxed series V (Lemma 90 and the bulk non-interference above). It remains to trap u_d between them.

Lower flank. Every pure-travel element is a group element, and by Proposition 85 its true and relaxed lengths coincide, so each pure-travel element of length d contributes to u_d ; hence $t_d \leq u_d$.

Upper flank. For every g one has $\ell_R(g) \leq \ell_T(g)$, since $\ell_T = \ell_R + 2c$ with $c \geq 0$. Hence $\{g : \ell_T(g) \leq d\} \subseteq \{g : \ell_R(g) \leq d\}$, so

$$\sum_{k \leq d} u_k \leq \sum_{k \leq d} v_k = O((\beta_2^{\text{rel}})^d),$$

the last equality because $v_k \sim C''(\beta_2^{\text{rel}})^k$ with $\beta_2^{\text{rel}} > 1$. Since $u_d \leq \sum_{k \leq d} u_k$, this gives $\limsup_d u_d^{1/d} \leq \beta_2^{\text{rel}}$.

The two gradings do not compare termwise by inclusion: an element of true length d with cycle number $c \geq 1$ has relaxed length $d - 2c < d$. The inequality $u_d \leq v_d$ holds for every computed $d \leq 32$ (for instance $v_8 = 91$ against $u_8 = 89$), but is not used.

Combining the flanks with $\lim_d t_d^{1/d} = \beta_2^{\text{rel}}$,

$$\beta_2^{\text{rel}} = \lim_d t_d^{1/d} \leq \liminf_d u_d^{1/d} \leq \limsup_d u_d^{1/d} \leq \beta_2^{\text{rel}},$$

so $\lim_d u_d^{1/d} = \beta_2^{\text{rel}}$, i.e. $\beta_2 = \beta_2^{\text{rel}}$. □

Corollary 92 (Exclusion of $3/2$). $\beta_2 < \frac{3}{2}$. In particular the orbit-growth rate of A396406 is not $3/2$.

Proof. By Theorem 91, $\beta_2 = 1.4916177871143742268\dots < 1.5$. The inequality is strict and quantitative: $\frac{3}{2} - \beta_2 = 0.00838\dots > 0$. $\square$

Remark 93 (Unconditionality). Theorem 91 and Corollary 92 do not use the metric theorem or Theorem 80; they use only the inequality $\ell_R \leq \ell_T$ and the dominant-pole analysis. Their inputs should nonetheless be named exactly. Three enter: the travel resolvent $G_0^T = \Sigma_0/(1 - \Sigma_1)$, whose telescoping is stated above but not derived in full here; the location of q^* , which is certified numerically; and the bulk non-interference, whose series Σ_1^{bulk} is defined only through the certificate script named above rather than in the text. The first and third are supplied by certificate rather than by a derivation printed in this paper, and a reader wanting a self-contained proof will have to reconstruct them. The squeeze uses only two inputs: the termwise inequality $\ell_R \leq \ell_T$ (immediate from $c \geq 0$, requiring no metric lower bound) and the cycle-freeness $c = 0$ of pure-travel runs (Proposition 85, a direct application of Euler’s trail theorem). It does *not* invoke the full metric equality $\ell_T = \ell_R + 2c$ with the covering verification, nor the analytic cosine asymptotic used in the transcendence results of this paper. The rate determination is therefore independent of those hypotheses.

Remark 94 (Finite-horizon ratio estimate). The consecutive-ratio estimate u_{d+1}/u_d converges to β_2 at rate $O(1/d)$ with a period-4 sign oscillation. On the 43 available terms $u_0, \dots, u_{42}$ the last such ratio is $u_{42}/u_{41} = 1.49692\dots$, and the geometric mean of the last eight, which averages out one full period of the oscillation, is $1.49987\dots$, within 0.01% of $3/2$. This finite-depth value overshoots the limit and does not determine the growth rate, which is fixed exactly by Theorem 91.

Remark 95 (Nature of the constant; transcendence of the number is open). The constant $\beta_2 = 1/\sqrt{q^*}$ is defined as the value of an explicit q -series equation $\Sigma_1(q) = 1$ and is *not* presented as a closed-form algebraic number. An integer-relation (PSLQ) search at 130 decimal digits finds no algebraic relation for q^* or for β_2 of degree ≤ 6 , consistent with the transcendental q -difference structure of the underlying generating function. Theorem 91 determines the growth *rate* only; *transcendence of the number β_2 as a real number is open and is not claimed here*. It is logically independent of the transcendence of the generating functions U and V over $\mathbb{Q}(x)$ treated elsewhere in this paper, and in particular Theorem 91 carries no dependence on the analytic cosine asymptotic used there.

A Finite-horizon structural exclusions

This appendix records, with full precision and explicit bounds, the empirical exclusions of low-complexity recurrence and algebraicity for the universal sequence

$$u_0, u_1, \dots, u_{42} \quad (\text{A396406}),$$

stated as Proposition 98 below. The data are the 43 exact integer terms $u_0, \dots, u_{42}$: the first 39 are the published terms of A396406, and $u_{39}, \dots, u_{42}$ come from the depth-42 breadth-first search, recorded with their provenance in `code/data/u_terms_43.txt` and discussed in Appendix K; they were recovered from the stored logarithms of that run, a recovery that reproduces all 39 overlapping published terms exactly, and they have not been regenerated by the growth breadth-first search. All of $u_0, \dots, u_{22}$ are independently re-derived, with no breadth-first search anywhere in the pipeline, from the metric formula of Theorem 62 by enumerating the normal form of Theorem 60 and counting by assigned word length. Throughout, $F(x) = \sum_{d \geq 0} u_d x^d$ denotes the ordinary generating series.

The purpose of this appendix is twofold. First, to state precisely what is, and what is *not*, established by these searches: each is a non-existence statement *within a stated finite complexity box on 43 terms*, and none of them can be promoted to an unconditional structural claim such as “ F is not D -finite.” Second, to assemble these bounded exclusions as the evidence that motivates the q -difference picture of §7.4 and Proposition 91, with the question of genuine transcendence deferred to the companion paper [12]. The conditional routes to transcendence that were explored before that companion settled the question are recorded in Appendix D, and further exclusions of the same kind, with their Lean certificates, in Appendices E and K.

Complexity classes and the search method

We test three nested structural hypotheses.

Definition 96 (Structural classes). Let $(u_d)_{d \geq 0}$ be an integer sequence with generating series $F(x)$.

- (a) *C-finite of order $\leq r$* : there exist constants $c_0, \dots, c_{r-1} \in \mathbb{Q}$, not all zero, with $u_{d+r} = \sum_{i=0}^{r-1} c_i u_{d+i}$ for all $d \geq 0$. Equivalently, F is rational with denominator degree $\leq r$.
- (b) *Holonomic (D -finite) of bidegree (ρ, δ)* : there exist polynomials $p_0, \dots, p_\rho \in \mathbb{Q}[d]$ of degree $\leq \delta$, not all zero, with $\sum_{i=0}^\rho p_i(d) u_{d+i} = 0$ for all $d \geq 0$.
- (c) *Algebraic of bidegree $(\deg_x, \deg_F)$* : there exists a nonzero $P \in \mathbb{Q}[x, y]$ with $\deg_x P \leq \deg_x$ and $\deg_y P \leq \deg_F$ such that $P(x, F(x)) = 0$ as a formal power series.

Each hypothesis, restricted to a fixed complexity box, is the existence of a nonzero rational vector in the kernel of an explicit integer matrix built from finitely many of the u_d (the Hankel/coefficient matrix of the relevant ansatz; for the algebraic case the rows are the power-series coefficients of $x^j F(x)^k$). A given box is *refuted on the available data* when the corresponding matrix has full column rank, i.e. its only kernel vector is 0, so that no relation of that shape can hold. We perform every rank computation as an exact, fraction-free integer nullspace search (no floating point), and we require the matrix to be *over-determined*, with strictly more independent constraints than unknowns, so that full column rank is a genuine obstruction and not an artifact of an under-constrained system.

Remark 97 (Detector validation). The same machinery, applied as a positive control, recovers the expected structure of two benchmark sequences from the matching number of initial terms: the order-2 C -finite recurrence and the generating-function relation of the Fibonacci numbers, and the second-order holonomic recurrence of the Catalan numbers. The detector therefore reports *presence* of structure when it is present at the tested complexity; consequently the negative reports below are properties of the data, not failures of the detector.

The exclusions

Proposition 98 (Finite-horizon structural exclusions). *On the 43 exact terms $u_0, \dots, u_{42}$ of A396406, each of the following coefficient matrices has full column rank, so the corresponding relation does not exist within the stated box:*

- (i) (No low-order C -finite recurrence.) *The sequence satisfies no constant-coefficient linear recurrence of order ≤ 21 ; 21 is the maximal order testable on 43 terms with a positive over-determination margin. Consequently, if F is rational, its denominator has degree > 21 .*
- (ii) (No low-complexity holonomic recurrence.) *The sequence satisfies no holonomic recurrence at any bidegree (ρ, δ) with*

$$(\rho + 1)(\delta + 1) \leq 32 \quad \text{and} \quad (\rho + 1)(\delta + 1) < 43 - \rho, \quad (6)$$

that is, at any over-determined cell of the box $(\rho + 1)(\delta + 1) \leq 32$. There are 106 such cells, and over them the over-determination margin ranges from 2 to 42. The second condition in (6) is not a convenience: the 13 cells of the box that violate it are underdetermined on 43 terms and exclude nothing, so the box without it claims more than any search can deliver (Remark 125). Consequently, if F is D -finite, its annihilating operator has complexity outside this region.

- (iii) (No low-degree algebraic equation.) The sequence satisfies no algebraic equation $P(x, F) = 0$ with $\deg_x P \leq 14$ and $\deg_F P \leq 6$: each of the 29 tested $(\deg_x, \deg_F)$ cells yields a full-column-rank coefficient matrix, over-determined by 7 to 34 equations. Consequently, if F is algebraic, it has degree exceeding this box.

Proof. For each class and each box the assertion is the statement that a single explicit integer matrix, formed from $u_0, \dots, u_{42}$ as in Definition 96, has trivial kernel. This is decided by an exact fraction-free Gaussian elimination over $\mathbb{Z}$, which returns the rank without rounding; the quoted over-determination margins are the excess of the number of rows (constraints) over the number of columns (unknown coefficients) in each case, and are all positive. Full column rank in every cell is equivalent to non-existence of a nonzero relation of the corresponding shape, which is the claim. The recovery of the Fibonacci and Catalan structures on the same machinery (Remark 97) certifies that the procedure detects relations when they are present.

Part (ii) additionally carries a Lean certificate. Substituting $n = d + \rho$ and $j = \rho - i$ turns $\sum_{i \leq \rho} p_i(d) u_{d+i} = 0$ into the system (8) at $(k, m) = (\rho, \delta)$: the change of variable is affine and invertible, so it carries polynomials of degree $\leq \delta$ in d onto polynomials of degree $\leq \delta$ in n , and the two coefficient spaces correspond. That reindexing is elementary and is not itself formalized. In the normalization of (8), the whole region (6) is machine-checked as `NoDFiniteCertificates.no_recurrence_in_box` (Appendix K), by kernel evaluation of modular rank certificates at the eleven maximal cells of the region, together with the downward propagation of non-existence along the componentwise order. Parts (i) and (iii) remain outside Lean. $\square$

Scope of the exclusions

The exclusions of Proposition 98 are *finite-horizon* in two independent senses, and both must be carried faithfully into any use of them.

Remark 99 (What is and is not established). First, each statement is a non-existence of relations of *bounded complexity*: a recurrence of order > 21 , a holonomic operator outside the region (6), or an algebraic equation of bidegree exceeding $(14, 6)$ is not excluded. The underdetermined cells of the box $(\rho + 1)(\delta + 1) \leq 32$ are outside the region for a reason that is not a gap in the search but a property of the data: on them no search of any kind can exclude anything. Second, every conclusion is drawn from exactly 43 terms: a relation that first fails to hold only beyond index 42 is invisible to the search. In particular Proposition 98 **does not** establish that F fails to be D -finite, nor that F is transcendental; it may not be cited as “ u_d is not D -finite.” What it does establish is that the sequence admits no *low-complexity* closed structure of any of the three tested kinds, so that any such structure, were it to exist, must be of complexity strictly beyond these boxes.

Remark 100 (Finite-depth ratio estimates are unreliable). The consecutive ratios u_{d+1}/u_d converge only at rate $O(1/d)$ and carry a period-4 sign oscillation; the estimate $1.49987\dots$ obtained from the 43 available terms (Remark 94) is a finite-depth overshoot of the true growth rate and must not be read as evidence for the value $3/2$. The growth rate is instead pinned exactly by the closed-form singularity analysis of the relaxed model, $\beta_2 = 1.49161778711\dots$ (Proposition 91), which excludes

3/2. The finite-horizon exclusions above and the exact β_2 are complementary: the former rule out low-complexity structure on the data, the latter supplies the value the data cannot.

Remark 101 (Role in the program). Proposition 98 is the empirical complement of the exact lamplighter transfer of §7.4. The absence of any low-complexity C -finite, holonomic, or algebraic relation on 43 terms is consistent with, and motivates, the q -difference functional equation satisfied by the universal series, whose catalytic dilation $t \mapsto q^2 t$ is precisely the mechanism that places A396406 outside the low-complexity D -finite and algebraic classes. The exclusions recorded here are stated as bounded empirical facts; the unconditional question of whether F is D -finite or transcendental is deferred to the companion paper [12].

B Reproducibility: Lean and modular certificates

This section records the precise provenance and trust base of every machine-verified claim in the paper. We distinguish three tiers of verification, in decreasing order of trust: (i) Lean 4 proofs discharged by the `ring` tactic and checked by the Lean kernel; (ii) Lean 4 computations discharged by `native_decide`, whose trusted base additionally includes the Lean compiler and the hand-written `partial` functions it compiles; and (iii) a multi-prime modular certificate executed outside Lean, in Python/NumPy and re-implemented in Rust, which intersects candidate collisions across several primes and then verifies all survivors exactly over $\mathbb{Q}(i)$. We state explicitly which results sit in which tier, and in particular we do *not* claim that the sharp depth-32 threshold (Theorem 20) is verified by the Lean kernel.

Throughout we use the generator dictionary fixed in §2: R_0, R_1 are the reflections in the two legs at the right angle and R_2 the reflection in the hypotenuse, acting on the right triangle placed at $A = (0, 0)$, $B = (a, 0)$, $C = (a, b)$ with $a, b > 0$. Here u_d^T denotes the breadth-first orbit-growth count at depth d for the right triangle T with positive unequal rational legs, u_d the universal sequence (OEIS A396406), Q_n the Schur-complement Gram matrix of §2, c_T the rotation denominator (norm of the leg-ratio numerator), and $\zeta_T = (a + bi)/(a - bi)$ the rotation number of T .

B.1 Exact symbolic and numerical computations

All computations use exact arithmetic; no floating-point step enters any asserted result. The symbolic normal form of Theorem 10 is verified in $\mathbb{Q}(a, b)$ with SymPy, and the Schur-complement determinant identity of `lean/with_mathlib/SchurGeneral.lean` is verified in the same way. The Gram determinant identity $\det Q_n = -\prod_i a_i^2$ (for the dimensions $n \leq 10$ used here) is verified symbolically in the polynomial ring $\mathbb{Z}[a_{k-1}, a_k, a_{k+1}, S_{k-1}, P_{k-2}]$. This material, and the rank-0 descents below, belong to the higher-dimensional companion [13] and are reproduced here only because the same scripts certify both papers; the faithfulness statement they support is that companion’s, not Theorem 64, which is about the word metric. The rank-0 descents on the Cremona curves 24a1, 32a2, 72a2 (for $\mathcal{V}_{1/4}, \mathcal{V}_{1/2}, \mathcal{V}_{3/4}$ respectively) are computed in SageMath by `EllipticCurve.rank()` together with a direct rational-point determination, in `code/mordell/`. The orbit-growth values are produced by an exact-rational breadth-first search through depth 25 and by a Rust disk-streaming breadth-first search through depth 42, the latter independently reproduced on a second machine.

B.2 Shape arithmetic

Four numerical statements of §3 and §9 are certified by `code/zeta_probe/tools/shape_arith` (Rust, exact integer arithmetic) and proved in `lean/with_mathlib/ShapeArith.lean` (twelve

theorems, no sorry): that e_T is even in both branches of its definition, hence that the four abstract pairs (61, 11), (37, 35), (53, 45), (73, 55) are realized by no rational-leg triangle (`eT_even`, `abstract_pairs_unreal`); the coefficient identity for the lamp profile of $-2(t+1)\mu_T$ and the two specialisations that separate the sign conventions, $(-10, 2, 2, -10)$ at $e_T = 6$ against $(-10, -22, -22, -10)$ at $e_T = -6$ (`profile_coeffs`); the four representations behind the shape count $18 + 4 = 22$ at $c_T \leq 72$ (`sixtyfive_four_shapes`, `shape_count_72`); and the size $3 \cdot 2 \cdot 2 = 12$ of the strand alphabet (`strand_alphabet`). The Rust program independently reproduces all nine (c_T, e_T) rows of the deviation-law table and finds no odd e_T among the 97,354 coprime leg pairs with legs at most 400.

The finite-state principle withdrawn in Remark 77 is settled in `lean/with_mathlib/MooreCriterion.lean` (six theorems, no sorry). `agree_of_same_machine` proves the version that is valid, namely that two readings of the *same* machine agreeing at every reachable state agree on every input; `not_agree_of_transition_cover` refutes the version that was used, by exhibiting two finite-state functions that agree on a set driving the first machine through each of its transitions and differ at length two.

The lower bound itself (Theorem 80, Remark 78) is proved in `lean/with_mathlib/Philipschitz.lean` and `EltBridge.lean` (`wordLength_eq_lRTrue_add_two_cTrue`, composing the descent lemma `exists_descent` with a strong induction on the corrected potential), with the span repair in `CorrectedSpan.lean`. `#print axioms` on the theorem lists only `propext`, `Classical.choice` and `Quot.sound`.

B.3 Tier (i): kernel-checked ring proofs

Two Lean 4 developments accompany the paper in the `lean/` subtree of the project repository. The Mathlib-dependent development `lean/with_mathlib/` contains the strongest *shape-uniform* machine-checked statements, all proved by the `ring` tactic and therefore reduced to a polynomial identity verified by the Lean kernel.

Proposition 102 (Kernel-checked length-10 relations). *In `SymbolicVerification.lean`, the theorems `rel1_symbolic` through `rel8_symbolic` assert that each of the eight length-10 word pairs of Table 1 evaluates to the same element of $\text{Aff}(\mathbb{Q}(a, b))$, i.e. over $\text{MvPolynomial}(\text{Fin } 2) \mathbb{Q}$. Each is discharged by `ring` and hence checked by the Lean kernel.*

Proof. Each relation is an equality of two products of affine matrices whose entries are polynomials in the legs a, b ; clearing the common denominator $(a^2 + b^2)^k$ reduces the claim to a polynomial identity in $\mathbb{Q}[a, b]$, which `ring` normalizes to $0 = 0$. No hypothesis $a \neq b$ or rationality is used in the identity itself; the relations hold over the full function field, hence simultaneously for every right triangle with positive unequal legs. $\square$

Proposition 103 (Kernel-checked Schur induction steps). *In `SchurGeneral.lean`, the four polynomial identities `schur_base_step`, `schur_k2_step`, `schur_inductive_step`, `schur_boundary_step` encode the algebraic skeleton of the tridiagonal recursion proving $\det Q_n = -\prod_{i=1}^n a_i^2$. All four are discharged by `ring` and checked by the Lean kernel.*

Remark 104. These two propositions are the only results whose Lean verification is entirely kernel-checked. They cover the algebraic core of the universal length-10 relations (§2) and of the Schur-complement identity (§2). Since 2026-08 they also include the recursion’s closing step $\det Q_n = -\prod a_i^2$ as a single statement over general n : `lean/with_mathlib/OrthoschemeDet.lean` assembles the two base cases, the interior recurrence and the boundary row into `det_Q_of_recurrence`, over an arbitrary commutative ring and with no positivity, ordering or characteristic hypothesis (axioms

`propext`, `Classical.choice`, `Quot.sound`; no sorry). The earlier restriction to concrete leg sequences in tier (ii) is therefore superseded. The Mathlib-free file `lean/RightTriangleReflection.lean` additionally re-checks the basic Coxeter relations $R_i^2 = 1$ and $(R_0 R_1)^2 = 1$ and the explicit affine-matrix value of each length-10 relation on the $(3, 4, 5)$ triangle.

B.4 Tier (ii): `native_decide` computations

The deeper finite verifications are evaluations of computable functions on fixed inputs, discharged by `native_decide`. Every result proved this way carries, alongside `propext`, a generated axiom recording the compiler evaluation: such results trust the Lean compiler and its evaluation of the compiled term, not only the kernel. The exact name is toolchain-dependent and both forms occur in this development: the Mathlib-free project (pinned to `v4.13.0`) reports the global `Lean.ofReduceBool`, while the Mathlib project (pinned to `v4.30.0`) reports a per-declaration axiom `name._native.native_decide.ax_1_1`. We flag this explicitly because it is a strictly larger trusted base than tier (i), and every statement below inherits it. We record the enlarged trust base before stating the results.

Remark 105 (Trust base of `native_decide`). A theorem proved by `native_decide` is correct provided the Lean kernel *and* the Lean native-code compiler *and* the hand-written `partial` functions invoked by the computation are correct. In the present developments those `partial` functions are the bivariate-polynomial determinant routine, the tracked-denominator affine-isometry arithmetic, and the cross-multiplication deduplication `dedupBy CAff.equivB`. This is a strictly larger trusted base than tier (i); every tier-(ii) result is flagged as such.

Proposition 106 (`native_decide` certificates). *The following hold by `native_decide`, with the trust base of Remark 105.*

- (a) (Determinant identity, concrete legs.) *In `lean/RightTriangleReflection.lean`, $\det Q_n = -\prod_{i=1}^n a_i^2$ for $n = 2, \dots, 10$ on concrete leg sequences, the case $n = 10$ being the 11×11 determinant on $(1, 2, 3, 5, 7, 11, 13, 17, 19, 23)$.*
- (b) (Direct orbit reconstruction.) *A layer-by-layer breadth-first search of the Cayley orbit on $(3, 4, 5)$ in exact rational arithmetic reproduces $u_0, \dots, u_{17}$ of **A396406** exactly, and the Fibonacci coincidence $u_n = F(n + 3)$ for $1 \leq n \leq 9$ together with the deficit $u_{10} + 8 = F(13)$.*
- (c) (Shape-uniform universality through depth 22.) *In `ComputableUniversality.lean`, the single theorem `universal_layers_through_22` asserts*

$$u_d^T = u_d \quad (0 \leq d \leq 22)$$

for every right triangle T with positive unequal legs, equivalently

$$\text{allLayerCounts } 22 = [1, 3, 5, 8, 13, 21, 34, 55, 89, 144, 225, 351, 554, 875, 1345, \\ 2066, 3203, 4971, 7574, 11543, 17683, 27108, 41067],$$

*i.e. **A396406** $a(0), \dots, a(22)$.*

Method. Part (c) is a single breadth-first sweep over canonical Coxeter words in which symbolic affine isometries over $\mathbb{Q}(a, b)$ are deduplicated by comparing numerators relative to the common denominator $(a^2 + b^2)^{22}$; equality of two such numerators as polynomials in $\mathbb{Q}[a, b]$ is decided exactly, so the layer counts hold simultaneously for all admissible T rather than for any specific triangle. The

hand-written computable types `CPoly` (bivariate polynomials, no `MvPolynomial` dependency), `CAff` (tracked-denominator affine isometries), and the deduplicator `dedupBy CAff.equivB` carry this computation; the cross-checks `slow_fast_agree_at_10`, `layer_10_eq_225`, and `layer_11_eq_351` pin the fast hashed common-denominator equivalence against the slow cross-multiplication equivalence at low depth. Verification consists of a successful `lake build`; the depth-22 `native_decide` evaluation completes in about 22 minutes. $\square$

Remark 107. Proposition 106(c) is the deepest *Lean-verified* universality statement: shape-uniform equality $u_d^T = u_d$ through depth 22. It is verified by `native_decide`, not by the kernel alone (Remark 105). The sharp threshold of depth 32 (Theorem 20) is *not* a Lean theorem; it is established by the modular certificate of §B.5.

B.5 Tier (iii): the multi-prime modular certificate

The depths beyond the reach of the Lean developments are certified by an exact modular computation outside Lean. We isolate its logical content as a one-sided exactness principle, then state how the certificate is made reliable at large depth.

Lemma 108 (One-sided exactness of the modular test). *Fix a depth d and a candidate rotation number $\zeta_T \in \mathbb{Q}(i)$ with denominator c_T . Let q be a prime with $q \equiv 1 \pmod{4}$ (so that $i \in \mathbb{F}_q$) and $q \nmid c_T$. If the reductions modulo q of two symbolic ball elements of depth $\leq d$, evaluated at the $\mathbb{F}_q$ -image of ζ_T , are distinct in $\mathbb{F}_q$, then the corresponding affine isometries are distinct over $\mathbb{Q}(i)$; hence the two ball elements are distinct orbit points.*

Proof. Reduction modulo a prime of good reduction is a ring homomorphism $\mathbb{Z}[i] \rightarrow \mathbb{F}_q$ (well defined since $q \equiv 1 \pmod{4}$ and $q \nmid c_T$ keep all denominators invertible). A ring homomorphism cannot separate equal elements, so distinct images force distinct sources. The implication runs in one direction only: equal images do *not* prove equality, only *flag* a candidate collision to be adjudicated separately. $\square$

Remark 109 (Reliability: multi-prime intersection and exact closure). A single prime is not a trust base: an unlucky prime q dividing the difference polynomial of a non-colliding pair registers a spurious modular collision, and at large depth one such prime can falsely flag thousands of pairs. The certificate therefore does *not* rest on any fixed small number of primes. A genuine collision survives reduction modulo *every* prime of good reduction, so the certificate *intersects* the candidate-collision sets across four independent primes $q \equiv 1 \pmod{4}$ and then verifies each surviving pair *exactly* in $\mathbb{Q}(i)$ by evaluating the two translation polynomials at ζ_T in rational arithmetic. The modular passes are a fast pre-filter; the exact $\mathbb{Q}(i)$ check on the survivors is the trust base. “Collision-free” verdicts are exact already at one prime, by Lemma 108; only flagged candidates require the exact closure.

By the effective universality theorem (Theorem 19), a deviation at depth d forces $c_T \leq 2d$, so at each depth only finitely many shapes can deviate, and the splitting/inertness conditions thin these to a sparse list of rotation numbers; the test of Lemma 108, closed by the exact $\mathbb{Q}(i)$ verification of Remark 109, then disposes of each survivor.

Proposition 110 (The depth-32 certificate). *For the sharp threshold (Theorem 20) the canonical certificate evaluates the full symbolic ball at depth 32 on every candidate rotation number with $c_T \leq 64$. Each candidate’s collision-pair set is intersected across the four independent primes*

$$2,000,000,033, \quad 1,900,000,097, \quad 1,800,000,049, \quad 1,700,000,009 \quad (\equiv 1 \pmod{4}),$$

and every surviving pair is verified exactly over $\mathbb{Q}(i)$; no candidate with $c_T \leq 64$ yields a genuine collision through depth 32. Combined with Theorem 19(ii) for the cofinitely many shapes $c_T > 64$, this proves universality through depth 32. The computation runs in `code/zeta_probe/`: a Python/NumPy pre-filter `certify.py` and the Rust certifier `certify38_rust/`, the latter exporting survivors to `exact_check.py` for the exact $\mathbb{Q}(i)$ adjudication. The Rust run extends to depth 36, reproducing the generic layer counts there and locating the first genuine deviation as the $(1, 2)$ triangle. It was independently reproduced on a second machine. The trust base is exact arithmetic over $\mathbb{Q}(i)$ on the multi-prime survivors; the Lean kernel is not involved.

Remark 111 (Provenance of the sharp depth-32 threshold). Theorem 20 (universality through depth 32, sharp, with $u_{33}^{(1,2)} = u_{33} - 30 = 3,848,354$) combines Theorem 19(ii) for the cofinitely many shapes $c_T > 64$ with the finite certificate of Proposition 110 for the sparse remaining candidates. Its verification is therefore tier (iii): a four-prime intersection closed by exact $\mathbb{Q}(i)$ arithmetic, executed in Python/NumPy and Rust. It is *not* a Lean-kernel result and is not claimed as one. The certificate extends the Lean bound of Proposition 106(c) and Remark 107 by ten further layers, at the cost of the enlarged, non-Lean trust base recorded here.

B.6 Toolchains, costs, and how to re-verify

The Mathlib-free Lean development is pinned to `leanprover/lean4:v4.13.0`; a clean `lake build` re-checks all of its theorems in about 20 seconds and needs no Mathlib cache. The Mathlib-dependent development is pinned to `leanprover/lean4:v4.30.0`; after the one-time Mathlib cache fetch the `ring` theorems of Propositions 102 and 103 rebuild in seconds, while the depth-22 `native_decide` computation of Proposition 106(c) completes in about 22 minutes. In each Lean development verification consists of a successful `lake build`: Lean is its own checker, with no separate test step. The modular certificate of Proposition 110 is re-run via `certify.py` (Python/NumPy pre-filter) and the Rust package `certify38_rust/`, with `exact_check.py` performing the exact $\mathbb{Q}(i)$ adjudication of any flagged survivors. The explicit witness words of the refutation of all-depths universality (Theorem 27) are verified in exact rational arithmetic by `code/zeta_probe/witness.py`, not in Lean.

Remark 112 (What is and is not formally certified). The trust boundary is as follows. Kernel-checked (tier i) are the eight universal length-10 relations over $\mathbb{Q}(a, b)$, the four Schur induction steps, and the structural exclusions of Appendices A(ii), D and E that carry Lean certificates: the Farkas witnesses of Proposition 121 and the modular rank certificates of Proposition 116, both discharged by kernel evaluation with no compiler-evaluation axiom. `native_decide`-checked (tier ii) are the concrete determinant identities $\det Q_n = -\prod a_i^2$ for $n = 2, \dots, 10$, the direct orbit reconstruction through depth 17, and the shape-uniform universality through depth 22. Modular-plus-exact certified (tier iii) is the sharp universality through depth 32 and the generic layer data through depth 36. The growth rate β_2 (Theorem 91) is unconditional: its proof uses only the inequality $\ell_R \leq \ell_T$ and the dominant-pole analysis of Σ_1 , with no analytic remainder input. The identification $c_{\text{pred}} = c$ needed for the lower bound of the metric theorem (Theorem 80) is now proved. An earlier version left it conditional: the connectivity defect's closed form is computed by the explicit transducer of Definition 74, which agrees with the true defect on the depth-24 enumeration (275,823 elements, held out at depths 19, 21, 23, 24) with 0 exceptions, but that certificate only establishes that the transducer computes the closed form, not that the closed form is the true defect (Remark 77). The identification is instead proved by an induction on a corrected potential, formalized in Lean and independently axiom-checked (Remark 78). With ℓ_R read as the closed form of Proposition 62, the metric theorem holds for every element (Lean theorem `metricAll`); only the relaxed-minimum

reading, that the closed form is the minimum over relaxed realizations, remains verified and not proved (Remark 63).

B.7 Index of machine-verified statements

Table 2 is the complete map from the statements of this paper to the Lean declarations that certify them. A statement absent from the table has no Lean certificate. Tier (i) entries are checked by the kernel alone; tier (ii) entries additionally trust the Lean compiler (Remark 105). File paths are relative to `lean/`; `wm/` abbreviates `with_mathlib/`.

Statement	Lean declaration	File	Tier
Thm 10, Prop 102	<code>rel1_symbolic-rel8_symbolic</code>	<code>wm/SymbolicVerification</code>	(i)
Prop 103	<code>schur_base_step,</code> <code>schur_inductive_step,</code> <code>schur_boundary_step</code>	<code>wm/SchurGeneral</code>	(i)
Rem 104	<code>det_Q_of_recurrence</code>	<code>wm/OrthoschemeDet</code>	(i)
Rem 35	<code>mu_root_abs_one</code>	<code>wm/DeviationLattice</code>	(i)
Prop 36, windows	<code>window_60, window_5637</code>	<code>wm/DeviationLattice</code>	(i)
Rem 117	<code>exists_nonzero_solution,</code> <code>le_of_exclusion,</code> <code>vacuous_card,</code> <code>guard_not_sufficient</code>	<code>wm/OverDetermination</code>	(i)
Rem 77	<code>agree_of_same_machine,</code> <code>not_agree_of_transition_cover</code>	<code>wm/MooreCriterion</code>	(i)
Thm 80, Rem 78	<code>wordLength_eq_lRTrue_add_two_cTrue,</code> <code>exists_descent</code>	<code>wm/Philipschitz,</code> <code>wm/EltBridge,</code> <code>wm/CorrectedSpan</code>	(i)
Shape arithmetic, §B.2	<code>eT_even,</code> <code>profile_coeffs,</code> <code>shape_count_72,</code> <code>strand_alphabet</code>	<code>wm/ShapeArith</code>	(i)
Prop 121	<code>no_linear_recurrence</code>	<code>wm/NoRecurrence</code>	(i)
Prop 116	<code>no_dfinite</code>	<code>wm/NoDFiniteCertificates</code>	(i)
Prop 98(ii)	<code>no_recurrence_in_box</code>	<code>wm/NoDFiniteCertificates</code>	(i)
Prop 114	<code>rotations_pairwise_distinct</code>	<code>wm/ModPStateInfinite</code>	(i)
Prop 124	<code>mod2_closed_form,</code> <code>eventually_three_periodic,</code> <code>not_purely_periodic,</code> <code>fib_recurrence_from_three,</code> <code>fib_recurrence_fails_at_two</code>	<code>PaperExtraMod2</code>	(i)
Prop 122(ii)	<code>monomial_count,</code> <code>nonlinear_search_underdetermined</code>	<code>PaperExtraCounts</code>	(i)
Prop 123, general form	<code>not_automatic_of_unbounded</code>	<code>PaperExtraCounts</code>	(i)
Grid counts of Prop 116	<code>searchGrid_count,</code> <code>searchGrid_overDetermined,</code> <code>count_depends_on_k_ge_one</code>	<code>PaperExtraCounts</code>	(i)
Prop 106(a)	<code>schur_n2-schur_n10</code>	<code>RightTriangleReflection</code>	(ii)
Prop 106(b)	<code>a_val_0-a_val_17,</code> <code>a_eq_fib_1-a_eq_fib_9,</code> <code>deficit_at_10</code>	<code>RightTriangleReflection</code>	(ii)
Prop 106(c)	<code>universal_layers_through_22</code>	<code>wm/ComputableUniversality</code>	(ii)

Table 2: Paper statement to Lean declaration. Absence from this table means no Lean certificate.

What is *not* in the table is the bulk of the paper’s mathematics, and that is a fact about scale, not about confidence. The Coxeter presentation and the Steinberg series of §2, the symbolic normal form of §4, the effective universality bound (Theorem 19), the translation-lattice and Mayer–Vietoris results of §7, the lamplighter structure, the metric theorem of §9, and the singularity analysis giving β_2 are pen-and-paper proofs with computational corroboration only; formalizing them would require

Coxeter theory, group homology, and analytic combinatorics well beyond what Mathlib currently supplies, with one exception: the lower half of the metric theorem (Theorem 80) *is* formalized, in `lean/with_mathlib/PhiLipschitz.lean` and `EltBridge.lean` (Remark 78), by an argument that turned out not to need the pen-and-paper machinery of this section. Two further items are outside Lean by construction: Theorem 20 and Proposition 110 are tier (iii).

C Scope of the remaining appendices

Appendices D–K collect partial results, conjectures, and computational evidence bearing on the program of this paper. With the exceptions named in the next paragraph, **nothing in them is claimed as a theorem**: every other statement there is explicitly conditional or empirical, and is labelled either as a *conjecture* (a target statement believed but not proved), as a *conditional proposition* (a deduction valid under a hypothesis that is itself only empirically verified at a finite horizon), or as a *numerical observation* (a finite computational record).

The exceptions are the finite and combinatorial statements that carry Lean certificates, listed in Table 2. Propositions 116, 121 and 124 are unconditional at the finite horizons they name, and Proposition 114 is unconditional outright. These are theorems, and the surrounding conditional material is not. Appendices A and B are not of this kind: their content is unconditional at the finite horizons they state.

D The transcendence program for β_2 : conjecture and three attack routes

Write

$$\beta_2 := \lim_{d \rightarrow \infty} u_d^{1/d}$$

for the growth rate of the universal right-triangle reflection sequence. By Theorem 91 this limit exists and equals

$$\beta_2 = 1/\sqrt{q^*} = 1.4916177871143742268\dots,$$

where $q^* = 0.44945363055894\dots$ is the least positive root of $\Sigma_1(q) = 1$; this is unconditional, and the singularity is a simple pole ($\Sigma_0(q^*) \neq 0$, Lemma 90). The consecutive ratios u_{d+1}/u_d are discussed in Remark 94. Only the arithmetic nature of β_2 remains open.

Conjecture 113 (Transcendence of β_2). *β_2 is transcendental over $\mathbb{Q}$.*

The conjecture as first stated also asserted that the generating function $U(t) = \sum_{d \geq 0} u_d t^d$ is transcendental over $\mathbb{Q}(t)$; that half is now proved, unconditionally, in [12]. Only the number β_2 remains open.

We collect three independent attack routes. Each rests on a different finite-horizon hypothesis that is empirically verified and structurally attackable; the routes are logically independent and each was aimed at the transcendence of $U(t)$, now proved by another route in [12]. They are recorded as history.

D.1 Route 1: Christol mod- p via the explicit cyclic-module structure

Let $\rho_{\text{abs}} : W \rightarrow Q \ltimes M$ be the abstract realization of Lemma 46. For each prime p , write

$$S_p := \{ (\pi(g), m \bmod p) : (g, m) = \rho_{\text{abs}}(w), w \in W \} \subset Q \times (M/pM).$$

Proposition 114 (Mod- p state space is infinite). *For every prime p , $|S_p| = \infty$. In particular the infinite family $\{((XY)^n, t_n) : n \geq 1\}$ (with t_n the translation part of $\rho_{\text{abs}}((R_0 R_2)^n) \bmod p$) consists of pairwise distinct elements of $Q \times (M/pM)$.*

Sketch. XY has infinite order in D_∞ , so the Q -components $(XY)^n$ are pairwise distinct. This alone suffices for infinity of S_p via projection. The stronger separation on the (M/pM) -coordinate, used in Route 1 below, follows from the witness monomial $(XY)^{n-1}X \cdot h$ surviving reduction modulo p by the cyclic-module relations $(Y + 1)h = (c + 1)h = 0$ of Lemma 39.

Numerical record. An exact-arithmetic Rust BFS in $\mathbb{F}_p[Q]$ enumerates $S_p(d) := \{\rho_{\text{abs}}(w) \bmod p : |w|_W \leq d\}$ at three primes:

p	$d_{\max}$	$ S_p(d_{\max}) $	β_p (ratio fit, last 10 depths)
3	42	65 428 964	1.4681
5	40	62 736 544	1.5048
7	40	65 724 148	1.5077

The implementation uses the fixed-degree truncation justified by $(Y + 1)h = (c + 1)h = 0$, which bounds the support of any reachable $m \in M/pM$ by $O(d)$ monomials.

Proposition 115 (U transcendence: Christol route, conditional). *Conditional on the Myhill–Nerode separation hypothesis (the states associated to $\{((XY)^n, t_n)\}_{n \geq 1}$ in the p -DFAO for $u_d \bmod p$ are pairwise distinct future-equivalence classes), $U(t)$ is transcendental over $\mathbb{Q}(t)$.*

Sketch. Under the separation hypothesis the p -DFAO has infinitely many states, hence (Allouche–Shallit Thm. 6.6.2) the p -kernel of $U(t) \bmod p$ is infinite [6], so by Christol [18, 19] $U(t) \bmod p$ is not algebraic over $\mathbb{F}_p(t)$. Adamczewski–Bell [3] lifts this to non-algebraicity of $U(t)$ over $\mathbb{Q}(t)$. The conclusion is transcendence of U , and it stops there; see Remark 119.

D.2 Route 2: D-finite exclusion

Proposition 116 (No D-finite recurrence at low complexity). *Let (k, m) satisfy $1 \leq k \leq 9$, $m \leq 7$ and the over-determination condition*

$$(k + 1)(m + 1) < 43 - k, \quad (7)$$

the right side being the number of equations available from $u_0, \dots, u_{42}$ at order k . There are 52 such pairs, and on none of them does a nonzero rational family $(c_{j,t})_{j \leq k, t \leq m}$ with

$$\sum_{j=0}^k \sum_{t=0}^m c_{j,t} n^t u_{n-j} = 0 \quad (k \leq n \leq 42) \quad (8)$$

exist; that is, the sequence admits no D-finite (P -recursive) recurrence of order k with polynomial-in- d coefficients of degree m on the 43 known terms.

Proof. System (8) is homogeneous, with $(k + 1)(m + 1)$ unknowns and $43 - k$ equations, so non-existence of a nonzero solution is exactly full column rank of its coefficient matrix. Both steps are machine-checked (Appendix K). First, non-existence propagates downward along the componentwise order on (k, m) , so only the six maximal pairs $(3, 7)$, $(4, 6)$, $(5, 5)$, $(6, 4)$, $(7, 3)$, $(9, 2)$ of the grid need treatment (`DFiniteReduction.no_dfinites_of_maximal`). Second, for each

of those the Lean kernel evaluates a left inverse, modulo the Mersenne prime $2^{31} - 1$, of a square submatrix cut out by an explicit choice of $(k + 1)(m + 1)$ rows; reduction modulo a prime can only lower rank, so a modular left inverse forces $\det \neq 0$ over $\mathbb{Z}$ and hence a trivial kernel over $\mathbb{Q}$ (`ModularRankCertificate.eq_zero_of_modular_left_inverse`). The assembled statement is `NoDFiniteCertificates.no_dfinitite`. No step uses `native_decide`. $\square$

Remark 117 (Why the guard is not optional). Condition (7) is what makes a negative result mean anything. Of the 72 pairs with $k \leq 9$ and $m \leq 7$, exactly 20 violate it, and on those the linear system is underdetermined, so a solution exists for *any* input sequence whatever and no exclusion can be read off. The extreme case is $(k, m) = (9, 7)$, with 80 unknowns against 34 equations. An earlier version of this proposition quantified over all 47 tested pairs without the guard and was therefore false as stated; the search method of Appendix A imposes the corresponding condition explicitly. The same remark applies to any extension of the search: raising k or m without also lengthening the sequence buys nothing.

The underlying fact is elementary and is machine-checked in `lean/with_mathlib/OverDetermination.lean`. A homogeneous system with strictly more unknowns than equations has a nontrivial solution for *every* coefficient matrix (`exists_nonzero_solution`), so a search over such a shape reports the existence of a relation whatever the input data are. Contrapositively, an exclusion that is correct forces unknowns $\leq$ equations (`le_of_exclusion`); that is the guard, and its failure convicts the claim. The same file proves the two counts quoted here, `vacuous_card = 20` of the 72 pairs and `252 > 38` for the nonlinear shape, and records that the guard is necessary but not sufficient (`guard_not_sufficient`): passing it still leaves the rank of the coefficient matrix to be computed.

Proposition 118 (*U* transcendence: D-finite route, conditional). *Conditional on extending Proposition 116 to all $(k, m) \in \mathbb{N}^2$, $U(t)$ is transcendental over $\mathbb{Q}(t)$.*

Sketch. By Stanley [38, Thm. 6.4.6], every power series algebraic over $\mathbb{Q}(t)$ is D-finite. Excluding D-finite at all (k, m) therefore excludes algebraic, and gives transcendence of U ; it says nothing about β_2 , by Remark 119. The natural structural target for the extension is a Gröbner-basis argument on the skew-polynomial ring $\mathbb{Q}\langle n, S \rangle$ acting against the cyclic-module relations $(Y + 1)h = (c + 1)h = 0$.

Remark 119 (The direction of the Flajolet–Sedgewick step, and what these routes do not give). Earlier versions of both routes ended with the sentence that Flajolet–Sedgewick [21, Thm. VII.7] “implies β_2 transcendental”. That inference is backwards. Theorem VII.7 says that an algebraic generating function has finitely many singularities, all algebraic branch points or poles, so that *algebraic U forces algebraic β_2* . Contrapositively it lets one deduce transcendence of U from transcendence of β_2 , which is the opposite of what is wanted here.

The converse is false, so no repair of the wording will do. Take

$$U_*(t) = \frac{1}{1 - 2t} + \sum_{d \geq 0} t^{2^d} \in \mathbb{Z}[[t]].$$

Its coefficients are 2^n plus the indicator that n is a power of two, so the growth rate is 2, an algebraic number, while the lacunary part gives the unit circle as a natural boundary and U_* is transcendental over $\mathbb{Q}(t)$. Transcendence of the series is thus compatible with an algebraic growth rate.

Consequently neither route above reaches β_2 , and neither does Route 3. What they reach, conditionally, is transcendence of U , and that is now known unconditionally by the companion paper [12], which supersedes them. Transcendence of β_2 itself remains open, and these appendices contain no approach to it.

D.3 Route 3: Mahler structure

Conjecture 120 ($U(t)$ is not a class-(1) Mahler function). *$U(t)$ is not a Mahler function of class (1) in the sense of Adamczewski–Faverjon [4]; for the general theory of Mahler functions see [27, 31, 9, 2], and for the transcendence machinery that a positive answer would feed, [30].*

Were Conjecture 120 established, the Adamczewski–Faverjon framework would yield a third independent route to transcendence of U , parallel to Propositions 115 and 118; by Remark 119 it would not reach β_2 .

D.4 Joint picture

The three conditionals are logically independent, and all three bear on the transcendence of U rather than of β_2 . In [12] the transcendence of U is conditional: on the travel invariance (T), on the junction non-vanishing (R-J), and on the transfer model (M). The routes recorded here are therefore retained as alternatives that remain open, not as superseded ones. The second half of Conjecture 113, transcendence of β_2 , is untouched by any of them.

E Structural exclusions for $U(t)$ at low complexity

The following finite linear-algebra exclusions on $u_0, \dots, u_{42}$ constrain the form of $U(t)$ at the verifiable horizon. They overlap Proposition 98 of Appendix A; the relation between the two is recorded in Remark 125.

Proposition 121 (No low-order linear recurrence). *The sequence $u_0, \dots, u_{38}$ satisfies no linear recurrence over $\mathbb{Q}$ of order ≤ 19 . (Equivalently: for every $k \leq 19$, the system $u_n = \sum_{j=1}^k c_j u_{n-j}$ has empty rational solution set on the available terms.)*

Proposition 122 (No nonlinear or modular structure at low complexity). *On $u_0, \dots, u_{42}$:*

- (i) *No D -finite recurrence with $k \leq 9$ and $m \leq 7$ (Proposition 116).*
- (ii) *Withdrawn. An earlier version asserted that no nonlinear polynomial relation $\sum_{\alpha} q_{\alpha}(n) u_n^{\alpha_0} \dots u_{n-w}^{\alpha_w} = 0$ holds with lookback $w \leq 5$, total u -degree $|\alpha| \leq 3$ and coefficient degree ≤ 2 . That search is vacuous: the monomials of total degree at most 3 in six variables number $\binom{9}{3} = 84$, so with three coefficient degrees there are 252 unknowns against the 38 equations available from $u_0, \dots, u_{42}$. An underdetermined system has solutions for every input, so nothing is excluded. No nonlinear relation of that shape is testable at all on 43 terms.*
- (iii) *No non-trivial modular linear recurrence at any modulus $m \in \{3, 4, \dots, 101\}$ over the tested range, and at $m = 2$ the order-2 Fibonacci recurrence $u_d \equiv u_{d-1} + u_{d-2} \pmod{2}$ holds for $d \geq 3$ but not at $d = 2$, where $u_0 + u_1 = 4 \equiv 0$ while $u_2 = 5 \equiv 1$.*

Proposition 123 (Coefficient-height gap). *$\log u_d = c \cdot d + o(d)$ with $c = \log \beta_2 = 0.39986129\dots$, this being forced by the growth rate of Theorem 91; ordinary least squares on $d \in [25, 42]$ returns 0.407927, with $R^2 \geq 0.999$, the gap from $\log \beta_2$ being the usual finite-horizon bias. An earlier version quoted $c \approx 0.404$, which matches neither: $e^{0.404} = 1.49780\dots$ against $\beta_2 = 1.49161\dots$*

The consequence to be drawn is only the weak one. Since u_d grows exponentially it is unbounded, so it is not the value sequence of any k -automatic integer sequence, automatic sequences taking finitely many values; no height-gap theorem is needed for that. In particular the earlier appeal to a height gap to exclude Mahler functions was misplaced: by Adamczewski, Bell and Smertnig [1] the

coefficient sequences of Mahler functions realise five distinct growth behaviours, all of them attained, so exponential growth excludes nothing.

Proposition 124 (Mod-2 algebraicity). $u_d \bmod 2$ is eventually 3-periodic, with pre-period 1: $u_d \equiv 1$ for $d \not\equiv 0 \pmod{3}$ or $d \leq 2$, and $u_d \equiv 0$ otherwise. It is not purely periodic, since $d = 0$ is $\equiv 0 \pmod{3}$ yet $u_0 \equiv 1$. The order-2 Fibonacci recurrence $u_d \equiv u_{d-1} + u_{d-2} \pmod{2}$ holds for $d \geq 3$, and fails at $d = 2$: $u_0 + u_1 = 4 \equiv 0$ while $u_2 = 5 \equiv 1$. Hence $U(t) \in \mathbb{F}_2[[t]]$ is algebraic over $\mathbb{F}_2(t)$ via a 4-state finite automaton.

Caveat. This is a characteristic-2 statement and does not lift to algebraicity over $\mathbb{Q}(t)$.

Remark 125 (Relation to Appendix A). Proposition 121 and Proposition 98(i) are exclusions of the same kind at different horizons. Proposition 98(i) is the stronger statement, order ≤ 21 on the 43 terms $u_0, \dots, u_{42}$, and rests on the fraction-free elimination of Appendix A, outside Lean. Proposition 121 is the weaker statement, order ≤ 19 on the 39 published terms, and is the one carrying a Lean certificate (Appendix K). The two are consistent, and neither is promoted by the other.

The guard of Remark 117 applies verbatim to Proposition 98(ii), and is the reason that part is stated on the over-determined cells of the box rather than on the whole box. Of the 119 cells with $(\rho + 1)(\delta + 1) \leq 32$, exactly 13 fail $(\rho + 1)(\delta + 1) < 43 - \rho$: the cells $(\rho, \delta) = (14, 1)$ and $(15, 1)$, with 30 and 32 unknowns against 29 and 28 equations, and the cells $(\rho, 0)$ for $21 \leq \rho \leq 31$. On those the system is underdetermined, so a nonzero solution exists for every input sequence whatever and a search proves nothing; they are excluded from the statement. An earlier version of the proposition named the plain box $(\rho + 1)(\delta + 1) \leq 32$ and therefore claimed more than the search covers.

F Universality class: scope and known examples

The mechanism behind universality in this paper, the cyclic-ideal identification $I_A \cap I_B = \mathbb{Z}[Q] \cdot h$ of Lemma 39 with explicit central generator h inside the linear quotient Q of a free product $W = A * B$, admits a natural class of free-product reflection systems.

Conjecture 126 (Open question: scope of the cyclic-ideal mechanism). *Characterise the pairs (W, ρ) for which the cyclic-ideal mechanism of Lemma 39 applies and yields universality of the orbit-growth sequence across the family of geometric realisations.*

Two examples are currently known.

Example A: right triangles. $A = V_4$, $B = \mathbb{Z}/2$, $Q = D_\infty \times \mathbb{Z}/2$, $h = (c - 1)(Y - 1)$. This is the case treated above (Lemma 39, Theorem 20).

Example B: hyperbolic two-right-angle quadrilaterals. $A = B = V_4$, $W = V_4 * V_4$, $Q = D_\infty \times \mathbb{Z}/2$, $h = c - 1$ (no $(Y - 1)$ factor needed: centrality of c alone suffices when both factors of the free product contain c). This example is recorded as a claim only: we have neither a proof that the mechanism applies nor a computational artifact for it, and it should not be relied on.

Cases where the mechanism is structurally blocked. Acute and obtuse triangle reflection groups have no central involution playing the role of c . Equilateral and other triangle groups with all finite Coxeter labels lack the free-product structure. The rectangle group $D_\infty \times D_\infty$ is a direct product, not a free product. Right-corner orthoschemes in dimension $n \geq 3$ carry cross-relations $(R_i R_j)^2 = 1$ between generators of distinct free-product factors, again destroying the $W = A * B$ structure; universality in this cyclic-ideal sense is therefore specific to dimension 2.

G Structural reading of universality (algebraic-channel BFS)

Fix $\rho_{a,b} : W \rightarrow \text{Aff}(\mathbb{R}^2)$. Let Y denote the linear part of $\rho_{a,b}(R_2)$ and v_2 its translation part. The identity $Yv_2 = -v_2$ holds for every (a, b) and is the image of a single canonical word $w_\star \in W$. Let N_{alg} be the normal closure of $w_\star$ in W .

Numerical observation 127 (Empirical algebraic form of universality). Through depth 22, on the four test right triangles $(3, 4)$, $(5, 12)$, $(1, 2)$, $(7, 24)$, and within the horizon in which universality itself holds (Theorem 20: proved to depth 32, failing at depth 33 for $(1, 2)$), the equivalence relation on canonical Coxeter words generated by $Yv_2 + v_2$ in $\mathbb{Z}[D_\infty]$ exactly recovers the affine equivalence relation. Equivalently: $N(T) = N_{\text{alg}}$ on the depth- ≤ 22 filtration for every tested T .

A parallel BFS verifies this with cumulative state count 118 938 at depth 22 in every channel: four channels for the four test triangles above and a fifth for the algebraic model N_{alg} against which they are compared. Reading: the eight length-10 affine relations of Theorem 10 are not sporadic length-10 coincidences but the first eight consequences of the single quadratic relation $Yv_2 + v_2 = 0$, made visible by the canonical-word filtration.

H Empirical evidence against finite confluent rewriting

Knuth–Bendix experiment. GAP KBMAG ran Knuth–Bendix completion on W together with the eight length-10 affine relations of Theorem 10; the procedure produced 32 730 derived rules in approximately 8 seconds of `kbprog` computation before hitting the equation-count cap, never reaching confluence. This is empirical evidence (not a proof) that W/N has no finite confluent rewriting system in shortlex on the standard generators.

Antolín–Ciobanu route (structurally blocked). Antolín and Ciobanu prove that the conjugacy-growth series of a non-elementary acylindrically hyperbolic group is *transcendental* [7]; an earlier version of this paragraph called it a rationality theorem, which inverts the result, and attributed it to the wrong journal. Either way it does not apply here, and for a reason that is about the group and not about the series: the image of $W/\ker(\rho_{\text{abs}})$ lies inside $Q \rtimes M \subseteq (\mathbb{Z} \wr \mathbb{Z}) \rtimes V_4$, a virtually metabelian group, which is not acylindrically hyperbolic. Note also that $\mathbb{Z} \wr \mathbb{Z}$ is *not* finitely presented, by Baumslag [8]; the earlier description of the ambient group as finitely presented was wrong, and is not needed for the conclusion.

I Class A and Class B in 3D

The 3D right-corner orthoscheme reflection groups fall into three faithfulness regimes by leg-symmetry. The Class C case, pairwise distinct legs, is treated in the n -dimensional companion [11], where the exclusion of rank-two relations is proved and faithfulness itself is stated as a conjecture; it is not a theorem, and nothing below assumes it.

Class A ($a = b = c$, S_3 -symmetric, polynomial growth). The $(1, 1, 1)$ cube-corner orthoscheme is the fundamental chamber of the affine Weyl group $\tilde{C}_3$. For $n \geq 1$, $u_n = (8n^2 + c_{n \bmod 5})/5$ with $c = (10, 12, 13, 13, 12)$, with asymptotic $u_n \sim (8/5)n^2$. The closed form, its integrality and its agreement with the Bott series of $\tilde{C}_3$ hold for $n \geq 1$; in particular $u_1 = 4$ and $u_2 = 9$. Order-7 linear recurrence with characteristic polynomial $(x-1)^3\Phi_5(x)$, obtained from the series $W_A(t) = N(t)/((1-t)^3\Phi_5(t))$ with $N(t) = 2 + 3t^2 + 3t^3 + 3t^4 + t^5 + 4t^6$. Two cautions about that

display. Its constant term is $N(0) = 2$, whereas an orbit-growth series has constant term 1, so as written it is not the growth series but that series plus one; the closed form for $n \geq 1$ is unaffected, and the discrepancy sits entirely in degree 0. And “parabolic Poincaré series” is a misnomer for it. This is a clean restatement of standard affine-Weyl-group material (Bourbaki [14, Ch. VI, §4]; Humphreys [23, §4.7]) and is recorded here for completeness.

Class B (exactly two equal legs, intermediate exponential rate). Through depth 18 the sequence on $(1, 1, 2)$ is 1, 4, 9, 18, 35, 67, 129, 249, 480, 925, 1783, 3437, 6625, 12770, 24591, 47298, 90948, 174939, 336544, with empirical ratio settling around ≈ 1.92 . No order- ≤ 9 linear recurrence over $\mathbb{Q}$ fits the depth-18 data. No closed form is known.

J Material recorded in the project repository

The following items belong to the same program and are recorded in the project repository rather than here:

- Numerical and structural analysis of the representation gap β_n vs. r_n across dimensions.
- Collision-class enumeration to depth 30, the structural sequence battery, the $\sqrt{3}$ -asymptotic of $\delta_d - 2z_d$, and the crossover $\delta_d > u_d$ for $d \geq 24$.
- $\{3, 4, 5\}$ -cascade observation for maximal outerplanar graphs on $n \leq 15$ vertices, the slack/denominator/precision analysis, and the cascade-existence conjecture.
- The standard 3-simplex remark and merged length-10-relation cascades from Pythagorean triples sharing a common edge length.
- Per-script hardware envelopes and runtime tables for the BFS implementations.

K Machine verification of the appendix propositions

The finite and combinatorial content of the propositions of Appendices D and E is machine-checked in Lean 4, in files additional to those of Appendix B. Two files in the Mathlib-free project `lean/` carry part of it, so they compile in core Lean with no dependency fetch; a clean `lake build` rechecks the whole project, including them, in about four minutes.

`lean/PaperExtraMod2.lean` discharges Proposition 124 on the 39 published terms: the closed form $u_d \bmod 2$, eventual 3-periodicity with pre-period 1, the failure of pure periodicity at $d = 0$, the order-2 Fibonacci recurrence for $d \geq 3$, and its failure at $d = 2$. It also records the agreement with Fibonacci for $1 \leq n \leq 9$ and the first deviation $u_{10} = 225$. These same two recurrence facts are the mod-2 half of Proposition 122(iii).

`lean/PaperExtraCounts.lean` verifies the search grid of Proposition 116: that the over-determination condition (7) selects exactly 52 pairs, and that every selected pair has strictly fewer unknowns than equations. It also verifies the arithmetic behind the withdrawal in Proposition 122(ii), namely $\binom{9}{3} = 84$ monomials, 252 unknowns and 38 equations, and it proves the general form of the argument used in Proposition 123: for any notion of automaticity whose sequences are finitely valued, an unbounded sequence is not automatic. The finitely-valued property appears as an explicit hypothesis rather than being assumed silently.

`lean/with_mathlib/NoRecurrence.lean` certifies Proposition 121 outright. For each order k with $1 \leq k \leq 19$ the system of $39 - k$ equations in k unknowns is shown to have no rational solution by exhibiting a Farkas witness: an integer vector w with $w^\top A = 0$ and $w^\top b \neq 0$, which forbids a

solution since any rational c would give $w^\top b = w^\top (Ac) = 0$. The witnesses were computed once in exact GMP arithmetic by the Rust tool `code/zeta_probe/tools/norec`; Lean re-runs none of that linear algebra and verifies only the two defining integer dot products, so a reader need not trust the search that produced them. All nineteen witnesses are small, at most 22 digits per entry, so the check is kernel evaluation and carries no compiler-evaluation axiom. The Farkas step itself is proved for arbitrary integer data, so it is visibly independent of this sequence.

`lean/with_mathlib/ModPStateInfinite.lean` certifies Proposition 114. The pairs $((XY)^n, t_n)$ lie in $Q \times (M/pM)$ and are distinct as soon as their first components are, so it suffices that XY has infinite order; neither the translation parts nor the prime p enter, which is why the conclusion is uniform in p . For the 3-4-5 triangle XY is multiplication by $(3+4i)/5$, and writing $(3+4i)^n = a_n + b_n i$ the rotation by n steps is trivial exactly when $b_n = 0$.

The standard argument factors $3+4i = (2+i)^2$ and $5 = (2+i)(2-i)$ in $\mathbb{Z}[i]$ and invokes unique factorization. That machinery is not needed. The recursion $(a_{n+1}, b_{n+1}) = (3a_n - 4b_n, 4a_n + 3b_n)$ fixes $(3, 4)$ modulo 5, since $3 \cdot 3 + 4 = 13 \equiv 3$ and $4 \cdot 3 + 3 \cdot 4 = 24 \equiv 4$; as $(a_1, b_1) = (3, 4)$, every b_n with $n \geq 1$ is $\equiv 4 \pmod{5}$ and in particular nonzero. The formalization records this one-line induction, the multiplicativity of the norm, the group law, and the resulting pairwise distinctness of the rotations.

For Proposition 116 the search grid is reduced rather than enumerated. `lean/with_mathlib/DFiniteReduction` proves that checking all 52 pairs is unnecessary: if $(k, m) \leq (k', m')$ componentwise then a nonzero (k, m) -recurrence yields a nonzero (k', m') -recurrence by padding the coefficient family with zeros. Two things make the padding work, and they pull in opposite directions: the ansatz grows, while the constraints *shrink*, since the (k', m') equations run over $k' \leq n \leq 42$ and that range sits inside the range $k \leq n \leq 42$ of the (k, m) equations. Non-existence therefore propagates downward, and it suffices to treat the maximal pairs. The grid has exactly six, $(3, 7)$, $(4, 6)$, $(5, 5)$, $(6, 4)$, $(7, 3)$, $(9, 2)$, and the file verifies by kernel evaluation that every one of the 52 is dominated by one of them.

The base cases are supplied as rank certificates by `code/zeta_probe/tools/nodfinite`. A D-finite recurrence is a nonzero solution of a *homogeneous* system, so non-existence means full column rank; for each pair the tool exhibits square submatrix rows together with an inverse modulo $2^{31} - 1$, and since reduction modulo a prime can only lower rank, full column rank there gives it over $\mathbb{Q}$. All 52 pairs were certified this way before the reduction was proved, so the six are not a weakening of the search. Importing them into Lean requires the passage from a modular left inverse to $\det M \neq 0$ over $\mathbb{Z}$, and thence to triviality of the rational kernel. That implication is proved in full generality, for an arbitrary size and an arbitrary modulus greater than one, in `lean/with_mathlib/ModularRankCertificate.lean` (`det_ne_zero_of_modular_left_inverse, eq_zero_of_modular_left_inverse`).

Two further files in the same directory instantiate the certificates against it and close the proposition. The first, `NoDFiniteData.lean`, carries the certificate data and the modular identities themselves. It has no Mathlib dependency, and its eleven checks `cert_k_m` are discharged by `decide +kernel`, so the arithmetic runs in the Lean kernel once, with axiom list `[propext]` and no compiler-evaluation axiom: this is tier (i) work, not tier (ii). The check recomputes the coefficient matrix from $u_0, \dots, u_{42}$ and the formula $n^t u_{n-j}$, so the search that produced the data is not trusted, only the data. The companion `NoDFiniteCertificates.lean` supplies the plumbing: the coefficient matrix as a square integer matrix, the column bijection $c \leftrightarrow (j, t) = (c \div (m+1), c \bmod (m+1))$ that turns a coefficient family into a vector, and the passage from the kernel's list arithmetic to `Finset.sum`. Its conclusion `no_dfinit` is Proposition 116 as stated, and its axiom list is `[propext, Classical.choice, Quot.sound]`.

The list layer exists for speed, not for mathematics. The eleven checks together are about 3.5×10^5 multiply-add-mod steps on `Nat` literals, which the kernel completes in well under a minute;

the same identity phrased directly as an equality of matrices over $\mathbb{Z}/p$ does not reduce in usable time. Six of the eleven cells are the maximal pairs of the grid of Proposition 116; the other five, $(0, 31)$, $(1, 15)$, $(2, 9)$, $(13, 1)$, $(20, 0)$, are what the narrowed box of Proposition 98(ii) adds, and `no_recurrence_in_box` covers the union of the two regions.

One convention is worth recording, since the statement of Proposition 116 leaves it implicit. The count 52 is for $k \geq 1$, that is for recurrences of order at least one. Admitting $k = 0$ gives 60 pairs and requiring $m \geq 1$ gives 50; only $1 \leq k \leq 9$, $0 \leq m \leq 7$ reproduces the published figure. The Lean file verifies both counts, so the convention is fixed by the formalization rather than by the prose.

What is not formalized is stated plainly in each file. Not formalized is the passage from eventual periodicity to algebraicity over $\mathbb{F}_2(t)$, which needs formal power series; nor the unboundedness of u_d , which is a statement about all d and rests on the growth rate of Theorem 91. Strict increase across the published range is verified, and is evidence for unboundedness rather than a proof of it. The regression numbers in Proposition 123 are numerics and are not formalized. The terms $u_{39}, \dots, u_{42}$, used by the propositions stated over $u_0, \dots, u_{42}$, are recorded with their provenance in `code/data/u_terms_43.txt`; they were recovered from the stored logarithms, a recovery that reproduces all 39 overlapping published terms exactly, and they have not been regenerated by the growth BFS. For Proposition 114 the identification of XY with the rotation by $(3 + 4i)/5$ is the setup of the 3-4-5 reflection group and is not re-derived in Lean; what is proved there is that the rotation so described has infinite order.

The two conditional routes, Propositions 115 and 118, are not formalized and are not candidates for it here: each is an implication whose antecedent is an open hypothesis and whose engine is Christol's theorem, which is not available in Mathlib. Stating them in Lean with that theorem assumed would record no more than the prose already does.

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